

Equity Strategy - China

China AI: from localization to monetization

Equity Strategy

From technology catch-up to ecosystem competition

China's AI localization is moving from policy ambition to commercial reality. Domestic capabilities and adoption are advancing despite gaps in leading-edge hardware, while improvements in model efficiency and system optimization could support commercial viability before full technological self-sufficiency. Competition is increasingly shifting from technology catch-up to ecosystem scale and monetization. Technology progress and high-profile listings are broadening China's investable AI universe, offering diverse opportunities across localization, global supply chains and applications.

Localization: gaining shares and closing gaps

China continues to make progress toward technology self-sufficiency, albeit unevenly across segments. In AI accelerators, domestic vendors' revenue market share rose to nearly 50% in 2025 and could approach 80% by 2028E, with Huawei leading in training capabilities and other local players gaining traction in inference. In memory, CXMT's global DRAM share rose from 3-4% in 2024 to 10% in 2Q26, while YMTC's NAND share increased from 7-8% to 14%. CXMT remains several years behind global leaders, while YMTC is closer to the NAND frontier. In semi equipment, domestic suppliers' market share reached nearly 30% in 2025, up from 22% in 2019, though significant gaps remain in lithography, ion implantation, and metrology/inspection. Meanwhile, Chinese open-weight LLMs are narrowing the capability gap with leading proprietary models.

Monetization: the AI value chain smile curve

We see China's AI stack evolving into a barbell profit structure, with the strongest economics concentrated at both ends of the value chain: strategic hardware bottlenecks such as AI accelerators, semi equipment, foundries, and memory benefit from high entry barriers and capital intensity, while cloud service providers and large AI platforms enjoy scale advantages, rich use case, and cross-selling opportunities. By contrast, standalone LLM labs and physical AI manufacturers face lower switching costs and more intense competition, limiting pricing power and profitability.

China AI stocks: three different themes to play

We group China AI-related stocks into three buckets with distinct risk-reward profiles. 1) China **AI localization beneficiaries** are aligned with policy priorities and offer diversification from the US AI cycle, but are well owned and richly valued. 2) **AI infrastructure exporters** are proven winners from global AI buildouts, though they are highly exposed to the US spending cycle with emerging sanction risks. 3) **AI application names** represent the ultimate monetization layer with long-term potential, but face near-term challenges from intense competition and the lack of a clear killer APP. If US CAPEX slows, exporters would likely face the greatest earnings and valuation pressure, localization plays could see multiple compression but create buy-the-dip opportunities, while selected deep-value application names may be more defensive.

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Refer to important disclosures on page 70 to 72.

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See glossary on page 68

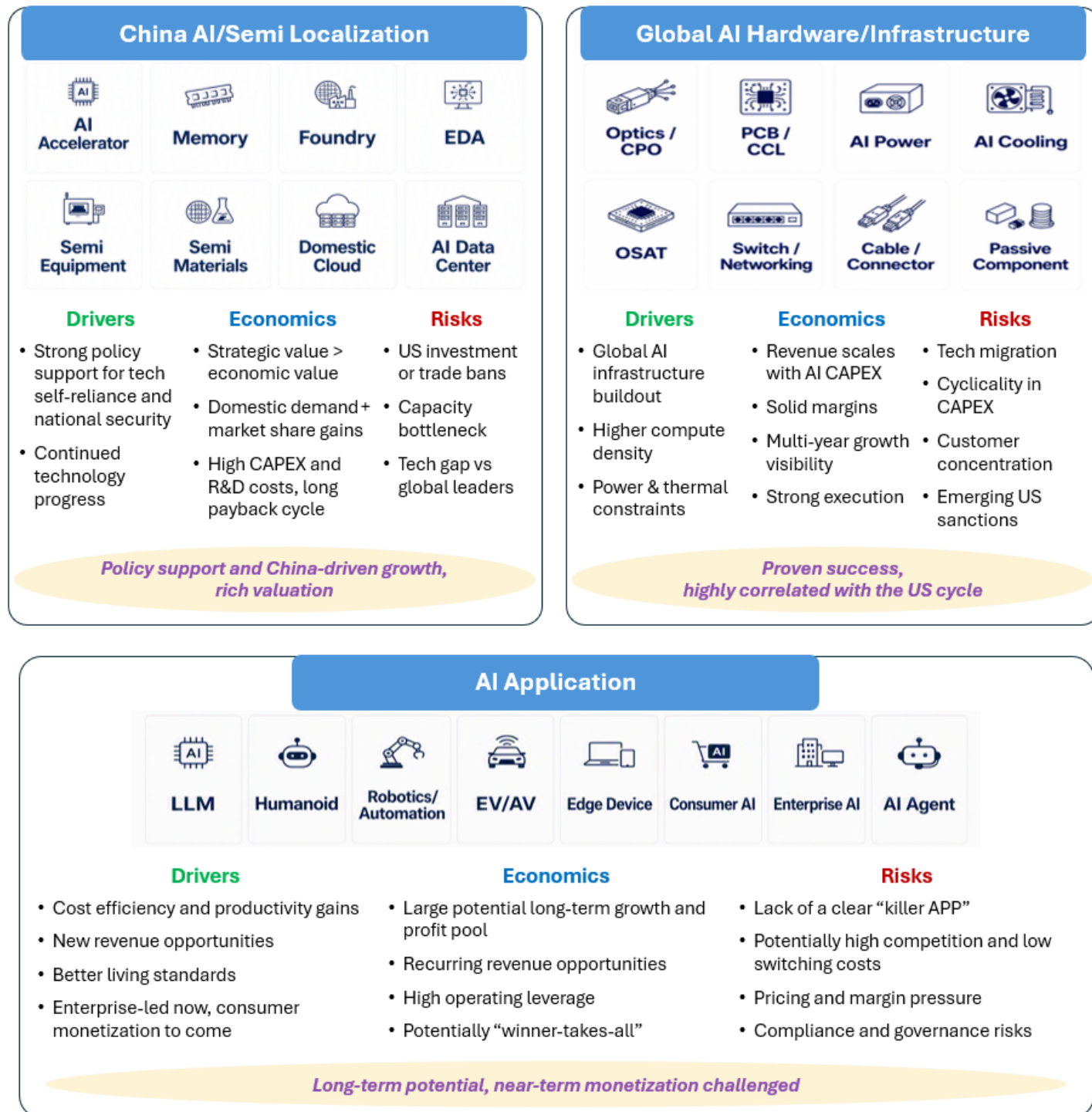
Contents

Localization: from chips to systems	4
Local champions gaining share across the AI stack	4
Closing the technological gap at different speeds	5
The next AI race is monetization	8
AI value chain economics: a smile curve	8
Three investment themes in China AI	12
Local, global and application: distinct risk-return profiles	12
The two ecosystems: not a simple decoupling story	14
China AI stock mapping	15
AI accelerator	18
Memory	23
Semiconductor equipment	30
AI model	34
Status quo of China AI model landscape	35
US and China bring distinct AI model development strengths	38
Smarter spend & AI segmentation: a cheap-and-capable market emerging	40
Token export: more complicated than just low-power cost advantage	45
AI monetization: enterprise-driven, cloud-centric, oversea ramping	46
Alibaba: a clear example of ecosystem-level AI monetization	47
Independent AI labs: fast near-term growth, training first, eager for capital, looking overseas	52
Software	55
Data center	57
Humanoid robot	59
Robust shipment volume growth in 2026	59
Competition landscape: US and China take the lead	60
From R&D to industrial & commercial applications	61
BOM cost trend: to reach USD16k in China in 2030E	65
Glossary	68

Exhibit 1: We categorize China AI related sectors/stocks into three groups with different investment opportunities and risks
China's AI ecosystem and investment framework

CHINA AI ECOSYSTEM AND INVESTMENT FRAMEWORK

From sovereign AI capability and infrastructure to real-world productivity



Source: BofA Global Research

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Localization: from chips to systems

AI localization in China is no longer just a policy ambition. Domestic capabilities continue to improve, local players are gaining market share, adoption is broadening, and the ecosystem is becoming more self-sustaining. While development remains uneven and China still lags global leaders at the technological frontier, progress is tangible and the long-term trajectory is increasingly evident.

China's response to hardware constraints is focused on system-level optimization. Rather than relying on a single breakthrough chip, China is leveraging larger domestic compute clusters, faster interconnects, software-hardware co-design, improved scheduling, and more compute-efficient models to narrow the performance gap. Recent deployments suggest that advanced models can be trained on domestic AI clusters, while new software frameworks are improving interoperability across multiple domestic accelerator platforms. As a result, the localization story is evolving from component substitution toward full-stack optimization.

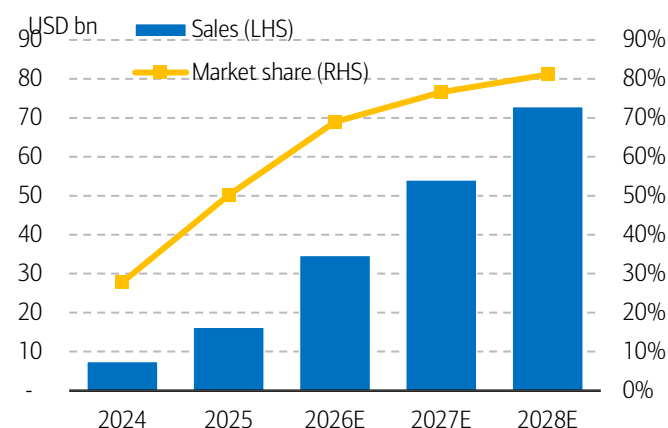
Local champions gaining share across the AI stack

In AI accelerators, Nvidia still led China's market with a 47% revenue share in 2025, but Huawei had already reached 35.5%, making it the clear domestic leader. Based on our estimates, Chinese vendors collectively captured about 50% of market revenue in 2025, vs less than 30% in 2024, and their share is expected to reach nearly 80% by 2028E.

In memory, China remains well behind the leading edge in HBM, but is gaining share rapidly in conventional DRAM and NAND. CXMT's global DRAM revenue share has risen from about 3-4% in 2024 to ~10% by 2Q26, while YMTC's share in NAND increased from 7-8% to ~14% over the same period. Although both firms remain concentrated in less advanced and consumer-oriented products, growing scale, customer adoption, and manufacturing experience should help accelerate technology catch-up.

Exhibit 2: Chinese AI accelerators' market share reached ~50% in 2025

Revenue and market share of domestic AI accelerator vendors in China

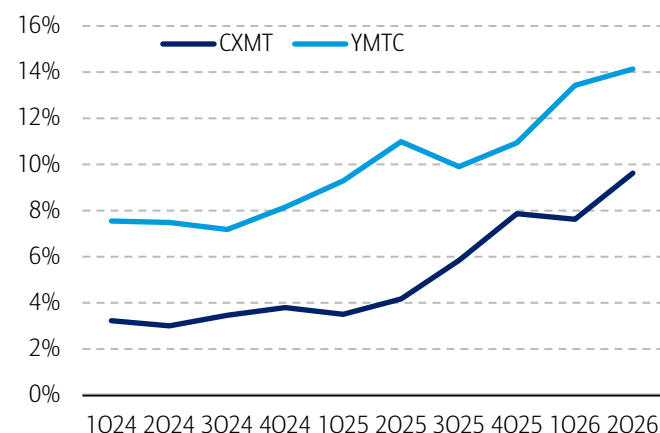


Source: Company reports, IDC, BofA estimates

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Exhibit 3: CXMT and YMTC gained market share in DRAM and NAND

Global revenue market share of Chinese memory firms



Source: Company reports, BofA estimates based on aggregate revenue of known suppliers

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Semi equipment is another major localization battleground. According to SEMI.org, China's equipment billings reached USD49.3bn in 2025, broadly flat YoY but up from USD19bn in 2020, implying a five-year CAGR of 21%. China remained the world's largest equipment market for the sixth consecutive year. Based on China's equipment import data, we estimate that the domestic suppliers' share reached around 30% in 2025, up from 22% in 2019. However, localization remains highly uneven: Chinese vendors have made notable progress in cleaning, etching, CMP and deposition, while advanced lithography, ion implanting, and metrology/inspection are still bottleneck areas.

Exhibit 4: Domestic suppliers achieved notable market shares, while lithography, ion implanting and metrology/inspection are still the bottlenecks

China WFE localization progress by equipment type

Equipment	Local suppliers' market share		Domestic players	Global leaders
	Prior to 2025	now		
Lithography	<5%	<5%	SMEE, Yuliangsheng	ASML, Nikon, Canon
Etching	near 30%	near 40%	AMEC, NAURA, E-Town	Lam Research, TEL, AMAT
Deposition	near 20%	>40%	NAURA, Piotech, Leadmicro, SiCarrier	AMAT, Lam Research, TEL
Ion implantation	<5%	10-15%	Wanye, CETC, NAURA	AMAT
PR development	near 10%	>10%	Kingsemi	TEL
CMP	>30%	>50%	Hwatsing	AMAT, EBARA
Cleaning & PR stripping	>35%	>50%	NAURA, ACM Research, Kingsemi, E-Town	SCREEN, TEL, Lam Research
Metrology/inspection	na	>10%	Skyverse, Jingce, SiCarrier	KLA, AMAT

Source: BofA Global Research estimates

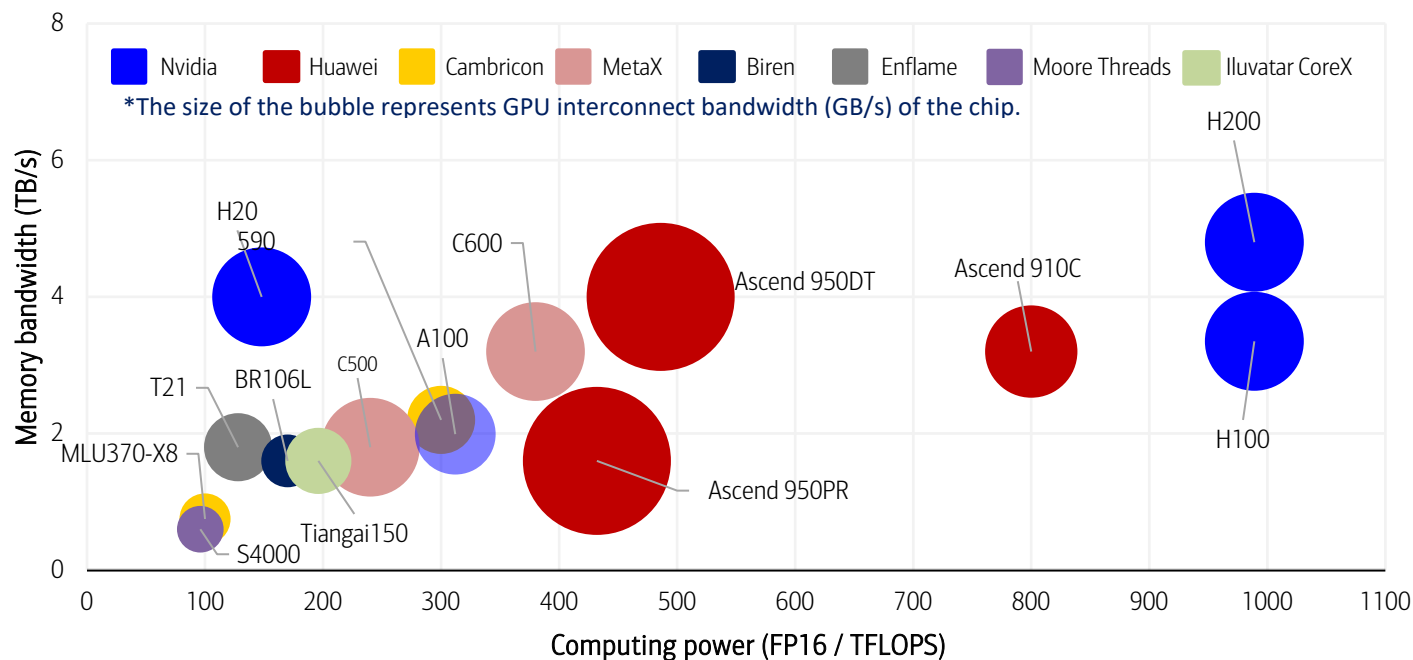
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Closing the technological gap at different speeds

Domestic AI accelerators are gaining ground, with Huawei leading in single-chip performance, clear roadmap and training capabilities. Other Chinese players are getting competitive in inference, while recent listings improved transparency and disclosure.

Exhibit 5: Domestic AI chips are narrowing the gap vs. Nvidia's H100 / H20 chips, especially Huawei

Selected AI accelerator performance across global and domestic vendors



Source: Companies, BofA Global Research; Note: Interconnect bandwidth for Iluvatar Tiangai-150 is not disclosed; values shown are based on estimates

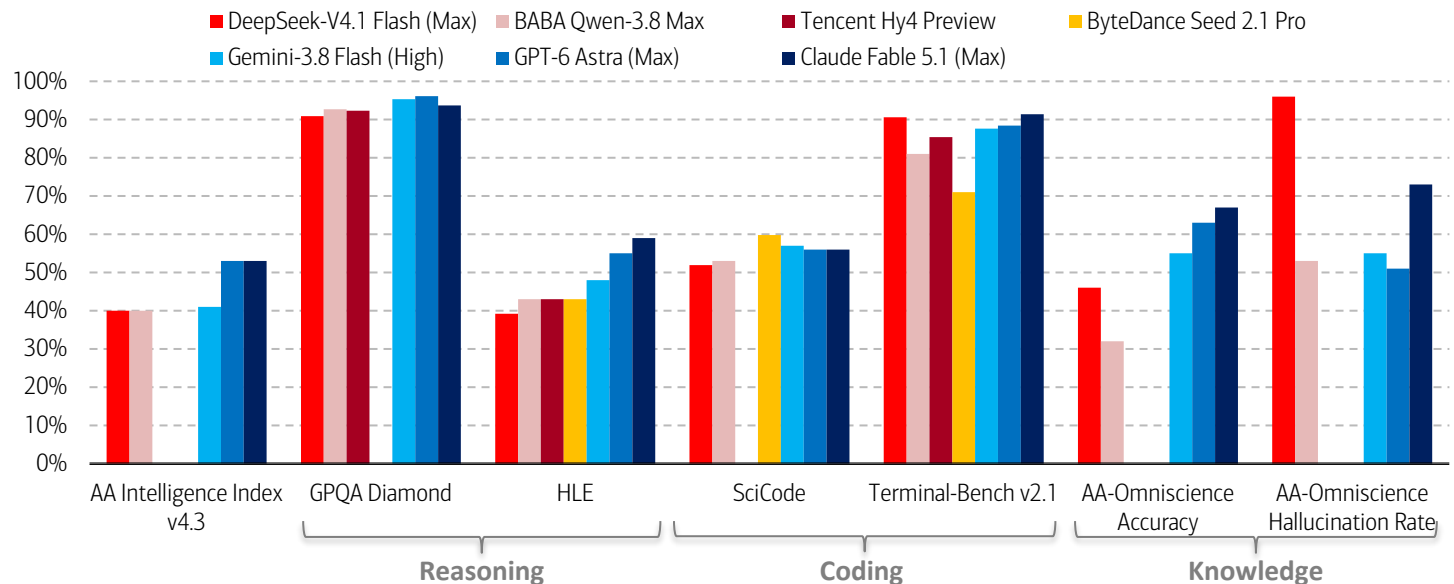
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Chinese open-weight LLMs are rapidly narrowing the gap with proprietary frontier models. Recent releases such as DeepSeek V4.1 and Kimi K3 have demonstrated competitive reasoning, coding, and agentic capabilities at materially lower inference costs. As a result, Chinese LLMs are carving out the “value-for-money” segment of the global AI market. While they may not consistently lead at the frontier, they are increasingly good enough for mainstream workloads and significantly cheaper to deploy.

At the same time, US officials and AI companies have accused some Chinese developers of using model distillation to accelerate progress, while Chinese firms and authorities argue that their advances reflect open-source innovation and engineering improvements. Regardless, the debate highlights how AI competition is extending beyond semiconductors into models, intellectual property, and ecosystem leadership.

Exhibit 6: Chinese models are getting competitive

Benchmark comparison of selected Chinese and US models



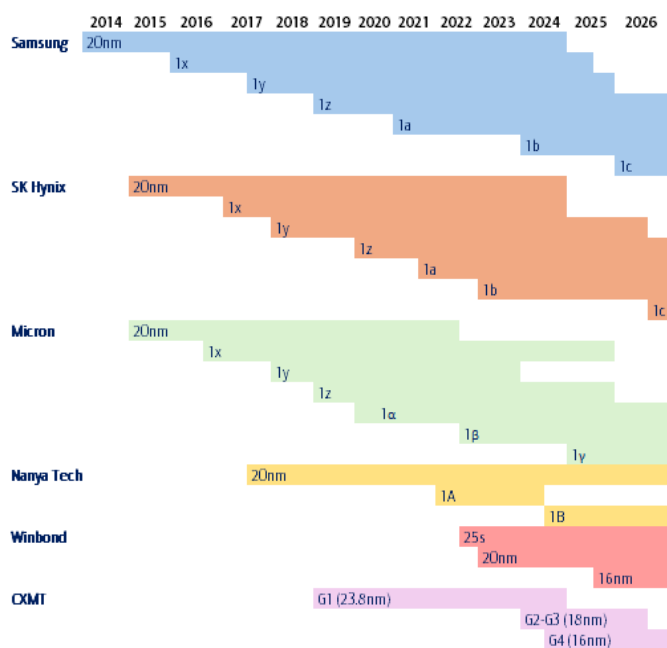
Source: OpenRouter, Artificial Analysis, Vals AI, company websites, BofA Global Research (Data as of September 11, 2026)

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In memory, CXMT is a few years behind global DRAM leaders but is increasingly competitive in mainstream products, having mass produced DDR5 and LPDDR6. Further progress to G5 depends on domestic equipment capabilities, while process and packaging constraints are likely to limit large-scale HBM output over the next 1-2 years. YMTC is advancing from 232-layer to 260+ layer NAND in 2026 and toward 300+ thereafter. It trails the global NAND frontier by roughly one year, with competitive layer density and proprietary Xtacking architecture, but still faces yield and scale challenges.

Exhibit 7: CXMT's G4 is similar to tier-1 suppliers' 1z technology

Major DRAM manufacturers' technology roadmap

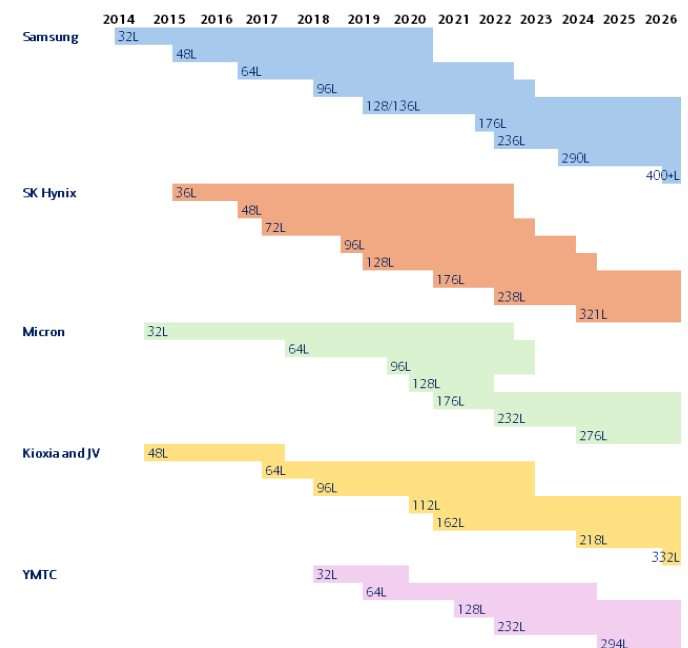


Source: Companies, media reports *Starting point means the mass production commencement date

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Exhibit 8: YMTC's technology level is close to tier-1 suppliers'

Major NAND manufacturers' technology roadmap



Source: Companies, media reports *Starting point means the mass production commencement date

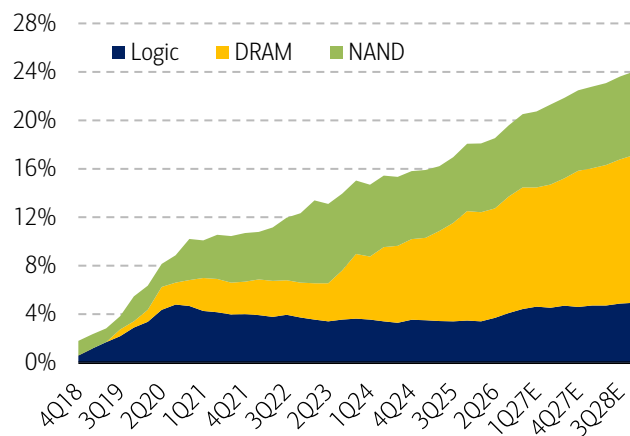
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Advanced-node capacity is expanding. CXMT's new Shanghai fab targets 40-50k wafers per month of HBM production from 2027, according to TrendForce. More broadly, advanced logic at 28nm-and-below and memory accounted for nearly 19% of China's total wafer capacity in 2Q26, up from 15-16% in 2024, based on our estimates. We expect this share to reach 23-24% by 2028E.

In physical AI, China is leveraging the same advantages that underpinned its rise in smartphones and EVs: manufacturing scale, supply-chain depth, rapid product iteration and the ability to generate large volumes of real-world deployment data. Based on our estimates, Chinese vendors accounted for nearly 95% of global humanoid robot shipments in 2025, led by AgiBot and Unitree. While China's share may moderate as Tesla, Figure and other global players scale production, it should remain the industry's largest manufacturing base over the next several years.

Exhibit 9: Fab capacity is moving toward advanced nodes and memory

Advanced logic (28nm-and-below) and memory as % of China's total wafer capacity in China fabs

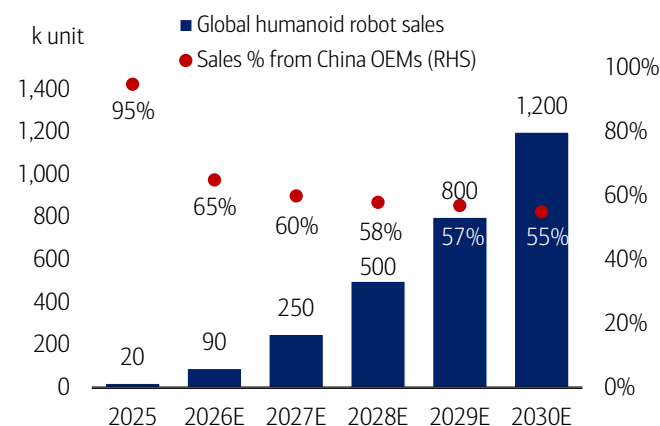


Source: BofA Global Research estimates

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Exhibit 10: China should account for nearly 60% market share in global humanoid robot sales in 2027-28E

BofA global humanoid robot shipment forecast



Source: BofA Global Research estimates

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The next AI race is monetization

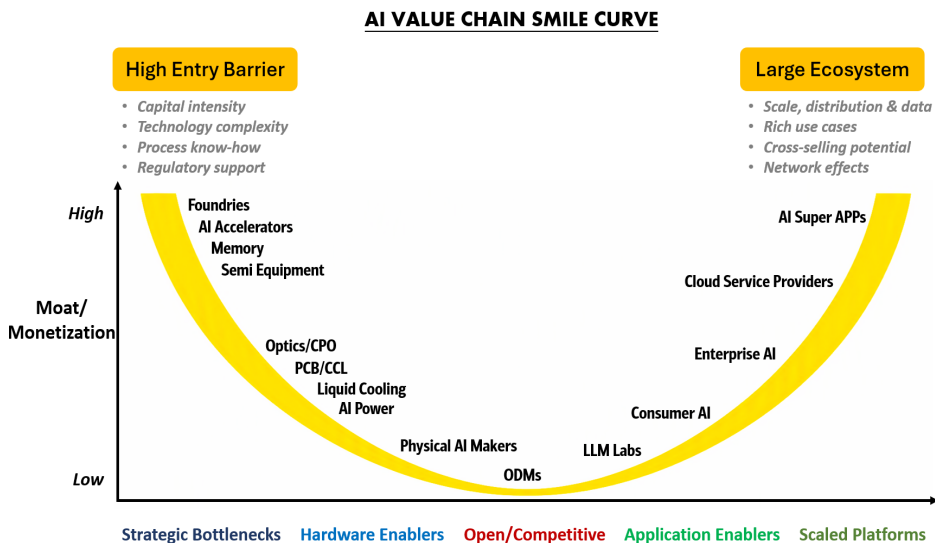
Western export controls have accelerated China's domestic substitution, but strategic importance does not guarantee commercial sustainability. Policy support can enlarge the addressable market, but it can also attract excessive capital and new entrants, leading to duplicated investment, price competition, and margin pressure. Ultimately, localization creates the opportunity, but only players that build durable moats and translate technology leadership into sustainable monetization will capture long-term value.

AI value chain economics: a smile curve

China's AI stack is developing a barbell profit structure, with the strongest economics concentrated at both ends of the value chain. On one side, strategic bottlenecks such as AI accelerators, semi equipment, foundries, and memory benefit from high capital intensity, long development cycles, deep technical know-how, and regulatory barriers, creating durable competitive moats. On the other side, scaled cloud and internet platforms benefit from diverse use cases, scale advantages, cross-selling opportunities and network effects that could support sustainable monetization. By contrast, the middle layers, including standalone LLM labs and physical AI manufacturers, could face lower switching costs, faster imitation, heavy investment and intense price competition, resulting in weaker pricing power and profitability. Long-term value creation could therefore concentrate in strategic bottlenecks and large-scale ecosystems.

Exhibit 11: Long-term valuation creation concentrates in strategic bottlenecks and large ecosystems

AI value chain smile curve: moat and monetization



Source: BofA Global Research

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Strategic bottlenecks: cyclical recovery + structural growth

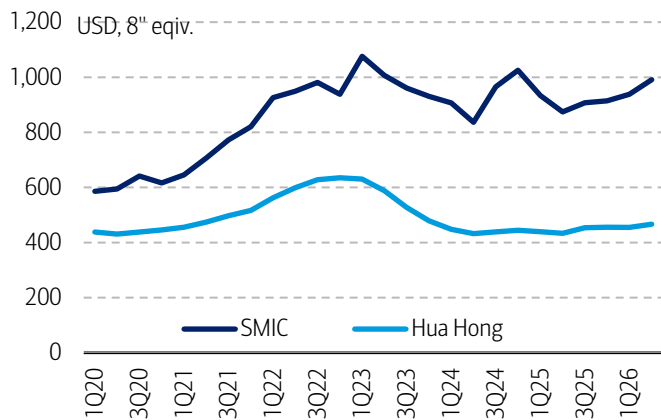
Chinese foundry pricing appears to have troughed. SMIC's blended average selling price rose from USD904 per 8-inch-equivalent wafer in 1H25 to USD965 in 1H26, while Hua Hong's ASP increased from USD437 in 1H25 to USD461 in 1H26. The recovery reflects high utilization, stronger AI-related specialty demand and selective price increases. In 2Q26, reported utilization reached 94% at SMIC and 103% at Hua Hong, supporting greater pricing discipline.

Semi equipment offers a structural domestic-substitution opportunity. US restrictions have accelerated the adoption of Chinese tools, with NAURA rising to fifth and AMEC to thirteenth among global semi-equipment suppliers by 2025 sales. NAURA and AMEC both faced gross-margin pressure in 2025 as they ramped and qualified new products, but both remain profitable and target long-term gross margins above 40%. The key

question is whether today's investment can translate into a larger installed base, recurring service and upgrade revenue, and rational pricing across domestic competitors.

Exhibit 12: Foundry pricing enjoyed modest improvement in 1H26

Blended ASP of Chinese foundries, USD/wafer (8" equivalent)

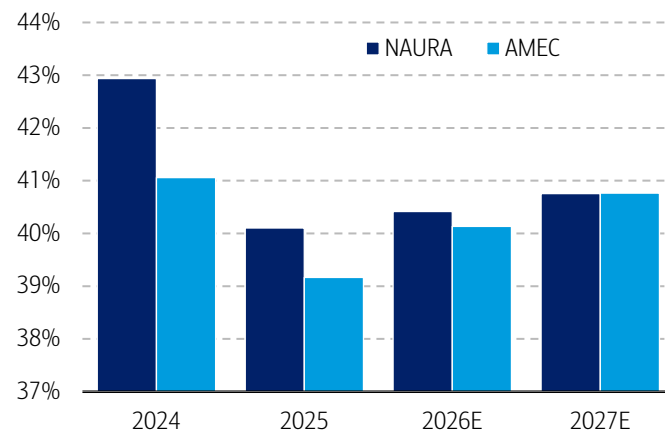


Source: Company reports

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Exhibit 13: Semi equipment players aim to stabilize GPM

Gross margin of leading Chinese semi equipment players



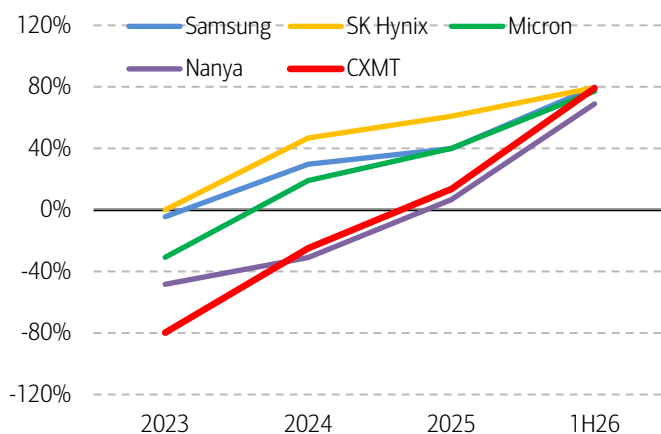
Source: Company reports, Bloomberg consensus

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Memory is more cyclical. CXMT's operating margin reached 79% in 1H26, broadly in line with global leaders and indicative of attractive economics at current pricing. However, memory profitability can reverse quickly as new capacity catches up with demand.

Exhibit 14: CXMT's OPM was on par with global leaders in 1H26

Leading DRAM players' operating margin comparison

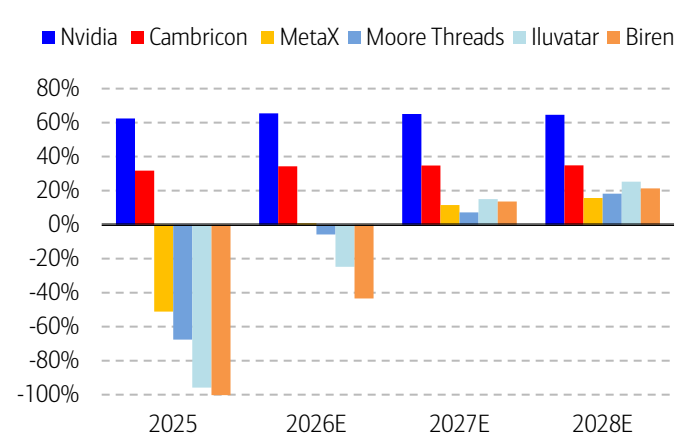


Source: Company reports

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Exhibit 15: Chinese AI chip companies' OPM remained low vs NVIDIA

Operating margin: NVDA vs listed Chinese AI accelerators



Source: Company reports, Bloomberg consensus

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Chinese AI accelerators are structurally attractive but are in the early stages of scaling. Nvidia continues to set the benchmark, with operating margins sustained above 60%, reflecting its dominant ecosystem and pricing power. Chinese accelerators are credible domestic-substitution beneficiaries, but profitability is highly dependent on scale and the AI investment cycle. Among the listed Chinese AI chip names, Cambricon is further along the commercialization curve, having turned profitable in 2025 and with operating margins expected to remain in the mid-30% range through 2028E, based on Bloomberg consensus. In contrast, MetaX, Moore Threads, Iluvatar, and Biren are still relatively early in the investment phase, with profitability expected to increase over time as revenue scales and operating leverage improves.

While localization supports rapid adoption, hardware earnings could become more volatile if AI CAPEX slows. The key question is whether vendors can build sufficient ecosystem adoption and scale to sustain margins while avoiding excessive price competition. Over the longer term, higher-margin software, systems, and services revenue will be critical to support profitability beyond the initial substitution cycle.

LLMs vs CSPs: token deflation, ecosystem monetization

China's LLM market has entered a deflationary phase. Registered GenAI services and applications reached over 980 by June 2026, over 15 times the number at end-2023. While the proliferation demonstrates China's depth in models, applications and talent, supply is running ahead of proven monetization, raising the risk of price competition.

China's LLM price war is evolving from blanket cuts to tiered pricing. DeepSeek and other vendors have driven token costs sharply lower, with DeepSeek V4.1 Flash priced at just USD0.15/USD0.60 per million input/output tokens during off-peak periods, versus USD10/USD50 for Anthropic Claude Fable 5.1. Meanwhile, leading models such as Kimi K3 and Qwen 3.8 Max continue to charge meaningful premiums for stronger reasoning, longer context windows, multimodal capabilities, and agentic workflows. Basic inference is increasingly commoditized, while frontier capabilities remain differentiated. The key economic metric is shifting from price per token to cost per completed task.

The current monetization path appears to be: models attract users and developers; APIs drive adoption and ecosystem dependency; agents, subscriptions and enterprise deployment lift ARPU; and cloud infrastructure and applications capture the larger profit pools. Cloud providers are already benefiting from AI demand, but their returns remain capital-intensive. Alibaba Cloud growth accelerated from 36% YoY in the Dec-25 quarter to 45% in Jun-26, supported by triple-digit AI product growth. Kingsoft Cloud's 1H26 revenue rose 34% YoY and AI cloud billings increased 85% to 40% of total revenue. However, Alibaba's CAPEX rose 75% in the latest quarter, while higher depreciation kept Kingsoft Cloud loss-making on a GAAP basis despite positive adjusted operating profit. These results highlight both the scale of AI opportunity and the capital intensity of AI infrastructure, particularly if frequent GPU and server replacement shorten asset lives.

Exhibit 16: Chinese models are much cheaper on token costs

Current pricing of major US and Chinese LLMs

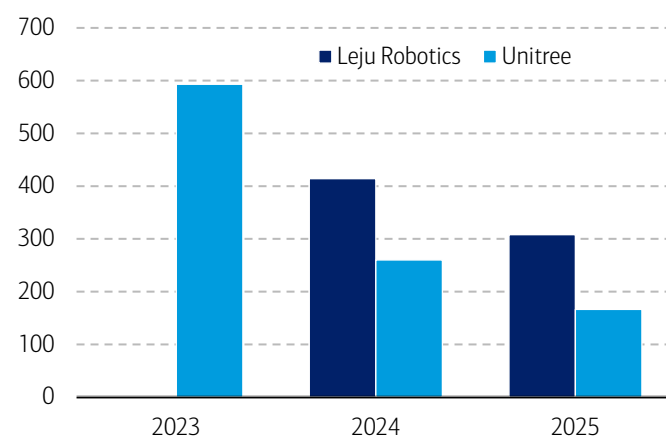
Region	Provider	Model	Input \$/M	Cached input \$/M	Output \$/M
China	DeepSeek	V4.1 Flash	0.2	0.003	0.6
China	Xiaomi	MiMo-V2.5-Pro	0.4	0.004	0.9
China	MiniMax	MiniMax M3	0.3	0.060	1.2
China	DeepSeek	V4 Pro	0.7	0.022	2.0
China	Tencent	Hunyuan4 Preview	0.8	0.040	2.5
China	ByteDance	Seed 2.1 Pro	0.9	0.180	4.5
China	Z.ai	GLM-5.3	1.4	0.260	4.4
China	Alibaba Qwen	Qwen 3.8 Max	2.0	0.200	6.0
China	Moonshot	Kimi K3	3.0	0.300	15.0
US	Google	Gemini 3.8 Flash	0.8	0.075	3.8
US	Google	Gemini 3.1 Pro Preview	2.0	0.200	12.0
US	xAI	Grok 4.6	2.0	0.500	6.0
US	OpenAI	GPT-5.6 Terra	2.0	0.200	12.0
US	OpenAI	GPT-5.6 Sol	4.0	0.400	20.0
US	Anthropic	Claude Fable 5.1	10.0	0.250	50.0

Source: Company reports

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Exhibit 17: Unitree's ASP fell >70% from 2023 to 2025

Annual ASP of Unitree and Leju humanoid robot products (RMB '000)



Source: Company reports, prospectus *Leju Robotics: ASP of Kuavo series

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Among AI application players, Alibaba appears well positioned, with Qwen driving Cloud demand, enterprise deployment and consumer applications while increasing utilization across its infrastructure and proprietary chips. Cloud platforms have structural advantages over standalone LLM providers through the control of compute, enterprise distribution and customer relationships, reducing deployment friction and enabling

cross-selling across models, agents and applications. However, sustainable value creation depends on translating strong demand and high utilization into attractive returns through disciplined CAPEX, durable pricing and longer asset lives. Independent LLM developers face an even tougher path, as they must convert low-priced usage into recurring revenue while funding heavy R&D and compute costs.

Humanoids: lower prices broaden adoption but expose weak economics

China's humanoid robot market is also entering an "involutionary" phase. The country had around 140 manufacturers and introduced more than 330 product models in 2025. Unitree's humanoid robot ASP fell by roughly 72% between 2023 and 2025, while Leju's Kuavo series ASP declined by 26% in 2025. Falling component costs, domestic substitution and vertical integration are driving prices down, although part of the ASP decline also reflects a mix shift toward smaller, lower-specification products.

Unitree is pursuing scale through lower-priced platforms, including the RMB29,900 R1 and USD13,500 G1. Yet over 70% of its 2025 humanoid sales went to universities and research institutions, versus less than 10% for commercial deployment. By contrast, UBTECH focuses on integrated industrial systems: full-size humanoid revenue reached RMB820mn in 2025 at a 54.6% gross margin, although heavy R&D and delivery costs weighed on overall profitability.

The investment case hinges on customer ROI rather than shipment growth. Sustainable returns require deployment at scale, high utilization, and recurring revenue that can offset hardware margin compression.

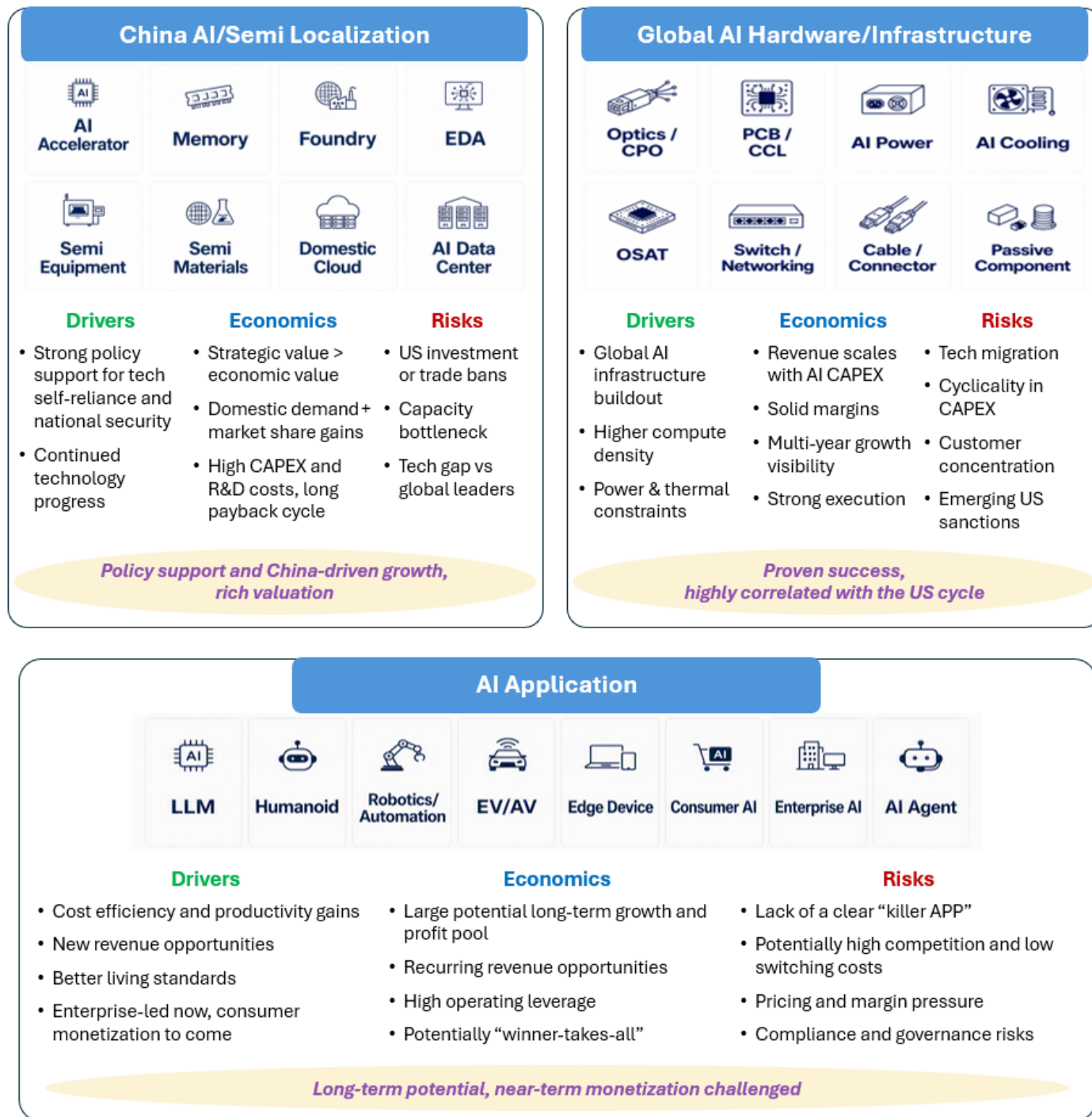
Three investment themes in China AI

Local, global and application: distinct risk-return profiles

Exhibit 18: We categorize China AI related sectors/stocks into three groups with different investment opportunities and risks
China's AI ecosystem and investment framework

CHINA AI ECOSYSTEM AND INVESTMENT FRAMEWORK

From sovereign AI capability and infrastructure to real-world productivity



Source: BofA Global Research

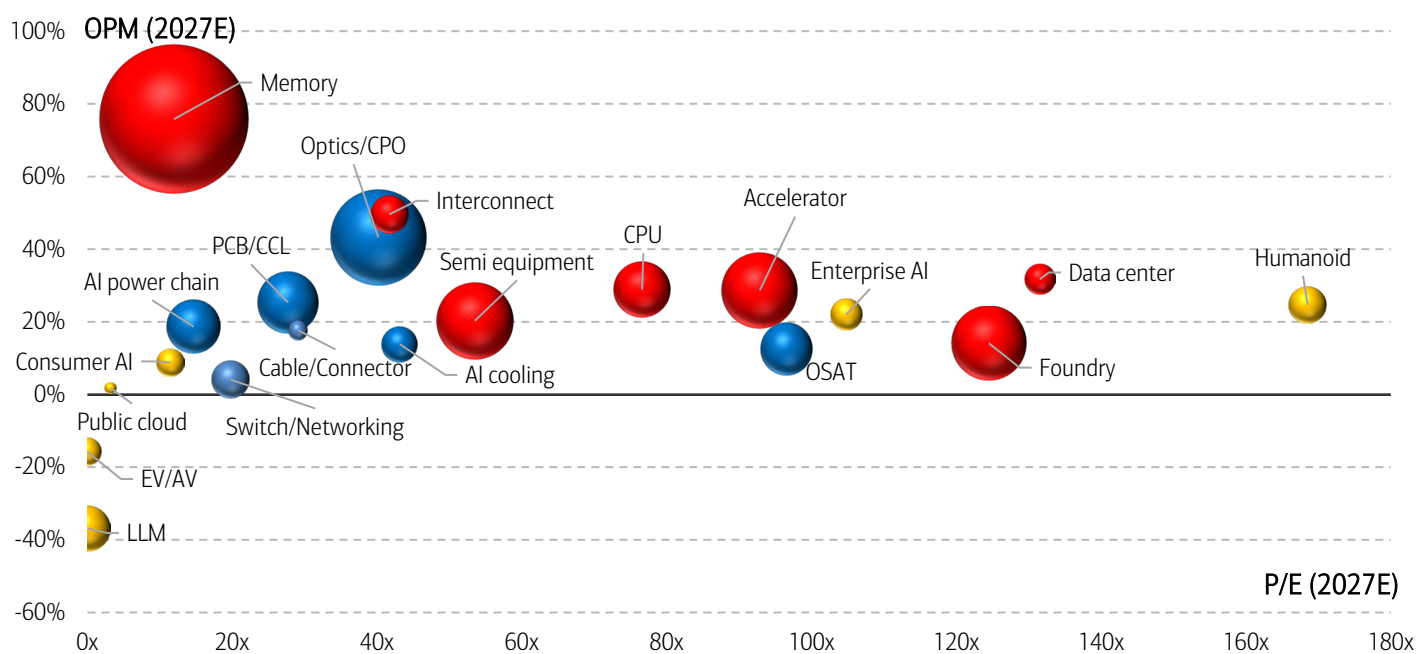
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We classify China AI-related stocks into three segments, each with distinct drivers, economics, and risks: (1) China AI/semi localization beneficiaries, (2) AI hardware and infrastructure exporters, and (3) AI application beneficiaries.

1. **China AI/semi localization beneficiaries:** these include semi equipment (eg NAURA, AMEC), AI accelerators (eg Cambricon), memory (CXMT), foundry (SMIC), EDA, etc. They are closely aligned with China's technology self-reliance and national security priorities and continue to make technological advances. For global investors, they offer diversification from the US AI CAPEX cycle, with growth mainly driven by China's import substitution and technology development, while retaining exposure to the broader AI investment theme. However, many of these firms still lag global leaders in key technologies, face capacity constraints from US export controls, and are subject to various US investment or trade restrictions. The long-term investment case is well recognized, with leading names already well owned by Asian and European investors and trading at elevated valuations of 50-150x 2027E P/E. Ownership among US investors remains relatively low, partly due to concerns on sanction risk. If the US AI cycle softens, these stocks could face sharp multiple compression but relatively limited earnings impact, potentially creating buy-on-dip opportunities for long-term investors.
2. **AI hardware/infrastructure exporters:** these include optical modules (eg Innolight), PCB/CCL (Victory Giant), liquid cooling, AI power supply, and other AI infrastructure suppliers and ODMs. They are primarily leveraged to the US AI CAPEX cycle and have benefited from strong export growth driven by hyperscaler investment. Supported by supply bottlenecks, technological leadership, and strong customer demand, many have delivered rapid earnings growth and relatively stable margins. Despite strong share-price gains in the past 3-4 years, valuations remain relatively reasonable at 20-40x P/E. However, their earnings and stock performance are more tied to the US AI spending cycle. Trade and policy uncertainty is also emerging as the US increases scrutiny of Chinese suppliers in strategic AI infrastructure segments, including optical modules and power equipment.

Exhibit 19: Many China AI localization plays are trading at elevated valuations

Operating margin, P/E and market cap of China AI sectors*



Source: Bloomberg, company reports *Based on data of the Top 1-3 representative stocks in each sector; bubble size is indicative of the aggregate market cap. Bubble colors: Red = China AI/semi localization plays; Blue = Hardware/infrastructure exporters to global AI CAPEX; Yellow = AI application sectors.

3. **AI application beneficiaries:** these include LLMs, humanoids, robotics, EVs/AVs, and consumer and enterprise AI applications. They represent the end-user monetization layer of the AI value chain, with significant long-term potential to boost productivity, reduce costs, and improve quality of life. However, near-term monetization remains uncertain, as the industry has yet to produce a clear “killer app” capable of driving mass adoption and sustainable profits. It also remains unclear whether the AI era will see the same network effects and winner-takes-all dynamics as previous internet platforms. Meanwhile, rapid technology diffusion, relatively low switching cost, and intense competition are already pressuring pricing and margins in segments such as LLMs and humanoid robotics. Therefore, investors are generally more cautious on these stocks given the near-term monetization uncertainty and execution risks. In a US AI slowdown, deep-value names (eg large internet platforms) could be relatively defensive given light investor positioning, undemanding valuations and reduced CAPEX pressure.

In a US AI slowdown, exporters would face pressure on both earnings and valuations. Localization plays could see multiple compression but may offer buy-the-dip opportunities, while deep-value AI application names may be relatively more defensive.

The two ecosystems: not a simple decoupling story

A key debate is whether China’s AI sector can replicate the smartphone and solar playbook, evolving from an import-substitution story into a global cost-disruption and export-share opportunity. However, increasing concerns over national security, sanctions, and data sovereignty suggest the global AI market is increasingly bifurcating into two strategically competing, albeit partially interoperable, ecosystems.

This is not a simple decoupling story. While the US has selectively permitted exports of certain AI technologies to China under licensing conditions (eg Nvidia H200 chips), it has also expanded scrutiny and restrictions across a growing range of sectors, from drones and routers to power inverters and robotics. In our view, the US is moving to prevent Chinese hardware and AI models from becoming embedded in critical US infrastructure.

Today, the US-led ecosystem dominates many of the highest-value layers of the AI stack, including advanced accelerators, EDA, semi equipment, hyperscale cloud, proprietary frontier models, and software platforms. Its competitive advantage extends beyond chip performance to ecosystem lock-in through software, developer tools, cloud infrastructure, and enterprise distribution.

China, by contrast, is emerging as a strong competitor in efficient deployment and physical AI, while steadily building a more self-reliant technology stack around domestic compute, open-weight models, industrial applications, and deep manufacturing capabilities. Although export controls will likely constrain China’s access to leading-edge training compute, it remains well positioned in cost-efficient inference, localized models, consumer AI, industrial automation, and robotics. China’s strategy is less about matching the best chips and more about integrating hardware, software, systems engineering, and manufacturing scale to deliver competitive AI solutions at lower cost.

If Chinese models, hardware and embodied-AI products (eg robots and edge devices) become sufficiently capable and meaningfully cheaper, international adoption could provide the scale needed to reinforce the economics of China’s domestic AI ecosystem.

China AI stock mapping

Exhibit 20: China's leading AI stocks offer diverse earnings and valuation profiles

Chinese stocks across AI value chain

Company	Ticker	Mkt cap		ADTV		P/E (x)		P/S (x)		Earnings YoY		GPM		OPM		ROE		Performance		
		USD bn	USD mn	26E	27E	26E	27E	26E	27E	26E	27E	26E	27E	26E	27E	26E	27E	QTD	YTD	Since '23
AI Accelerator		186.3	1,315	794.7	176.8	42.2	21.0	140.0	269.9	51.9	52.7	14.0	23.4	19.7	30.1	-40.9	2.8	681.4		
Cambricon #	688256 CH	100.8	2,248	120.0	56.7	41.3	18.8	186.5	111.7	53.3	53.6	34.9	35.1	37.9	50.2	-32.6	18.3	1,088.7		
MetaX #	688802 CH	28.3	237	1,824.1	228.9	52.3	26.7	110.7	697.3	56.6	55.9	0.8	11.5	0.4	6.2	-42.7	-18.1	353.9		
Moore Threads	688795 CH	24.5	260		446.7	48.5	30.2	79.4	259.8	63.7	63.7	-5.8	7.2	-1.8	3.4	-51.1	-40.4	206.6		
Biren	6082 HK	10.6	78		49.7	29.1	7.8	95.4	241.4	49.9	50.1	-43.4	13.6	-0.7	9.0	-47.6	63.6	63.6		
Iluvatar #	9903 HK	11.1	112		74.9	27.7	10.7	23.9	333.2	38.7	50.7	-26.1	14.7	-10.1	16.5	-85.5	-23.4			
Dosilicon	688110 CH	7.3	417	11,577	1,157.7	43.3	33.4	102.2	900.0			-0.1	3.7	0.2	2.1	-45.7	-15.8	221.1		
Jingjia Micro	300474 CH	3.7	91			26.9	21.2	50.0	25.0	46.8	47.3	-10.1	-6.3	-1.2	-1.0	-21.7	-34.1	-32.9		
Memory		616.8	3,265	144.2	24.1	9.7	6.3	7,656.8	66.6	79.0	79.6	67.3	69.5	76.4	57.3	421.7	446.4	30.1		
CXMT #	688825 CH	548.2	3,720	19.6	12.0	10.3	6.3	9,253.3	63.6	85.4	85.5	73.9	75.8	86.4	63.6	526.1	526.1			
GigaDevice	603986 CH	38.1	3,149	22.3	16.8	10.0	7.7	575.8	32.5	61.6	59.1	50.3	51.3	43.8	36.0	-55.4	69.7	294.7		
GigaDevice	3986 HK	38.1	247					575.8	32.5	61.6	59.1	50.3	51.3	43.8	36.0	-63.2	169.6			
Ingenic	300223 CH	9.6	821	38.3	29.8	9.1	7.1	347.5	28.5	44.0	42.8	24.9	25.0	12.1	11.7	-48.7	20.4	98.5		
Ingenic	3223 HK	9.6	10					347.5	28.5	44.0	42.8	24.9	25.0	12.1	11.7	-6.2	-6.2			
Puya	688766 CH	8.5	600				4.9				56.0			42.6		45.6		-49.6	203.5	678.1
Dosilicon	688110 CH	7.3	417	11,577	1,157.7	43.3	33.4	102.2	900.0			-0.1	3.7	0.2	2.1	-45.7	-15.8	221.1		
Fudan Micro	688385 CH	5.1	165	38.7	32.9	7.5	6.0	72.5	17.6	58.0	59.6	24.3	21.6	16.7	12.5	-26.8	-29.2	35.7		
Fudan Micro	1385 HK	5.1	23	15.3	13.9	7.4	6.1	417.1	9.8	56.9	58.3	20.0	21.2	16.5	14.5	-22.5	-46.8	73.1		
Foundry		166.7	923	110.6	79.7	9.5	7.7	133.1	35.6	21.6	23.1	10.6	13.0	4.6	5.7	-34.1	15.6	226.9		
SMIC #	688981 CH	90.2	1,407	111.9	88.6	7.9	6.6	67.0	26.3	25.2	26.4	15.8	17.7	5.5	6.5	-27.7	-6.5	116.6		
SMIC	981 HK	90.2	1,005	45.3	34.9	8.0	6.7	98.9	29.6	24.6	26.3	15.3	16.9	5.9	7.0	-32.0	-15.0	205.9		
Hua Hong Semi #	688347 CH	45.1	824	287.6	196.2	15.5	12.1	262.5	46.6	16.2	17.6	3.7	7.0	2.7	3.9	-36.2	98.9	403.8		
Hua Hong Semi	1347 HK	45.1	623	126.2	82.5	15.5	12.4	253.1	53.1	16.4	18.4	2.4	6.7	2.9	4.1	-50.9	42.3	464.7		
Nexchip	688249 CH	12.2	253	60.0	39.9	6.1	5.0	65.3	50.4	27.0	27.3	8.6	10.6	4.6	5.8	-41.9	3.3	99.9		
Nexchip	2249 HK	12.2	19					65.3	50.4	27.0	27.3	8.6	10.6	4.6	5.8	-17.5	-17.5			
United Nova	688469 CH	10.3	221	114.1	81.5	6.9	5.5	193.8	40.0	16.4	20.8	3.6	8.0	4.7	6.2	-22.3	23.2	64.1		
YDME	688172 CH	9.0	178													-49.7	45.5	128.4		
EDA		11.9	145	263.5	142.2	27.9	22.6	144.8	76.6	83.8	84.1	10.8	15.4	3.3	5.0	-28.7	-11.0	3.7		
Empyrean #	301269 CH	6.9	187	250.5	143.5	28.7	23.0	214.5	74.6	89.3	88.6	9.1	14.5	3.1	5.0	-29.6	-20.8	-20.2		
Primarius #	688206 CH	2.8	60	432.2	197.8	35.6	30.0	8.8	118.4	86.3	86.3	10.0	15.4	1.9	3.5	-19.0	3.2	65.1		
Semitronix #	301095 CH	2.2	125	83.1	64.9	15.6	11.8	102.2	28.0	63.5	67.2	17.3	18.1	5.6	6.8	-38.6	1.3	-1.4		
Semi Equipment		278.0	725	102.8	64.0	18.8	13.3	84.7	49.3	43.2	43.9	20.0	21.8	16.7	19.9	-28.0	86.8	278.7		
NAURA #	002371 CH	69.0	1,083	63.9	42.9	9.6	7.3	35.7	49.0	40.3	40.9	15.8	18.0	17.3	20.8	-27.8	39.1	252.7		
AMEC #	688012 CH	47.2	1,220	75.1	62.1	20.0	14.3	105.9	21.1	40.1	40.8	21.0	21.7	16.1	16.5	-29.4	80.8	222.2		
Piotech #	688072 CH	28.2	683	92.8	65.1	21.6	14.7	120.7	42.6	39.9	40.6	23.8	23.4	22.4	25.1	-22.0	96.8	317.0		
Changchuan	300604 CH	25.7	916	84.0	54.1	22.8	15.8	51.1	55.3	56.4	57.5	27.5	29.5	29.1	32.2	-21.7	163.9	607.6		
ACM Research	688082 CH	19.9	300	78.1	52.9	16.1	12.0	18.2	47.6	48.7	49.3	20.3	22.6	12.4	16.3	-35.7	57.8	169.3		
Hwatsing	688120 CH	18.9	420	84.9	63.3	22.2	16.8	42.9	33.9	42.4	42.5	25.4	26.0	16.2	18.7	-20.7	138.7	326.4		
Skyverse	688361 CH	16.2	401	403.8	189.8	39.6	25.3	365.0	112.7	49.4	50.0	8.7	12.7	5.5	10.5	-28.0	102.5	317.4		
Huafeng Test & Control	688200 CH	10.6	254	77.8	61.0	36.6	27.4	73.4	27.7	74.0	73.6	43.6	43.8	17.4	19.4	-32.5	176.7	333.7		
E-Town	688729 CH	10.2	125	74.3	48.3	12.0	8.7	36.3	53.8	39.9	40.9	16.1	19.4	8.8	11.8	-46.1	-5.2	174.3		
Kingsemi	688037 CH	9.6	307	373.6	163.6	27.7	19.2	151.7	128.1	36.5	37.9	7.9	12.6	6.1	12.7	-26.9	116.2	249.5		
Circuit Fabology	688630 CH	9.7	320	100.6	61.5	27.8	18.3	94.6	63.8	42.8	44.9	27.2	29.5	22.5	29.6	-24.9	207.7	393.1		
Circuit Fabology	9630 HK	9.7	30					94.6	63.8	42.8	44.9	27.2	29.5	22.5	29.6	-31.5	41.0			
Leadmicro	688147 CH	7.0	166	138.7	93.5	17.1	13.3	60.2	48.4	34.0	34.5	12.5	14.6	11.0	14.5	-37.8	60.4	159.6		
Wanye	600641 CH	4.4	306			12.5	9.6	47.5	24.7			-9.0	-5.4	-0.8	-0.6	-21.4	77.6	83.0		
PNC	603690 CH	1.3	117													-33.3	-24.9	-7.0		
Interconnect		34.5	1,027	65.2	43.5	32.2	21.1	66.6	49.9	64.6	65.9	44.1	49.9	22.7	25.9	-42.0	99.7	112.1		
Montage #	688008 CH	34.5	1,887	60.9	41.7	32.4	21.4	66.4	46.2	65.3	65.8	44.8	49.6	23.4	26.0	-39.3	59.8	224.2		
Montage	6809 HK	34.5	167	69.5	45.3	32.0	20.8	66.8	53.6	63.8	66.0	43.4	50.2	22.0	25.8	-44.8	139.7			
CPU		83.5	1,270	118.6	76.6	24.3	16.7	80.2	54.7	57.4	57.3	27.4	28.8	17.6	21.8	-39.2	-1.7	201.7		
Hygon #	688041 CH	77.8	1,352	118.6	76.6	24.3	16.7	80.2	54.7	57.4	57.3	27.4	28.8	17.6	21.8	-39.3	0.2	217.5		
Loongson	688047 CH	5.7	168													-37.8	-27.2	-13.0		
Public Cloud		276.6	1,420	19.4	11.4	1.7	1.4	-17.0	85.3	38.9	39.3	6.0	11.0	8.5	10.2	14.5	-24.2	49.4		
Alibaba	9988 HK	271.7	1,438	17.5	11.5	1.7	1.4	-12.2	51.6	39.2	39.7	6.0	11.2	8.7	10.4	15.5	-24.3	48.6		
Alibaba	BABA US	271.8	1,438	21.9	11.6	1.7	1.4	-13.4	88.5	39.2	39.7	6.0	11.2	8.7	10.4	13.9	-24.7	47.9		
Kingsoft Cloud #	3896 HK	2.8	75		3.2	1.5	1.1	34.4	1,577.5	15.0	16.3	-0.8	1.7	-6.2	-3.5	6.3	-10.5	145.3		
Kingsoft Cloud	KC US	2.9	18		3.2	1.5	1.2	-885.2	1,575.9	15.0	16.3	-0.8	1.7	-6.2	-3.5	5.2	-7.9	151.9		
UCloud	688158 CH	1.9	205													-23.5	0.1	71.0		
Data Center		22.7	297	49.2	189.2	8.0	6.4	111.9	-8.0	34.6	32.9	29.4	27.3	14.0	11.9	-13.7	1.3	180.8		
Range Tech #	300442 CH	14.9	529	34.7	25.5	12.7	9.9	-40.0	36.2	46.0	46.1	44.4	42.6	21.0	21.5	-26.5	15.7	147.5		
GDS	9698 HK	6.1	26	73.5	403.4	3.2	2.9	242.6	-81.8	22.6	18.1	13.9	11.1	9.3	1.6	5.7	-11.4	229.6		
GDS #	GDS US	6.1	70	73.8	407.6	3.2	2.9	403.1	-81.9	22.6	18.1	13.9	11.1	9.3	1.6	1.4	-12.8	233.8		
VNET #	VNET US	1.7	54		69.3	1.0	0.8	-72.1	134.9	21.7	22.6	8.7	10.0	-12.9	2.8	-25.9	-29.6	107.7		
Optics/CPO		443.8	2,203	55.8	25.5	15.8	8.5	224.0	106.4	42.7	44.0	35.								

Exhibit 20: China's leading AI stocks offer diverse earnings and valuation profiles

Chinese stocks across AI value chain

Company	Ticker	Mkt cap		ADTV		P/E (x)		P/S (x)		Earnings YoY		GPM		OPM		ROE		Performance		
		USD bn	USD mn	26E	27E	26E	27E	26E	27E	26E	27E	26E	27E	26E	27E	26E	27E	QTD	YTD	Since '23
DS Precision	002384 CH	49.9	2,447	44.1	17.8	5.2	3.2	451.6	147.5	23.5	29.7	13.7	21.6	29.9	45.8	-30.3	116.0	924.7		
TFC #	300394 CH	41.8	2,090	80.1	48.3	31.7	19.2	81.1	65.8	53.1	52.1	45.0	44.5	45.6	51.7	-15.0	77.8	689.7		
Yunajie Semi #	688498 CH	31.3	954	257.3	143.9	117.9	64.3	340.3	78.8	73.8	75.2	52.7	52.5	28.9	35.4	-10.6	280.8	1,545.3		
Accelink	002281 CH	20.8	1,387	89.2	48.0	8.2	5.3	71.7	85.7	24.1	24.9	10.5	13.7	14.1	22.7	-34.5	140.7	498.0		
RoboTechnik	300757 CH	15.0	576	431.0	56.5	56.6	14.7	446.3	662.2			18.4	31.2	7.1	14.9	-2.9	157.7	936.5		
Huagong Tech	000988 CH	15.0	1,469	49.2	36.2	5.2	3.8	46.5	35.9	22.1	22.1	11.7	11.0	17.3	18.5	-45.7	26.5	240.5		
CIG	603083 CH	10.2	744	45.7	23.3	6.1	3.0	419.1	96.5	31.4	35.0	26.1	36.6	17.8	62.8	-15.1	61.4	480.7		
CIG	6166 HK	10.2	153	18.2	8.2	6.7	3.7	498.9	122.7	30.0	33.0	20.6	25.7			-27.4	17.4			
Hshijia Photons	688313 CH	9.9	619	95.5	54.1	20.5	13.0	104.4	76.6	37.3	40.4	24.5	29.3	40.3	46.5	-20.5	65.7	1,059.6		
Yunnan Germanium	002428 CH	8.8	1,012	204.1	69.3	34.3	20.1	1,443.3	194.4	32.5	41.8	19.4	33.1	15.3	25.4	-24.9	184.3	634.9		
T&S	300570 CH	6.6	596	92.0	48.7	17.9	10.2	69.6	88.8	36.0	38.0	21.7	24.2	25.7	40.1	-20.8	69.0	413.0		
PCB/CCL		310.6	982	33.9	19.0	5.5	3.8	148.2	80.7	32.1	35.3	22.6	26.5	26.3	32.6	-25.3	80.0	578.1		
DS Precision	002384 CH	49.9	2,447	44.1	17.8	5.2	3.2	451.6	147.5	23.5	29.7	13.7	21.6	29.9	45.8	-30.3	116.0	924.7		
Shennan Circuits #	002916 CH	39.5	622	48.1	31.1	8.1	6.0	65.9	54.6	31.3	33.6	18.9	21.5	27.7	32.7	-14.9	68.8	635.5		
WUS Printed Circuit #	002463 CH	36.1	1,256	40.8	25.3	8.8	5.7	57.3	61.4	38.5	39.7	27.9	30.3	31.1	35.3	-17.7	73.0	489.8		
Victory Giant	300476 CH	35.8	1,814	26.4	14.0	7.7	4.3	73.8	88.3	37.4	40.0	29.5	32.4	26.5	31.3	-34.7	-20.9	1,145.7		
Victory Giant	2476 HK	35.8	197			73.8	88.3	37.4	40.0	29.5	32.4	26.5	31.3			-44.1	-7.0			
Avary	002938 CH	30.7	506	40.2	26.3	4.5	3.5	38.7	52.8	24.2	25.9	12.5	15.0	14.0	19.3	-15.7	77.3	321.6		
China Jushi	600176 CH	28.1	1,182	26.2	17.5	7.4	6.0	119.4	50.1	46.3	53.5	37.1	44.3	20.7	25.0	-35.4	176.7	407.9		
Kingboard Laminates	1888 HK	19.1	477	18.2	12.0	4.3	3.2	228.2	52.1	34.3	37.2	33.0	38.1	37.5	40.1	-52.0	264.0	684.3		
Kinwong	603228 CH	15.6	480	43.6	23.3	5.2	3.7	87.3	86.7	26.8	30.9	14.0	18.6	16.6	25.0	44.4	44.2	388.2		
Shengyi Electronics #	688183 CH	15.7	404	38.0	24.9	7.2	5.0	81.1	52.6	34.3	35.0	22.1	23.2	39.9	41.8	-4.9	32.2	1,025.7		
Sinoma Sci & Tech	002080 CH	13.4	449	31.1	21.7	2.6	2.3	65.3	43.6	21.7	24.8	11.2	14.8	12.6	15.0	-40.3	48.8	256.8		
Delton Technology	001389 CH	11.6	241	36.3	20.0	8.0	4.7	97.7	81.6	37.9	38.1	27.5	28.4	36.1	38.7	-27.7	110.1	880.6		
Delton Technology	1989 HK	11.6	37			97.7	81.6	37.9	38.1	27.5	28.4	36.1	38.7			-43.4	68.3			
Defu Tech	301511 CH	10.5	519	95.2	34.9	3.9	3.0	557.1	172.6	9.5	14.5	6.3	11.6	14.7	29.8	-22.0	213.9	579.9		
Olympic Circuit	603920 CH	4.6	244	44.1	25.0	4.6	3.5	2.1	76.3	21.4	25.1	12.0	16.4	10.0	15.6	-11.6	-11.9	146.6		
AI Power		163.4	477	25.1	16.9	3.0	2.3	45.9	39.0	29.0	29.7	15.9	17.2	19.0	21.1	-19.5	3.3	122.8		
NARI Tech #	600406 CH	26.3	270	19.3	17.2	2.3	2.1	9.7	12.2	25.5	25.2	14.1	14.0	16.5	17.0	-3.8	-0.2	6.1		
Sungrow #	300274 CH	26.1	1,418	12.9	10.6	1.9	1.6	0.8	22.1	30.7	30.5	17.6	18.1	24.3	23.9	-46.9	-50.4	40.1		
Jereh #	002353 CH	18.4	398	34.8	17.0	6.0	3.4	30.5	105.3	33.3	37.3	21.5	26.1	14.1	24.2	-20.9	71.2	349.4		
Sieyuan Electric	002028 CH	16.2	324	25.6	18.7	3.7	2.9	30.4	36.8	31.1	31.5	17.4	18.3	22.9	25.0	-19.5	-9.9	171.0		
Deye	605117 CH	16.1	281	19.0	14.9	4.6	3.7	78.4	27.8	37.3	37.1	28.2	28.1	42.8	41.4	-19.8	39.7	204.9		
TBEA	600089 CH	13.6	475	13.4	10.7	0.8	0.8	16.3	25.6	19.6	20.2	8.6	9.6	8.7	10.1	-18.2	-17.1	38.2		
Megmeet	002851 CH	10.3	563	94.1	43.1	5.4	3.9	383.0	118.4	25.5	27.7	6.5	10.4	10.1	17.8	-29.0	31.9	387.7		
Chint Electrics	601877 CH	8.0	119	10.0	8.9	0.8	0.7	17.2	13.4	25.8	26.5	12.8	14.2	11.2	11.5	-0.8	-8.9	24.8		
Hongfa	600885 CH	8.0	98	23.9	19.1	2.4	2.1	20.2	25.2	33.6	34.1	15.4	16.1	16.0	17.7	0.7	15.5	83.5		
TGOOD	300001 CH	5.1	146	22.3	17.6	2.0	1.7	20.6	27.0	27.5	27.8	10.9	12.1	16.2	17.5	-14.1	27.2	64.2		
Jinpan	688676 CH	4.3	187	33.8	23.0	3.4	2.7	31.6	47.0	26.4	28.0	11.1	13.2	14.9	19.4	-26.2	-30.4	79.8		
Henan Pinggao Electric	600312 CH	3.9	75	18.7	15.2	1.9	1.7	24.0	22.6	24.6	25.6	12.8	14.0	11.6	12.8	9.0	11.8	58.8		
XJ Electric	000400 CH	3.1	85	16.4	13.3	1.2	1.1	7.4	23.2	22.3	23.2	8.9	10.0	9.5	10.9	-5.9	-19.8	-2.8		
Hopewind	603063 CH	2.1	183	23.1	18.2	3.0	2.5	12.8	26.5	37.3	36.8	14.4	15.6	10.6	12.1	-43.6	-9.7	21.3		
Sineng Electric	300827 CH	2.0	128	21.1	16.1	1.9	1.6	24.5	31.2	24.0	24.4	10.2	11.1	13.6	15.4	-23.2	-30.7	10.6		
AI Cooling		30.8	428	67.3	43.1	6.9	5.4	328.7	53.5	25.8	27.1	11.9	13.9	19.2	23.8	-15.2	27.8	373.1		
HEC #	600673 CH	14.0	328	49.4	39.5	6.0	5.6	611.1	25.0	23.7	25.2	13.0	15.1	16.2	17.0	-7.1	39.0	343.8		
Envicool #	002837 CH	11.4	634	73.5	43.2	8.2	5.4	103.2	70.1	28.5	29.5	12.2	14.0	25.8	34.2	-24.8	-27.1	385.4		
Shenling #	301018 CH	5.4	253	100.4	52.3	6.6	4.6	73.7	92.0	25.7	26.8	8.6	10.3	13.2	19.3	-16.0	114.3	423.1		
OSAT		92.3	920	122.6	82.0	14.3	10.8	37.1	41.2	21.2	21.9	10.4	12.0	6.3	8.1	-33.7	246.9	70.4		
SJ Semi #	688820 CH	35.5	644	228.6	145.4	31.2	23.2		57.3	30.4	31.3	13.8	16.7	5.5	7.9	-34.0	549.5			
JCET #	600584 CH	17.8	1,710	54.0	39.1	2.8	2.5	47.9	38.0	15.1	15.7	6.1	7.2	7.4	9.3	-35.4	82.0	125.9		
Tongfu Micro #	002156 CH	13.0	1,217	46.2	41.7	2.8	2.4	60.0	10.8	15.1	15.6	8.7	8.5	10.8	11.4	-24.5	52.2	148.8		
Shenzhen Kaifa/Payton	000021 CH	8.0	817	40.7	34.2	3.1	2.7	22.7	19.1			11.3	11.7			-47.2	34.9	114.6		
Tianshui Huatian	002185 CH	8.1	861	63.5	48.1	2.7	2.3	20.5	32.1	14.4	15.1	2.9	4.0	4.8	6.0	-26.7	49.3	94.0		
Forehope Elec.	688362 CH	4.0	288	113.7																

Exhibit 20: China's leading AI stocks offer diverse earnings and valuation profiles

Chinese stocks across AI value chain

Company	Ticker	Mkt cap USD bn	ADTV USD mn	P/E (x)		P/S (x)		Earnings YoY		GPM		OPM		ROE		Performance		
				26E	27E	26E	27E	26E	27E	26E	27E	26E	27E	26E	27E	QTD	YTD	Since '23
Luxshare	2475 HK	59.3	47					21.5	36.6	12.1	12.9	5.5	6.4	20.2	22.3	-11.2	-11.2	
Dingtong Precision #	688668 CH	5.1	226	63.4	40.4	11.5	7.7	139.7	57.1	32.1	32.2	16.6	17.0	20.2	24.5	-33.7	99.8	539.9
FIT Hong Teng	6088 HK	4.8	58	18.1	12.4	0.9	0.7	68.2	45.9	20.2	21.3	7.1	8.5	8.8	13.2	-27.8	3.6	334.7
Woer Heat-Shrinkable #	002130 CH	3.3	136	15.5	11.8	2.0	1.7	13.6	30.9	31.4	31.4	17.0	18.6	21.4	23.1	-9.8	-35.1	130.6
Woer Heat-Shrinkable	9981 HK	3.3	5					13.6	30.9	31.4	31.4	17.0	18.6	21.4	23.1	-21.1	-55.2	
Suzhou Recodeal	688800 CH	2.8	155	42.4	30.5	4.7	3.5	57.3	39.1	23.9	24.9	11.7	12.8	15.4	19.2	-31.5	11.1	201.6
Passive Component		53.1	528	59.7	41.5	17.4	13.2	60.3	42.4	42.4	44.7	30.3	33.0	15.6	19.0	-4.3	113.7	182.3
Three-Circle	300408 CH	37.3	967	60.4	43.4	19.2	14.5	56.9	39.3	45.7	47.5	33.9	36.5	16.9	20.4	-24.4	177.3	340.3
Three-Circle	6951 HK	37.3	33	53.4	38.6	19.7	14.9	48.9	38.2	45.7	47.5	33.9	36.5	16.9	20.4	24.5	24.5	
Fenghua Advanced	000636 CH	9.8	885	96.6	54.9	9.1	7.3	142.0	76.0	21.2	28.4	10.6	14.9	5.3	8.4	-20.3	253.8	328.0
Sunlord	002138 CH	5.9	293	33.3	25.1	4.9	4.0	18.2	32.5	35.8	36.5	18.4	19.8	16.7	19.3	-32.5	42.3	93.3
LLM		50.8	562			42.9	14.4	12.9	-42.0	29.0	34.3	-103.4	-36.8	2.8	33.2	-62.8	394.1	
Zhipu/ Z.AI #	2513 HK	40.4	626			48.3	16.1	16.7	-49.9	29.8	34.7	-89.3	-28.6	0.4	41.4	-67.7	485.2	
MiniMax #	100 HK	10.4	317			22.4	7.7	-1.4	-11.4	25.9	32.6	-157.7	-68.9	12.0	1.4	-43.7	42.2	
Humanoid		33.3	511	257.9	168.5	55.2	32.4	97.9	95.4	56.4	56.3	20.9	24.7	10.5	15.6	176.1	174.0	-2.3
Unitree #	688836 CH	28.3	584	303.3	162.4	63.4	37.0	105.8	86.7	58.8	58.4	26.4	28.6	13.2	17.8	211.5	211.5	
UBTech #	9880 HK	5.0	98		203.1	8.5	5.9	53.3	144.8	42.8	44.4	-10.4	2.2	-4.6	3.0	-25.4	-39.3	-15.6
EV/AV		421.9	787	15.1	23.8	1.8	1.5	11.9	40.7	22.5	23.0	9.3	11.4	19.2	21.0	-12.3	-11.3	62.2
CATL	3750 HK	222.1	296	20.2	16.3	2.3	1.9	29.9	23.7	24.5	24.8	17.2	17.8	25.4	26.4	-26.3	3.6	96.2
CATL	300750 CH	222.1	1,974	14.8	11.8	2.3	1.9	28.4	25.1	24.5	24.9	18.3	18.7	24.7	25.6	-19.2	-12.2	108.2
BYD	1211 HK	105.3	313	15.7	12.2	0.8	0.7	20.2	29.0	18.6	19.1	5.2	6.0	14.5	16.2	9.1	-16.7	14.0
BYD	002594 CH	105.3	643	19.2	14.7	0.8	0.7	22.0	30.3	18.7	19.3	5.2	5.8	14.5	16.4	6.2	-13.4	31.3
Geely	175 HK	22.2	166	6.9	5.7	0.4	0.3	22.5	21.4	17.5	17.7	5.4	5.8	20.5	21.3	-4.3	-7.4	101.3
Chery	9973 HK	18.7	13	6.3	5.0	0.4	0.3	0.4	26.1	16.0	16.3	5.6	6.1	33.6	30.0	3.7	-15.2	-17.9
Li Auto	2015 HK	12.2	95		32.7	0.7	0.6	-203.5	209.0	13.2	16.1	-5.2	0.5	-4.2	2.5	0.1	-28.9	-68.6
Li Auto	LI US	12.2	55		32.2	0.7	0.6	-200.0	209.0	13.2	16.1	-5.2	0.5	-4.2	2.5	0.2	-30.5	-68.6
XPeng #	9868 HK	9.9	113		403.5	0.8	0.6	-672.5	104.6	19.2	19.3	-4.7	-0.7	-11.4	0.3	-20.3	-49.1	-28.8
XPeng	XPEV US	9.8	102		396.7	0.7	0.6	-181.0	104.7	19.2	19.3	-4.7	-0.7	-11.4	0.3	-22.7	-49.6	-29.9
NIO	9866 HK	9.2	32		36.8	0.5	0.4	94.9	300.9	18.0	18.2	-0.7	1.0	1.4	25.7	-26.0	-30.0	-61.0
NIO	NIO US	9.0	182		36.5	0.5	0.4	94.0	300.9	18.0	18.2	-0.7	1.0	1.4	25.7	-28.9	-29.4	-60.3
Horizon Robotics	9660 HK	8.3	115			10.1	6.5	29.7	24.2	63.9	61.7	-61.0	-25.4	-11.8	-17.2	0.7	-52.5	3.0
Leapmotor	9863 HK	6.2	54	14.1	8.0	0.4	0.3	460.0	77.4	13.6	14.4	2.4	3.4	18.2	24.7	4.0	-26.2	0.4
Momenta #	6880 HK	5.2	10			8.6	5.2	28.6	99.5	73.3	74.3	-18.4	-6.0	26.4	1.6	-41.8	-41.8	
Pony AI	2026 HK	2.8	8					-30.9	-0.3	16.6	22.9	-189.2	-84.7	-17.5	-20.8	-6.3	-56.4	
Pony AI #	PONY US	2.9	39			19.2	9.2	-30.9	-0.3	16.6	22.9	-189.2	-84.7	-17.5	-20.8	-4.0	-54.0	
Consumer AI		637.0	1,254	14.6	94.6	4.0	3.7	5.3	10.6	59.2	59.4	31.3	30.7	18.6	17.5	-0.8	-23.3	46.3
Tencent	700 HK	509.2	1,701	12.6	11.9	4.1	3.7	5.5	5.8	57.0	57.2	32.1	31.0	18.7	17.1	2.1	-25.9	53.8
NetEase	9999 HK	76.4	185	12.9	11.1	4.2	3.9	1.3	15.9	69.2	69.6	38.4	38.6	22.8	22.0	-7.2	-11.1	42.9
NetEase	NTES US	76.5	106	12.9	11.2	4.2	3.9	6.8	15.9	69.2	69.6	38.4	38.6	22.8	22.0	-6.4	-11.4	37.7
Kuaishou #	1024 HK	16.4	326	12.6	11.7	0.8	0.8	-58.3	7.6	50.5	50.6	5.7	6.1	8.3	8.5	-28.4	-52.6	-42.4
Iflytek	002230 CH	14.0	281	87.6	63.3	3.0	2.6	23.3	38.3	42.4	42.6	2.6	3.3	4.9	6.2	-4.3	-22.3	-15.3
Kunlun Tech	300418 CH	8.2	469		7,173.3	5.4	4.5	67.9	101.5	71.2	71.7	-7.4	0.7	-4.0	0.9	6.2	2.8	14.8
Bilibili	9626 HK	6.5	85	15.4	12.0	1.3	1.2	13.9	28.3	37.4	38.5	7.8	9.5	12.2	14.4	-7.8	-37.1	29.7
Bilibili	BILI US	6.5	54	15.8	12.3	1.3	1.2	11.5	28.3	37.4	38.5	7.8	9.5	12.2	14.4	-9.2	-37.1	27.0
Intsig Information	688615 CH	2.6	97	31.1	24.5	7.7	6.2	25.0	27.0	87.0	87.2	26.0	26.4	17.3	19.7	-20.9	-44.4	62.2
Meitu #	1357 HK	2.2	50	12.1	9.5	3.1	2.5	27.9	27.6	73.1	73.2	26.9	29.0	18.8	20.4	4.9	-43.2	15.9
Wondershare	300624 CH	1.4	81	251.7	97.2	5.6	5.0	141.9	158.9	91.2	91.3	4.4	4.5	1.9	6.8	-4.5	-29.9	-26.6
Enterprise AI		29.5	204	65.0	96.7	10.3	8.3	137.7	59.9	72.7	70.4	21.9	19.9	11.8	12.6	7.4	-8.8	-11.6
Kingsoft Office #	688111 CH	15.6	319	33.8	38.4	14.6	12.3	66.0	-12.0	86.2	80.4	39.6	31.7	20.7	16.3	6.5	-26.1	-27.8
Yonyou #	600588 CH	4.8	90		374.8	3.3	3.0	65.1	117.5	51.1	53.1	-4.9	0.8	-6.8	0.1	2.2	-29.3	-47.3
Xunce	3317 HK	4.1	90	285.4	82.0	11.4	7.0	257.6	248.3	61.6	62.7	6.7	11.6	3.0	13.5	-20.9	90.8	100.7
Kingdee #	268 HK	3.4	53	53.4	29.2	2.8	2.4	343.2	82.9	68.2	69.2	3.6	6.9	4.9	8.0	27.0	-43.4	-33.9
Marketingforce	2556 HK	1.7	37	29.9	18.4	2.5	1.7	307.1	62.1	44.1	47.6	7.2	10.2	16.8	21.8	60.4	39.8	12.3

Source: Bloomberg # Top 1-3 representative stocks used for margin and valuation calculations in the previous exhibit *market data and consensus estimates as of Sep 15th, 2026

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AI accelerator

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The technology gap between China and global semiconductor leaders varies significantly by segment. For AI accelerators, Chinese vendors have been trying to catch up AI accelerator performance by upgrading system-level infrastructure for interconnecting, particularly for inference, but remain behind overseas peers such as Nvidia in single chip performance due to lack of advanced node manufacturing capability. Interconnect and memory-interface chips represent the narrowest gap, with Montage already competing at the technology frontier. For other segments, power semiconductors and MCUs currently exhibit relatively narrow technology gaps, with domestic suppliers gaining market share and becoming increasingly competitive in mainstream applications. In contrast, high-end memory (particularly HBM and leading-edge DRAM) remain the challenging segment, given the significant advantages held by global leaders in architecture, manufacturing capabilities, ecosystems and customer qualification.

Exhibit 21: Both China and US have strong capabilities in terms of chip design

Major semiconductor companies in Mainland China and US by sub-sector

	Mainland China	U.S.	
Semiconductor	CPU		Analog
	Loongson (688047 CH) Hygon (688041 CH)	Intel (INTC US)* AMD (AMD US)*	SG Micro (300661 CH) Southchip (688484 CH) 3Peak (688536 CH) Texas Instrument (TXN US)* Analog Devices (ADI US)*
	GPU		Power semis
	Cambricon (688256 CH) Iluvatar CoreX (9903 HK)* MetaX (688802 CH)* Moore Threads (688795 CH) Biren (6082 HK) HiSilicon (Not listed)	Nvidia (NVDA US)* AMD (AMD US)* Intel (INTC US)*	InnoScience (2577 HK)* NCE Power (605111 CH)* Silan (600460 CH)* Yangjie Technology (300373 CH) CR Micro (688396 CH) StarPower (603290 CH) ON Semi (ON US)* Diodes (DIOD US) Vishay (VPG US)*
	Memory		MCU
	GigaDevice (603986 CH)* ChangXin Memory (688825 CH) Yangtze Memory (Not listed)	Micron (MU US)*	GigaDevice (603986 CH)* Sino Wealth (300327 CH) Fudan Micro (688385 CH) Chipsea (688595 CH) Microchip (MCHP US)*
	Interconnection		
	Montage (688008 CH)*	Rambus (RMBS US) Astera Lab (ALAB US)*	

Source: BofA Global Research; Note: (*) covered by BofA Global Research

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Domestic market: catching up with Nvidia, but the gap remains

The AI accelerator market remains one of the semiconductor categories where the technology gap between China and the US is most pronounced, particularly in the training of frontier models. Nvidia's competitive advantage extends well beyond GPU silicon itself, encompassing leading accelerator architectures, advanced packaging and HBM integration, NVLink / NVSwitch scale-up networking, and complete rack-level systems. For example, the GB300 NVL72 system connects 72 Blackwell Ultra GPUs through a 130TB/s NVLink fabric and provides over 20TB of aggregate GPU memory with 576TB/s of memory bandwidth. Beyond hardware, Nvidia benefits from a strong software moat built around the CUDA ecosystem, which provides mature development

tools, compilers and optimized libraries that simplify deployment and maximize performance across AI and HPC workloads.

Although domestic AI accelerators still lag advanced GPUs provided by Nvidia, we see solid progress from domestic vendors with a gradually narrowing gap vs global leaders. Among domestic vendors, **Huawei** clearly stands out both in terms of single-chip performance and the visibility of its technology roadmap. Huawei's Ascend series demonstrates a consistent upgrade path, from current Ascend 910B / 910C to upcoming 950, 960 and 970, moving steadily toward FP8/FP4 precision, higher memory capacity and bandwidth.

Beyond Huawei, several Chinese vendors have also delivered competitive products. **Cambricon's** MLU series (e.g., MLU590) provides solid FP16/INT8 performance with competitive memory bandwidth, making these products suitable for large-scale inference and fine-tuning workloads. **T-Head** (Zhenwu series) and **MetaX** (C500/C600) also stand out with strong memory configurations and improving compute density. In addition, **Iluvatar CoreX** is gradually advancing its roadmap through its Tiangai series, with Tiangai 300 (launched in Jul-2026) and Tiangai 400 (TBD), suggesting a transition toward higher compute density and next-generation architectures.

Overall, while Huawei leads the domestic peers in performance, roadmap clarity, and training capability, a broader group of domestic vendors (e.g., Cambricon, T-Head, Iluvatar CoreX, and MetaX) are building credible and increasingly competitive alternatives, particularly for inference-centric deployments, in our view.

Exhibit 22: Single-chip hardware performance, software stack, and interconnect capability all play a role in determining its actual performance

What determines AI chip / AI cluster performance

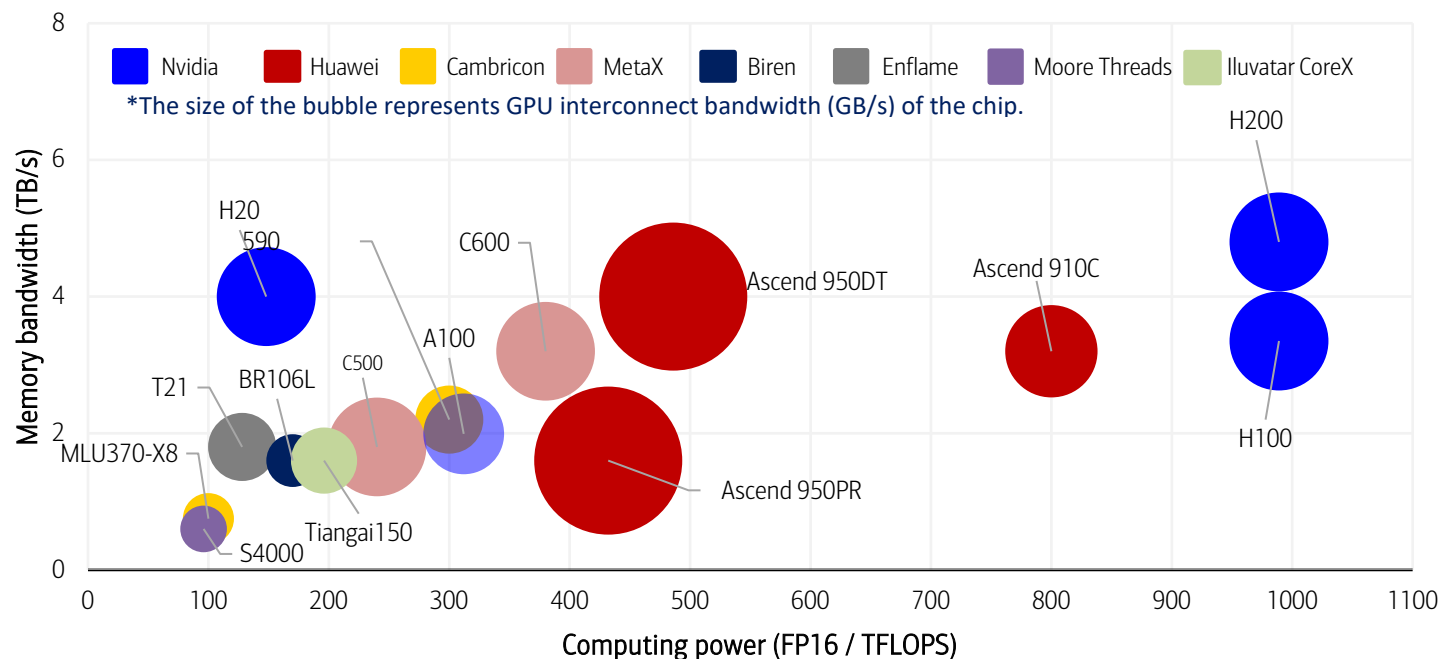
	Hardware	Software	Interconnect			
Significance	Determines baseline chip performance	Determines actual performance utilization & deployment efficiency of chips	Determines performance of clusters for AI training / inference			
Core performance metrics	- Compute density (TFLOPS/TOPS) - Memory capacity (GB HBM / LPDDR) - Memory bandwidth (TB/s) - Power efficiency / Thermal Design Power (TDP)	- Supporting mainstream toolchain (PyTorch, etc.) / toolchain maturity - Operator library coverage - CUDA compatibility - Migration cost	- Network bandwidth / inter-card bandwidth (GB/s) - Cluster scaling efficiency - Latency			
Illustrative example	- NVIDIA H20: Computing power: 148 TFLOPS (FP16) Memory spec: 96GB HBM3 Memory bandwidth: 4.0TB/s TDP: 350W - Huawei Ascend 910B: Computing power: 376 TFLOPS (FP16) Memory spec: 64GB HBM2e Memory bandwidth: 1.6TB/s TDP: 400W	- NVIDIA CUDA: Supporting full series of NVIDIA GPUs Full coverage of PyTorch toolchains Migration cost nearly zero - Huawei CANN: Requiring CANN plug-ins for PyTorch Lacking compatibility with other apps Relatively high migration cost	- NVIDIA H20 cluster: Network bandwidth: 900GB/s (NV Link) AI training throughput ~70% of H800 GPU cluster (4,096 cards), per media report - Huawei Ascend 910B cluster: Network bandwidth: 392GB/s (HCCS) AI training throughput ~50% of H800 GPU cluster (4,096 cards), per media report			
Performance equation	Computing power × Memory capacity × Memory bandwidth × Software efficiency factor × Interconnect scaling factor = Cluster performance					

Source: BofA Global Research

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Exhibit 23: Domestic AI chips are narrowing the gap vs. Nvidia's H100 / H20 chips, especially Huawei

Selected AI accelerator performance across global and domestic vendors



Source: Companies, BofA Global Research; Note: Interconnect bandwidth for Iluvatar Tianguai-150 is not publicly disclosed; values shown are based on estimates.

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Exhibit 24: Huawei leads in terms of the key performance metrics among domestic AI chip vendors

AI chip spec comparison

Company	Product	Launch year	Process	Computing power	Memory	Memory bandwidth	PCIe protocol	Interconnect bandwidth	TDP
Nvidia	B300 NVL (individual GPU)	2025	4nm	FP16: 2.5 PFLOPS FP8: 5 PFLOPS FP4: 15 PFLOPS	279GB HBM3e	8.0 TB/s	PCIe 6.0	1.8 TB/s (NVLink 5.0)	Up to 1,400W
	B200 NVL (individual GPU)	2024	4nm	FP16: 2.5 PFLOPS INT8: 5 POPS FP4: 10 PFLOPS	186GB HBM3e	8.0 TB/s	PCIe 5.0	1.8 TB/s (NVLink 5.0)	Up to 1,200W
	B100	2024	4nm	FP16: 1,750 TFLOPS INT8: 3.5 POPS FP4: 7 PFLOPS	192GB HBM3e	8.0 TB/s	PCIe 5.0	1.8 TB/s (NVLink 5.0)	Up to 700W
	H200 SXM	2023	4nm	FP16: 989 TFLOPS INT8: 1,979 TOPS	141GB HBM3e	4.8 TB/s	PCIe 5.0	900 GB/s (NVLink 4.0)	Up to 700W
	H100 SXM	2022	4nm	FP16: 989 TFLOPS INT8: 1,979 TOPS	80GB HBM3	3.35 TB/s	PCIe 5.0	900 GB/s (NVLink 4.0)	Up to 700W
	A100 SXM	2020	7nm	FP16: 312 TFLOPS INT8: 624 TOPS	80GB HBM2e	2.039 TB/s	PCIe 4.0	600 GB/s (NVLink 3.0)	400W
	H20	2023	4nm	FP16: 148 TFLOPS INT8: 296 TOPS	96GB HBM3	4.0 TB/s	PCIe 5.0	900 GB/s (NVLink 4.0)	400W
Huawei	Ascend 970	2028E	Not disclosed	FP8: 4 PFLOPS FP4: 8 PFLOPS	288GB	14.4 TB/s	Not disclosed	4 TB/s	Not disclosed
	Ascend 960	2027E	Not disclosed	FP8: 2 PFLOPS FP4: 4 PFLOPS	288GB	9.6 TB/s	Not disclosed	2.2 TB/s	Not disclosed
	Ascend 950DT	2026E	7nm	TF32: 243 TFLOPS FP16: 486 TFLOPS INT8: 973 TOPS FP8: 973 TFLOPS MXFP4: 1,946 TFLOPS	144GB HiZQ 2.0	4 TB/s	PCIe 5.0	2,016 GB/s (Unified Bus 2.0)	Not disclosed
	Ascend 950PR	2026	7nm	TF32: 216 TFLOPS FP16: 432 TFLOPS INT8: 865 TOPS FP8: 865 TFLOPS MXFP4: 1,730 TFLOPS	128GB HiBL 1.0	1.6 TB/s	PCIe 5.0	2,016 GB/s (Unified Bus 2.0)	600W

Exhibit 24: Huawei leads in terms of the key performance metrics among domestic AI chip vendors

AI chip spec comparison

Company	Product	Launch year	Process	Computing power	Memory	Memory bandwidth	PCIe protocol	Interconnect bandwidth	TDP
	Ascend 910C	2025	7nm	FP16: 800 TFLOPS	128GB HBM2e	3.2 TB/s	PCIe 5.0	784 GB/s	Not disclosed
	Ascend 910B	2023	7nm	FP32: 94 TFLOPS FP16: 376 TFLOPS	64GB HBM2e	1.6 TB/s	PCIe 5.0	392 GB/s (HCCL)	400W
Cambricon	MLU690	2026E				Not disclosed			
	MLU590	2024	7nm			Not disclosed			
	MLU370-X8	2022	7nm	FP32: 24 TFLOPS BF16: 96 TFLOPS FP16: 96 TFLOPS INT16: 128 TOPS INT8: 256 TOPS	48GB LPDDR5	614.4 GB/s	PCIe 4.0	200 GB/s (MLU-Link)	250W
	MLU370-X4	2021	7nm	FP32: 24 TFLOPS BF16: 96 TFLOPS FP16: 96 TFLOPS INT16: 128 TOPS INT8: 256 TOPS	24GB LPDDR5	307.2 GB/s	PCIe 4.0	Not disclosed	150W
	MLU370-S4	2021	7nm	FP32: 18 TFLOPS BF16: 72 TFLOPS FP16: 72 TFLOPS INT16: 96 TOPS INT8: 192 TOPS	24GB LPDDR5	307.2 GB/s	PCIe 4.0	Not disclosed	75W
Biren	BR30X	2028E				Not disclosed			
	BR20X	2026E				Not disclosed			
	BR166M / BR166L	2025	7nm	FP16: c.340 TFLOPS INT8: c.680 TOPS	Not disclosed	Not disclosed	Not disclosed	576 GB/s (BLink)	550W / 600W
	BR106M / BR106L	2023	7nm	FP16: 170 TFLOPS INT8: 340 TOPS	32GB HBM2e	819 GB/s	PCIe 4.0	256 GB/s (BLink)	400W
	BR104	2022	7nm	FP32: 128 TFLOPS BF16: 512 TFLOPS INT8: 1,024 TOPS	32GB HBM2e	1.14 TB/s	PCIe 5.0	192 GB/s (BLink)	300W
	BR100	2022	7nm	FP32: 256 TFLOPS BF16: 1,024 TFLOPS INT8: 2,048 TOPS	64GB HBM2e	2.3 TB/s	PCIe 5.0	512 GB/s (BLink)	550W
Moore Threads	Huashan	2025	Not disclosed	FP4 to FP64	Not disclosed	Not disclosed	Not disclosed	1,314 GB/s (MTLink 4.0)	Not disclosed
	S5000	2024	Not disclosed	FP8: 1,000 TFLOPS (peak)	80GB	1.6 TB/s	PCIe 5.0	784 GB/s (MTLink)	Not disclosed
	S4000	2023	7nm	FP32: 25 TFLOPS TF32: 50 TFLOPS FP16: 100 TFLOPS INT8: 200 TOPS	48GB	768 GB/s	PCIe 5.0	240 GB/s (MTLink)	450W
MetaX	C700	2026E				Not disclosed			
	C600	2025	12nm	FP16: 300 - 400 TFLOPS	144GB	Not disclosed	Not disclosed	Not disclosed	Not disclosed
	C500	2024	7nm	FP16: 240 TFLOPS INT8: 480 TOPS	64GB HBM2e	1.8 TB/s	PCIe 4.0	896 GB/s (MetaXLink)	350W
T-Head	J900	2028E				Not disclosed			
	V900	2027E				Not disclosed			
	Zhenwu M890	2026	Not disclosed	Training: FP32 / BF16 / FP16 Inference: FP8 / INT8 / FP4	144GB	Not disclosed	Not disclosed	800 GB/s	Not disclosed
	Zhenwu 810E	2026	Not disclosed	Not disclosed	96GB HBM2e	2,765 GB/s	PCIe 5.0	700 GB/s	400W
	Hanguang 800	2019	12nm	FP16: 205 TFLOPS INT8: 825 TOPS	96GB HBM2e	Not disclosed	Not disclosed	Not disclosed	280W
Iluvatar CoreX	Tiangai 400	TBD				Not disclosed			
	Tiangai 300	2026				Not disclosed			
	Tiangai 150	2023	Not disclosed	Not disclosed	64GB HBM2e	1.6 TB/s	PCIe 4.0	Not disclosed	350W
	Tiangai 100	2021	7nm	Not disclosed	32GB HBM2e	1.2 TB/s	PCIe 4.0	Not disclosed	250W

Source: Companies, PaddlePaddle, Gitee AI, Cloud Hin, EE Times China, Hotchips, IT Home, China Daily, CSDN

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Exhibit 25: Huawei leads in Super Node development among domestic AI accelerator vendors

Super Node: Nvidia vs domestic AI Accelerator vendors

Company	Super Node		Launch time	Number of GPUs	Total computing power	GPU interconnection	Operating power
Nvidia	Y	GB200 NVL72	2024	72 * B200	FP8: 360 PFLOPS FP4: 720 PFLOPS	NVLink 5.0 + NVSwitch	125kW - 135kW (est.)
		GB300 NVL72	2025	72 * B300	FP8: 360 PFLOPS FP4: 1,080 PFLOPS	NVLink 5.0 + NVSwitch	140kW
		Vera Rubin NVL72	2026	72 * Rubin GPU	FP8 training: 1,260 PFLOPS FP4 inference: 1,800 PFLOPS	NVLink 6.0 + NVSwitch	~190 kW (Max Q) ~230 kW (Max P)
		Vera Rubin NVL144	2H26E	72 * Rubin GPU packages (each contains 2 compute chips)	FP8 training: 1.2 EFLOPS FP4 inference: 3.6 EFLOPS	NVLink 6.0 + NVSwitch	250kW
		Vera Rubin NVL576	2H27E	144 * Rubin GPU packages (each contains 4 compute chips)	FP8 training: 5 EFLOPS FP4 inference: 15 EFLOPS	NVLink 7.0 + NVSwitch	650kW
		Vera Rubin POD (NVL1152)	TBC	1,152 * Rubin GPU	TBC	TBC	Not disclosed
Huawei	Y	Atlas 900 A3 SuperPoD	2025	384 * Ascend 910C	FP16: 307.2 PFLOPS	HCCS	559kW
		Atlas 950 SuperPoD	2026	8,192 * Ascend 950DT	FP8: 8 EFLOPS FP4: 16 EFLOPS	UnifiedBus + UBOE	Not disclosed
		Atlas 960 SuperPoD	4Q27E	15,488 * Ascend 960	FP8: 30 EFLOPS FP4: 60 EFLOPS	UnifiedBus + UBOE	Not disclosed
MetaX	Y	C500 64-card SuperNode	2025	64 * C500		MetaXLink	Not disclosed
		Xijing S600 SuperNode	2026	64 * C600		MetaXLink	Not disclosed
Biren	Y	LightSphere 128	2025	128 * BR166L	Not disclosed	Silicon Photonic OCS	Not disclosed
Moore Threads	Y	MTT C256	2025	256 * S5000	FP8: 256 PFLOPS (est.)	MTLink 3.0	Not disclosed
T-Head	Y	Panjiu AL128	2026	128 M890		ICN Switch 1.0	Not disclosed

Source: Companies, BofA Global Research

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Overseas expansion has potential in future but remains at an early stage

We see overseas expansion becoming a longer-term strategic direction for Chinese AI accelerator vendors, but commercial traction outside mainland China remains limited so far. The domestic market is still absorbing most available capacity, supported by localization policies, restricted NVIDIA supply and rapid AI-infrastructure build-out.

Huawei appears to be making the first meaningful breakthrough in overseas AI accelerator commercialization. In Aug-2026, Bloomberg reported that Huawei is bidding for an Egyptian government AI data-center project involving 2,008 Ascend accelerators, including 1,408 Ascend 950-series chips for an AI training cloud and another 600 Ascend 950 or 910B chips for two inference clusters, with a proposed 12-month deployment schedule. While the tender has yet to be awarded, we view the project as an important proof point for Huawei's overseas strategy, potentially serving as the first large-scale overseas deployment of Ascend accelerators and a reference case for sovereign AI infrastructure projects across emerging markets. Huawei is also structurally advantaged by its broader AI infrastructure stack, leveraging its international cloud, telecom, enterprise networking, storage and data center customer relationships to bundle Ascend with Atlas servers, CloudMatrix clusters, Huawei Cloud and software services.

Other Chinese AI accelerator vendors remain at an earlier stage of overseas expansion. Biren has established a Hong Kong regional headquarters and R&D center to support international development, although disclosed customer wins remain largely China-focused. MetaX has achieved limited deployments at AI computing platforms in Hong Kong and is actively building overseas ecosystems through research and industry partnerships, but overseas revenue contribution remains immaterial.

Overall, we believe the key opportunity for Chinese AI accelerators lies in system-level capability, inference economics and chip availability rather than single-chip performance. However, overseas adoption remains constrained by 1) CUDA's developer ecosystem advantage, 2) weaker support networks, 3) geopolitical and cybersecurity concerns, and 4) supply-chain restrictions. As such, overseas expansion offers strategic optionality but is unlikely to become a major revenue driver for most vendors in the near term.

Memory

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The global memory semiconductor market size was USD223bn in 2025. We expect it to reach nearly USD1tn in 2026 and near USD1.4tn by 2028, driven by demand from AI infrastructure, cloud computing, data centers, smartphones, and automotive electronics. China is one of the largest memory semiconductor markets in the world. According to the Semiconductor Industry Association (SIA), memory chip sales in China came in at about USD59bn in 2025, representing around 26% of the global market.

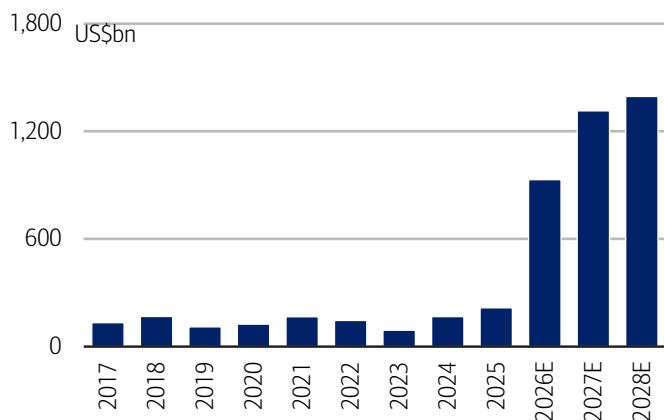
To date, China's memory chip supply still relies heavily on overseas suppliers. For example, China imported USD114bn worth of memory chips in 2025, accounting for 27% of its total semiconductor imports. This figure increased to USD145bn and represented 49% of total semiconductor imports in 1H26.

The current memory semiconductor ecosystem in China mainly consists of (1) major domestic memory chip IDMs (integrated device manufacturers), such as ChangXin Memory Technologies (CXMT), Yangtze Memory Technologies (YMTC), and SwaySure; (2) memory chip design companies ("fabless" firms), such as GigaDevice and Ingenic; (3) memory module manufacturers, including Longsys, Biwin, and Techwinsemi; and (4) suppliers of auxiliary memory-related chips, such as NAND flash controller vendors (e.g., Maxio and Yingren) and DRAM interface chip providers (e.g., Montage).

Besides domestic names, several major overseas memory companies also maintain significant manufacturing footprints in China. For example, Samsung Electronics operates a major NAND flash fabrication complex in Xi'an; SK Hynix operates DRAM production facilities in Wuxi and NAND flash manufacturing operations in Dalian; and Micron maintains memory packaging, testing, and module manufacturing operations in Xi'an.

Exhibit 26: Historically USD80bn (downcycle) to USD170bn (upcycle) p.a.; market size likely USD1.4tn by 2028 mainly driven by AI-related new applications

Global memory market size

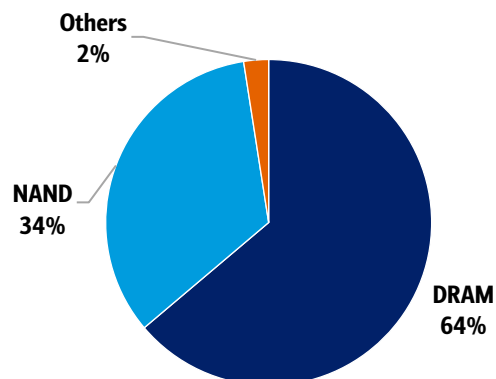


Source: BofA Global Research estimates

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Exhibit 27: DRAM accounted for nearly 2/3 of global memory market vs NAND flash about 1/3; other types (NOR flash, SRAM, etc) only 2%

Global memory market mix by chip type (2025)

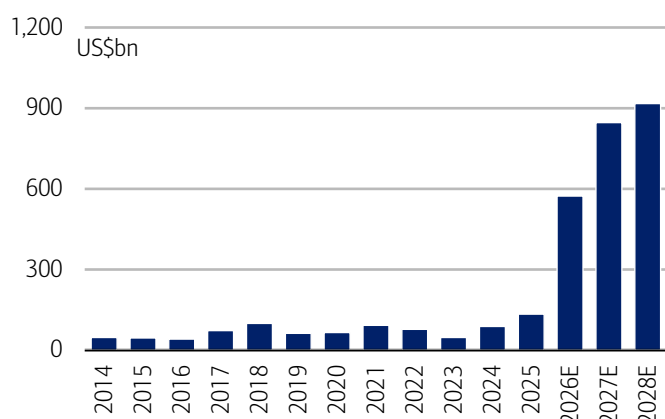


Source: WSTS, BofA Global Research

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Exhibit 28: Global DRAM market size to surpass USD500bn in 2026 vs sub-USD100bn p.a. prior to 2025 due to boosting AI demand

Global annual DRAM market size

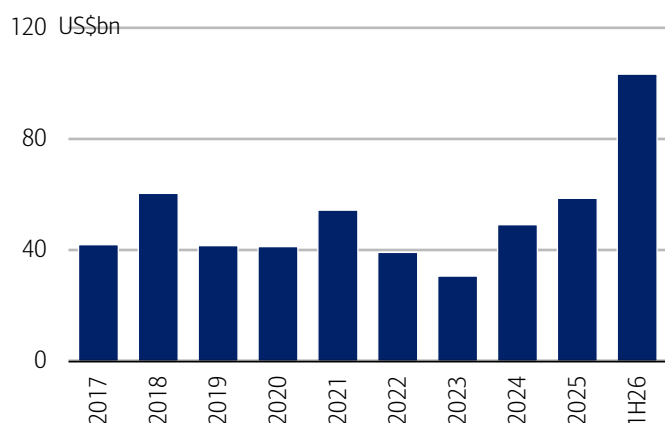


Source: BofA Global Research estimates

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Exhibit 30: Memory chip billings in China previously USD40-60bn but already exceeded USD100bn in 1H26

China memory market size

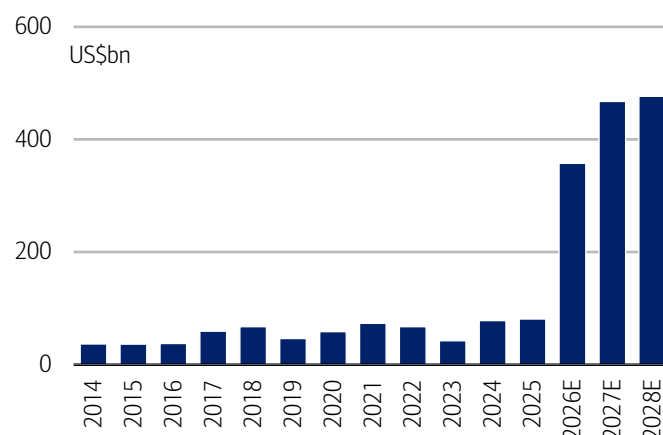


Source: WSTS, BofA Global Research

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Exhibit 29: Global NAND flash sales likely over USD350bn in 2026 vs historical range of approximately USD40bn-80bn per year

Global annual NAND market size

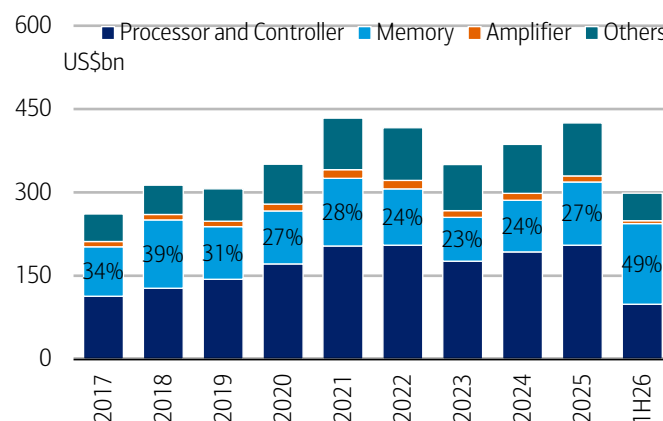


Source: BofA Global Research estimates

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Exhibit 31: Memory chip imports to China about USD100bn p.a. on average prior to 2026 (some were re-exported), representing c.25%-40% of China's semis imports

China IC imports breakdown by segment



Source: China General Administration of Customs

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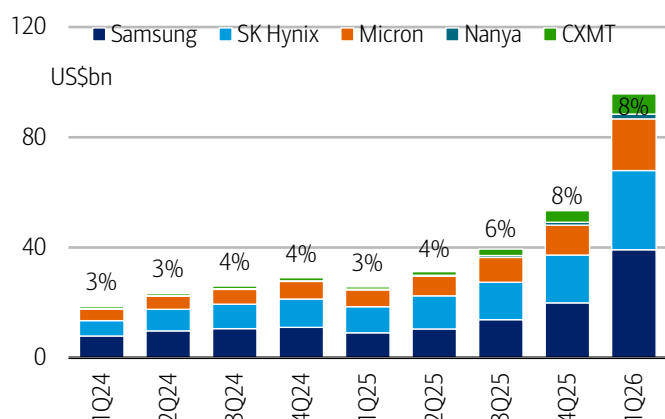
DRAM

Dynamic Random Access Memory (DRAM) is used to store, write, and read data. In electronic devices, DRAM acts as a cache and temporary memory layer between the permanent storage device (hard disk drives or solid-state drives) and the processor (CPU, etc.), storing frequently accessed data for rapid retrieval. Historically, the DRAM market was cyclical, with annual sales hovering between c. USD40bn and USD90bn per year over the past decade. Driven by AI demand, particularly high-bandwidth memory (HBM) used in AI training and inference, global DRAM sales surged to USD134bn in 2025 (+52% YoY) and are expected to be over USD900bn by 2028, implying a 90% CAGR over 2025-28.

The global DRAM market is dominated by three companies, namely Samsung Electronics, SK Hynix, and Micron, which collectively account for more than 90% market share by revenue. CXMT is the fourth-largest DRAM chipmaker globally, with 2025 revenue of USD8.6bn and a 6% global market share, followed by Nanya Tech (USD2.5bn revenue and 2% market share) and Winbond (USD0.9bn revenue and 1% market share).

Exhibit 32: CXMT global market share rose from just 3-4% in 2024/25 to 8% by 1Q26

Global DRAM market share by revenue

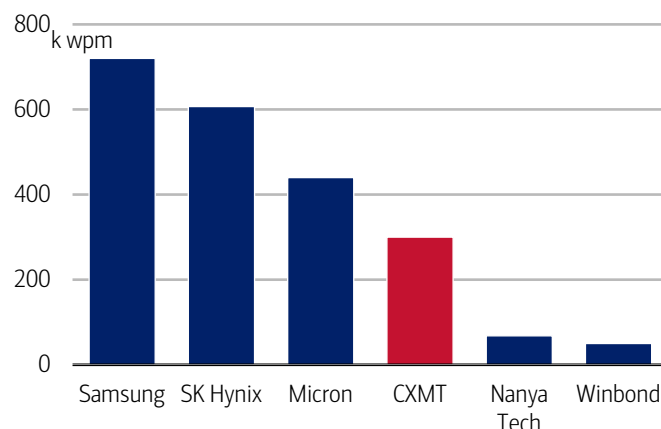


Source: Companies, BofA Global Research estimates

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Exhibit 33: We expect CXMT's DRAM capacity to be c.300k wpm by 2026, still behind global top-3 suppliers

DRAM wafer capacity by company (2026-end)



Source: Companies, BofA Global Research estimates

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CXMT began constructing its first DRAM fab in Hefei, China, in March 2017. So far, it has constructed three fabs, including two fabs (Fab A1 and A2) in Hefei and one fab in Beijing. The company is also building a new fab in Shanghai. According to media reports (Nikkei, 3 February 2026), CXMT's total wafer capacity was nearly 300k wpm, as of end-2025. In comparison, global DRAM production capacity came in at about 1.9mn wafers per month (wpm), as of end-2025, based on our estimates. Samsung Electronics, SK Hynix, and Micron had capacities of nearly 660k, 550k, and 410k wpm, respectively, as of end-2025. Nanya Tech's capacity was approximately 60-70k wpm.

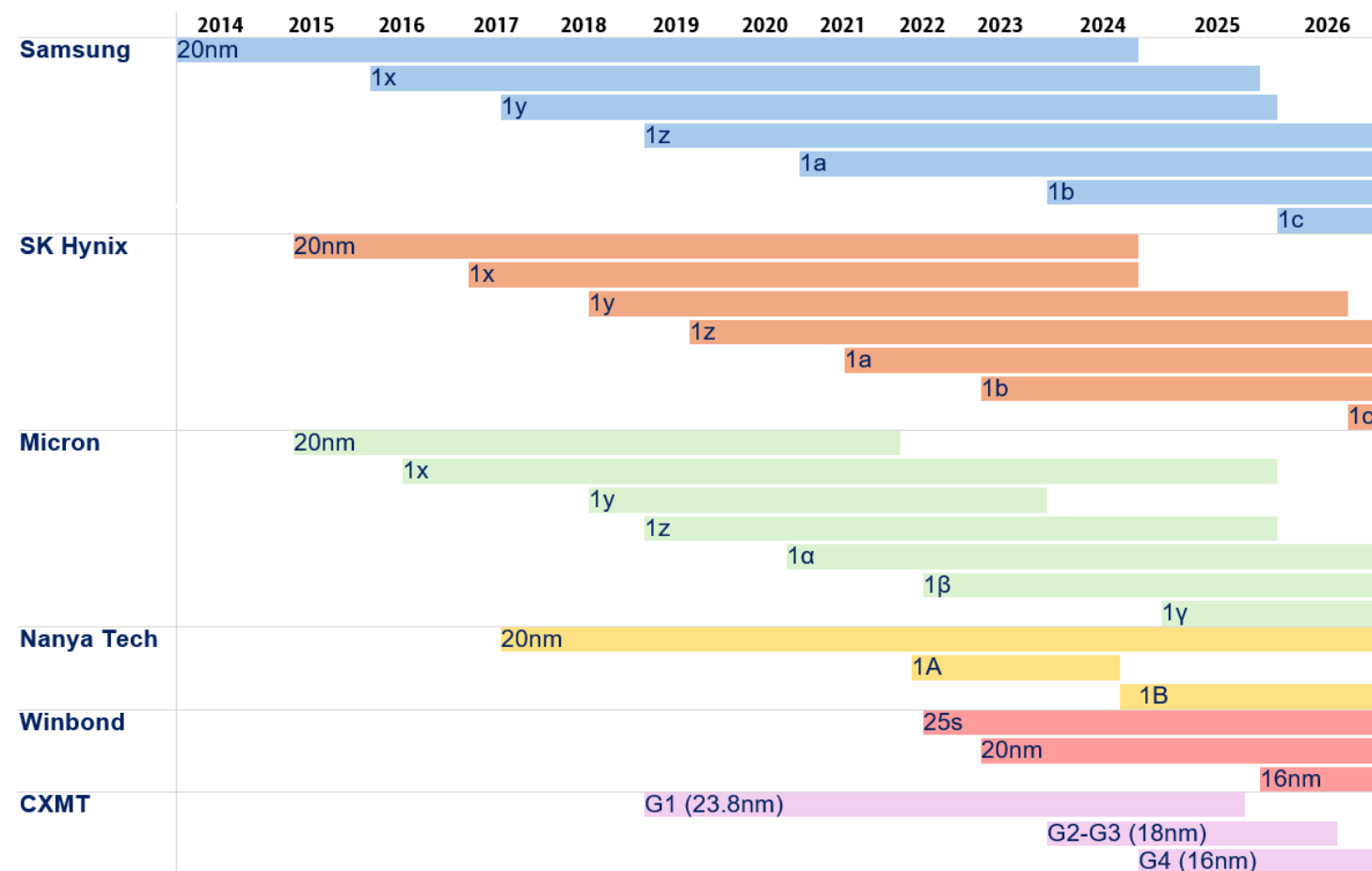
Samsung, SK Hynix, and Micron have begun mass production of DRAM chips based on their sixth-generation 10nm-class techy platforms. The node size is around 11-12nm, referred to as 1c nm by Samsung and SK Hynix, and 1γ (1-gamma) by Micron. CXMT has developed four generations of DRAM process technologies. According to TechInsights' teardown analysis, CXMT's initial DRAM node (G1) was a 24nm-class process, similar to the industry's 2y node, followed by its G2 and G3 generations, which were 18nm-class (1x nm) nodes, and its latest G4 generation (16nm-class/1z nm). CXMT stated that its first-generation node has been phased out, and that the G3 and G4 nodes are already in mass production. The company is also developing its fifth-generation platform, which adopts more advanced multi-patterning technologies to improve memory density and array performance.

By DRAM generation, CXMT introduced its first 8Gb DDR4 DRAM product in September 2019. Since then, the company has begun mass production of LPDDR4X, DDR5, and LPDDR5/5X products, while production of its in-house DDR4 products has been discontinued. According to the company, mass shipments of its fifth-generation DRAM products (DDR5 and LPDDR5/5X) started in 2023. The company disclosed that DDR5 and LPDDR5 accounted for 40% of its total revenue in 1H25. Meanwhile, high-density products, namely 12Gb LPDDR4X and 64Gb DDR5, contributed 9% and 10% of total sales, respectively. In September 2026, CXMT announced that it had begun mass production of LPDDR6 (its sixth-generation DRAM product), which has been adopted in Xiaomi's latest flagship smartphone, the Xiaomi 18 Fold.

Benefiting from booming AI-related demand, the global DRAM market has been in a "super-cycle" since 2H23. The overall DRAM ASP per bit (including HBM and conventional DRAM) was near USD12/GB in 2Q26, almost five times higher than the level of less than USD2/GB in 4Q23.

Exhibit 34: Samsung, SK Hynix, Micron migrating to the 6th generation 10nm-level node; CXMT's technology level similar to tier-1 suppliers' in 2019-20

Major DRAM manufacturers' technology roadmap

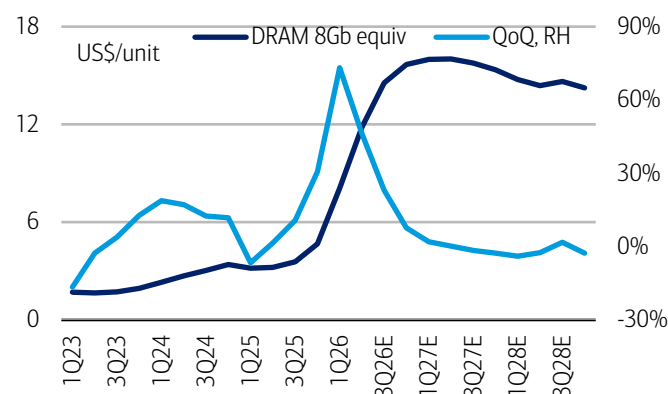


Source: Companies, Trendforce, BofA Global Research estimates

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Exhibit 35: DRAM “super-cycle” likely to extend through 2027/28

DRAM ASP and QoQ trend

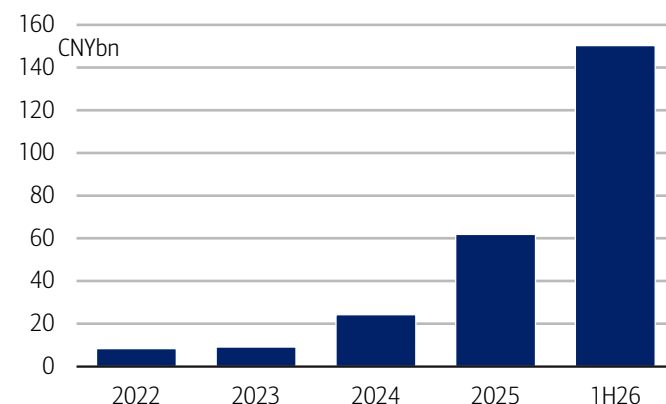


Source: BofA Global Research estimates

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Exhibit 36: CXMT 2025 sales were CNY61.8bn, +156% YoY; 1H26 sales reached CNY150bn, over 2x vs the whole year of 2025

CXMT – Annual sales



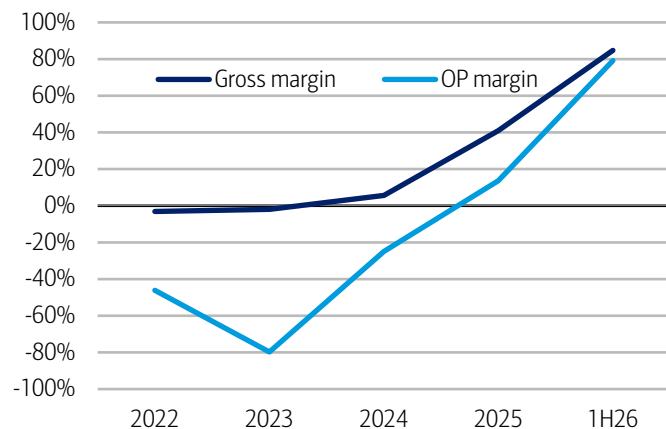
Source: Company

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Supported by surging DRAM prices and ongoing capacity expansion, CXMT's revenue surged from CNY9bn in 2023 to CNY62bn in 2025. Its 1H26 revenue reached CNY150bn, more than double its full-year 2025 revenue.

Exhibit 37: CXMT's gross/OP margin reached 85%/79% in 1H26 vs 41%/14% in 2025 and 6%/-25% in 2024

CXMT – Gross margin and OP margin

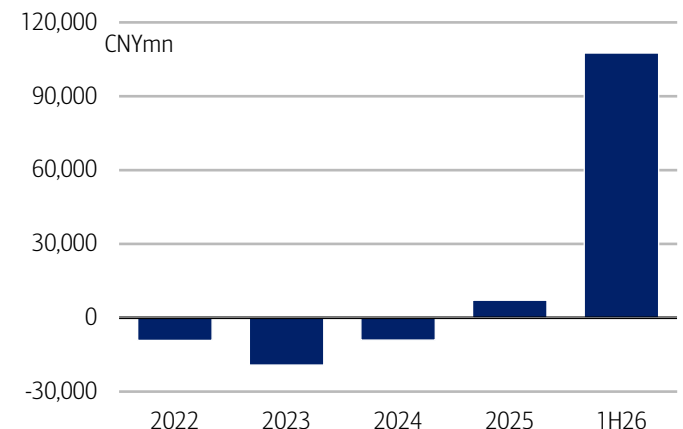


Source: Company

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Exhibit 38: CXMT turn profitable in 2025; CNY108bn net profit generated in 1H26 as DRAM prices rallied

CXMT – Net profit (including minority interest)



Source: Company

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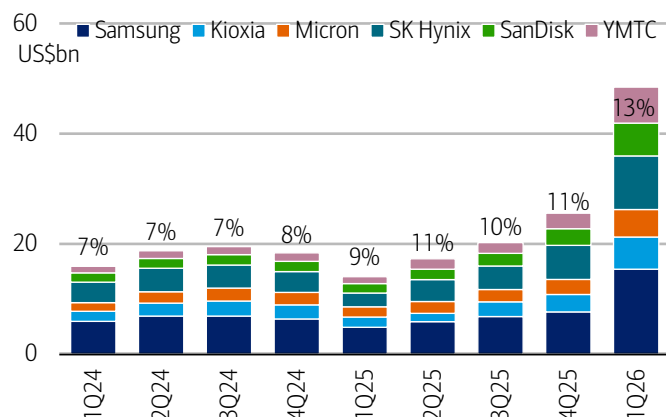
NAND flash

NAND flash (Not-And gate-based) is a type of semiconductor that retains data even when power is removed, making it suitable for permanent data storage in smartphones, PCs, data centers, and other electronic devices. The global NAND flash market size was about USD80bn per year in 2024/25 (vs. a historical range of USD40-60bn per year). We estimate the market could grow to more than USD350bn in 2026 and potentially exceed USD470bn by 2028.

The global NAND flash market is led by Samsung Electronics, SK Hynix, Kioxia, SanDisk, and Micron. YMTC was established in 2016 and has become one of the major NAND flash suppliers globally (with an estimated 13% market share in 1Q26, in our view).

Exhibit 39: YMTC's NAND flash market share around 13% in 1Q26, vs less than 10% prior to 2025

Global NAND flash market share by revenue

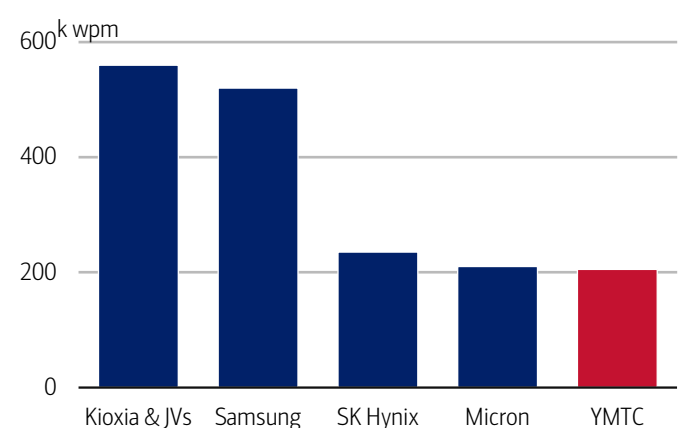


Source: Companies, BofA Global Research estimates

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Exhibit 40: We expect YMTC's NAND flash capacity to be around 200k wpm by 2026-end, likely similar to SK Hynix' and Micron's

NAND wafer capacity by company (2026-end)



Source: BofA Global Research estimates

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YMTC has planned three fabs for its Wuhan campus, each designed with a capacity of 100k wpm. Construction of Fab 1 started in 2016, and equipment move-in began in April 2018. YMTC announced the groundbreaking of Fab 2 in June 2020. Construction of the Fab 2 shell was divided into two phases (Fab 2A and Fab 2B), both of which have been completed. YMTC established a subsidiary in September 2025 to manage the Fab 3

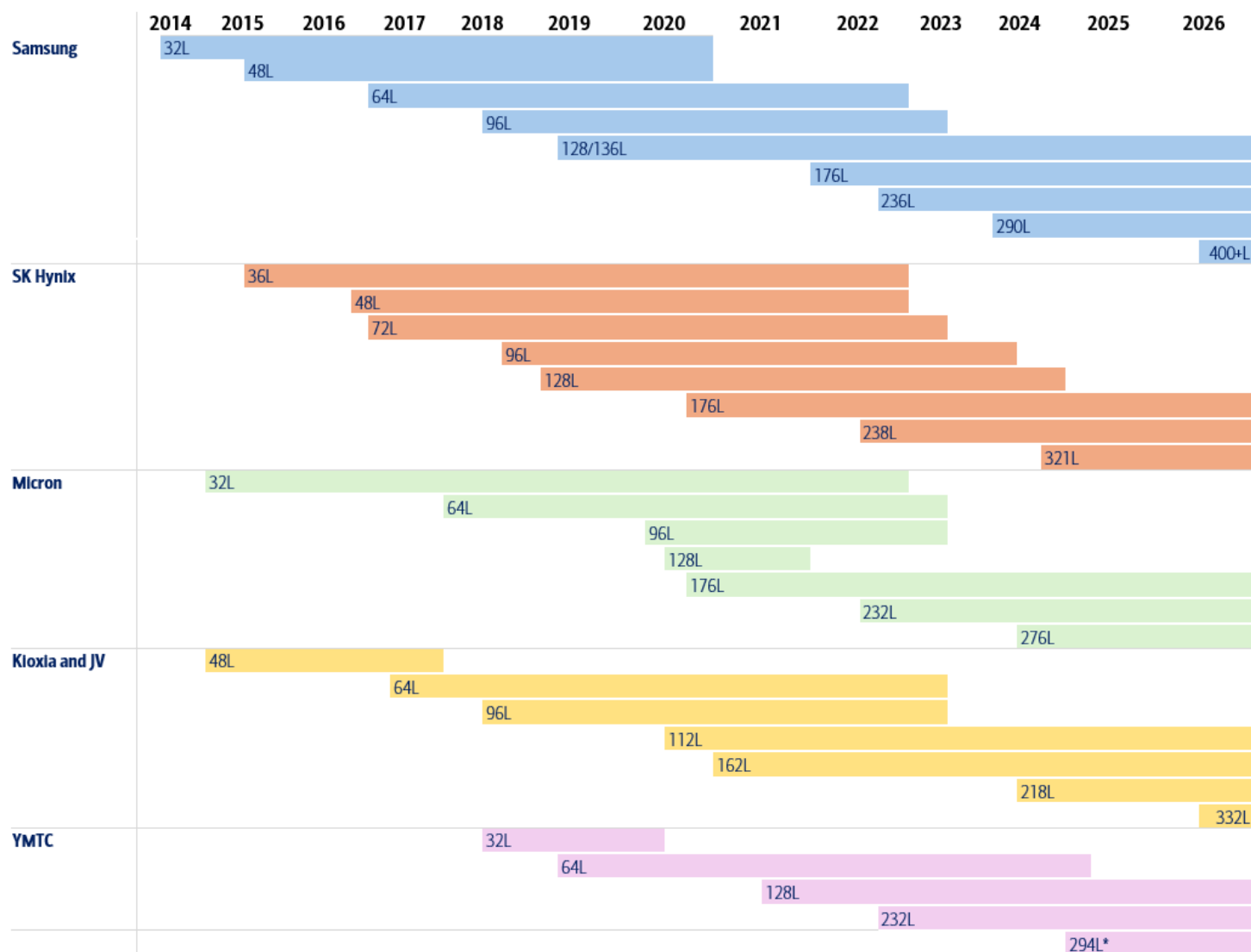


project. According to local media (Changjiang Daily, January 2026), construction of the Fab 3 shell had been completed, and the fab was undergoing cleanroom preparation. We estimate YMTC's total wafer capacity (excluding XMC) could expand to nearly 200k wpm by the end of 2026. This would be comparable to the NAND flash production capacities of Micron and SK Hynix (approximately 210-235k wpm each), although still below those of Samsung (c.520k wpm) and the Kioxia-SanDisk joint venture (c.560k wpm).

Before YMTC was formally established, XMC (also known as Wuhan Xinxin, now a subsidiary of YMTC) had worked with Spansion to co-develop 3D NAND flash. YMTC introduced its first 32-layer MLC NAND chip in August 2018 and then 64-layer/128-layer products in 2019 and 2021 respectively. Through 2024-25, YMTC introduced several chips based on the 4th generation Xtacking architecture, including X4-9060 (a 160-layer TLC NAND), X4-9070 (232-layer TLC), X4-6080 (232-layer QLC). Media (Digitimes, etc) reported in September 2025 that YMTC had begun producing 267-layer 3D NAND and was preparing designs for 300+ layers.

Exhibit 41: Global major NAND flash makers moving to 300/400-layer technologies in 2025-26 while YMTC mainly relying on 128/232-layer generations for now due to the restricted access to advanced manufacturing tools and technologies

Major NAND flash manufacturers' technology roadmap



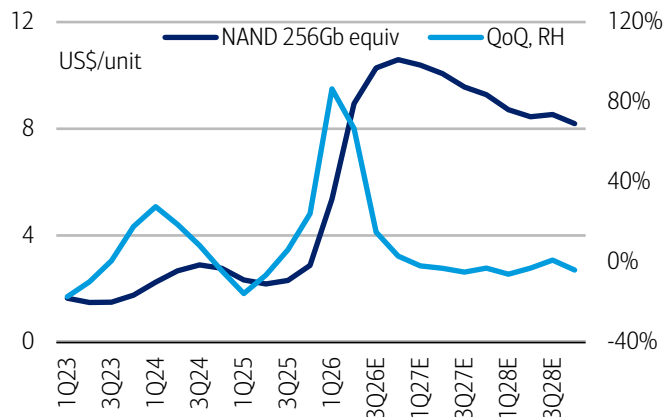
Source: Companies, TrendForce, BofA Global Research estimates *Per media reports (TechInsights, etc.), YMTC 5th generation 3D NAND chips have already stacked up to 294 layers but only 232 layers are active

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Similar to DRAM, NAND flash chips also saw a rapid surge in prices since 2023 – the average price per GB was about USD0.28 in 2Q26 vs about USD0.05 in 4Q23. YMTC's total revenue more than tripled from CNY19bn in 2023 to CNY63bn in 2025. The company reported its NAND flash revenue was CNY45bn in 1Q26, 5x of which in 1Q25. Meanwhile, its gross margin reached 77% in 1Q26, largely improved from 32-35% in 2024/25 and 5% in 2023.

Exhibit 42: NAND flash price likely to correct mildly in 2027/28 but still elevated level compared to the historical level

NAND ASP and QoQ trend

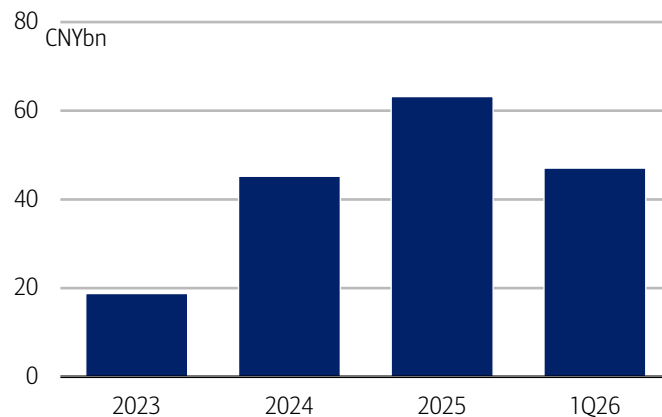


Source: BofA Global Research estimates

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Exhibit 43: YMTC 2025 sales CNY63bn, more than tripled vs CNY19bn in 2023; 1Q26 sales already CNY47bn

YMTC – Annual sales

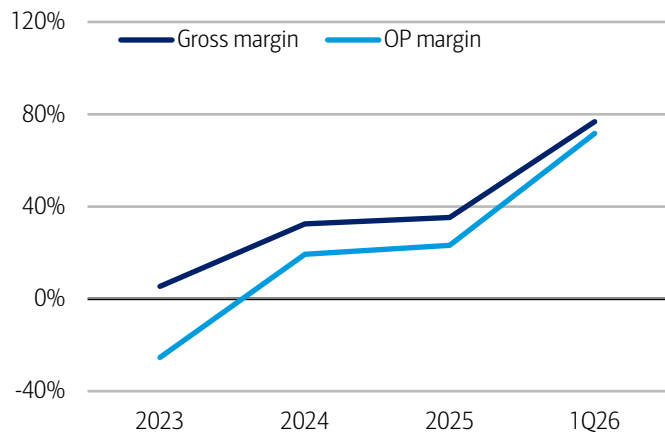


Source: Company

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Exhibit 44: GPM/OPM surged to 77%/72% in 1Q26 along with the NAND flash upcycle

YMTC – Gross margin and OP margin

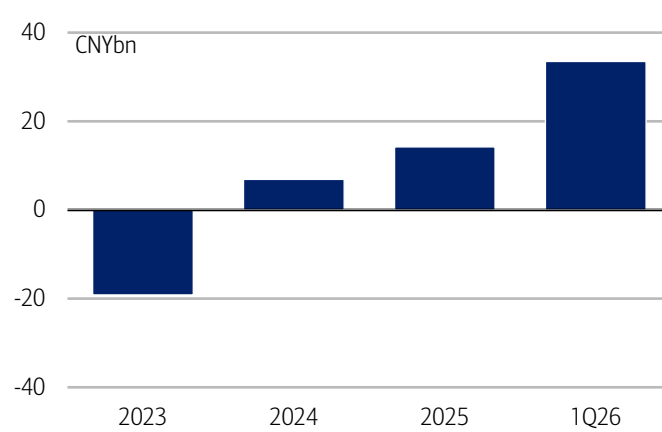


Source: Company

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Exhibit 45: 1Q26 net profit (RMB33.5bn) already exceeding 2024/25 full year profits

YMTC – Net profit (including minority interest)



Source: Company

BofA GLOBAL RESEARCH

Semiconductor equipment

Dai Shen >>

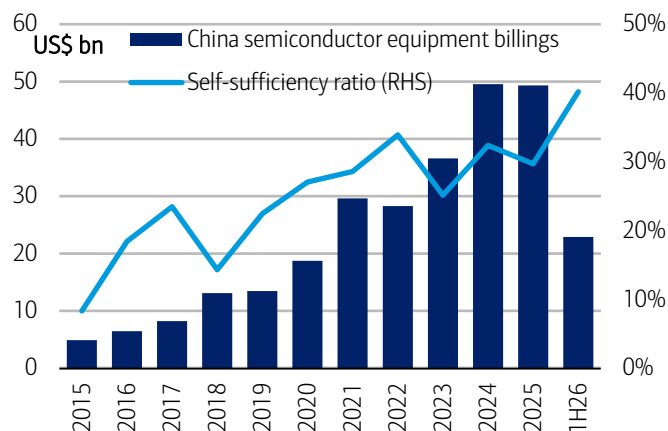
Merrill Lynch (Hong Kong)

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China is one of the world's largest semiconductor equipment markets and demand expands rapidly due to the fast-growing semiconductor manufacturing capacity in the country. According to SEMI.org, the semiconductor equipment billing in China was nearly USD50bn p.a. in 2024 and 2025, representing 42%/37% of the global market. Within the overall semiconductor equipment, the front-end wafer fabrication equipment (WFE) demand in China was around USD38-39bn (2024/25), accounting for about one-third of the global total. We believe the WFE demand in China could be more than USD50bn by 2028 and even USD60bn+/- in 2029-30.

Exhibit 46: China semis equipment market size was near USD50bn in 2024/25; local companies' market share climbed to 40% in 1H26

China semiconductor equipment demand and self-sufficiency ratio

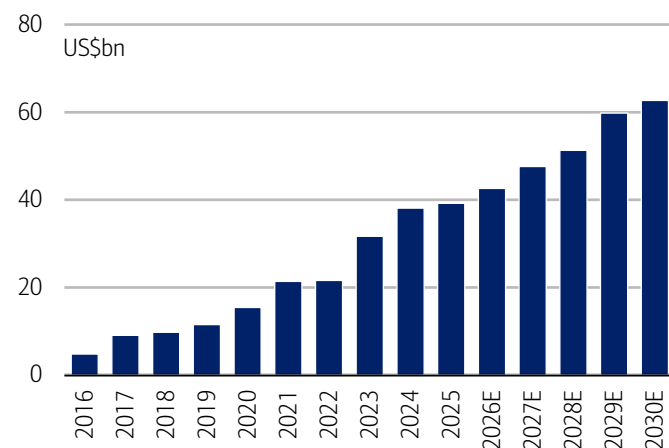


Source: SEMI.org, BofA Global Research

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Exhibit 47: Front-end equipment demand is likely to exceed USD40bn in 2026, and USD60bn in 2029-30

China WFE market size



Source: BofA Global Research estimates

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Prior to 2020, China's semiconductor manufacturing capacity was largely concentrated in 6-inch and 8-inch fabs operating mature process technologies, primarily 40nm and above. We observed a more meaningful expansion of 12-inch wafer capacity in 2020-25. We believe this capacity expansion and capital spending trend will continue, driven by:

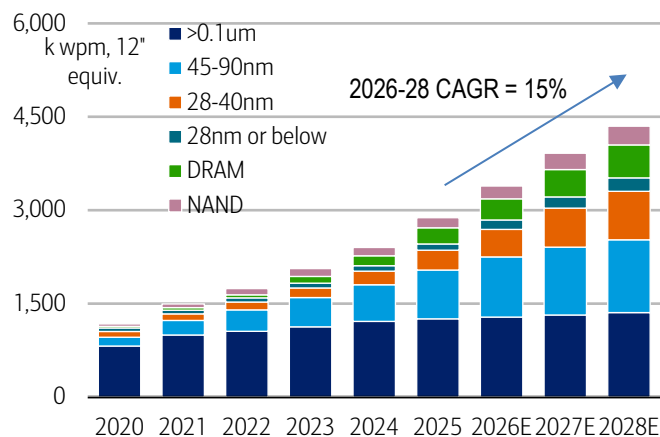
1. Technology migration toward mainstream and advanced process nodes (28nm and below) by domestic chipmakers such as SMIC, Huali IC, and Pengxinwei.
2. Continued expansion of memory production capacity in China, including investments by CXMT, YMTC, and SwaySure.
3. Ongoing additions of mature-node manufacturing capacity, which remains widely used for microcontrollers, analog ICs, power semiconductors, and connectivity chips serving applications such as AI and data centers, automotive, consumer electronics, and industrial markets.

Nevertheless, wafer capacity expansion in China is subject to policy and regulatory risks. For example, the US MATCH Act (Multilateral Alignment of Technology Controls on Hardware Act) was introduced in April 2026 in the US House of Representatives and remains subject to further legislative approval. The proposed legislation seeks to further restrict Chinese semiconductor manufacturers' access to advanced chipmaking equipment, including immersion DUV lithography systems, as well as maintenance services, spare parts, and technical support for installed equipment. If implemented, the

Act could pose a downside risk to the expansion plans and ongoing operations of major Chinese semiconductor manufacturers, including SMIC, CXMT, YMTC, Hua Hong, and Huawei, potentially affecting overall semiconductor equipment demand in China.

Exhibit 48: Future growth more driven by mainstream/advanced nodes, e.g. below-40nm logic, DRAM/NAND, etc.

China semiconductor fabs' capacity by node (year-end based)

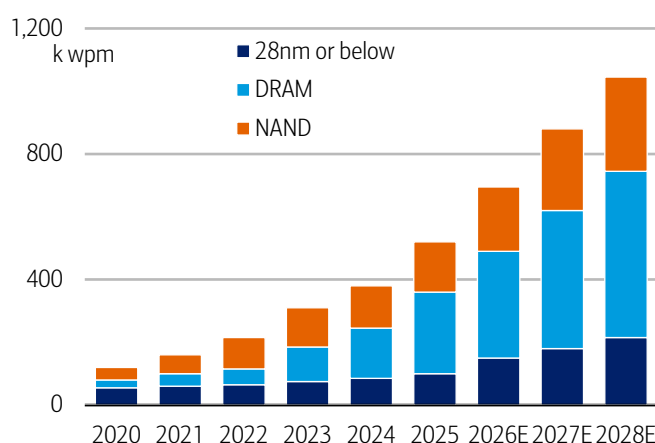


Source: BofA Global Research estimates

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Exhibit 49: We expect 28nm or below logic and memory capacities to double from c. 500k wpm in 2025 to over 1000k wpm by 2028

Advanced wafer capacity by Chinese fabs



Source: BofA Global Research estimates

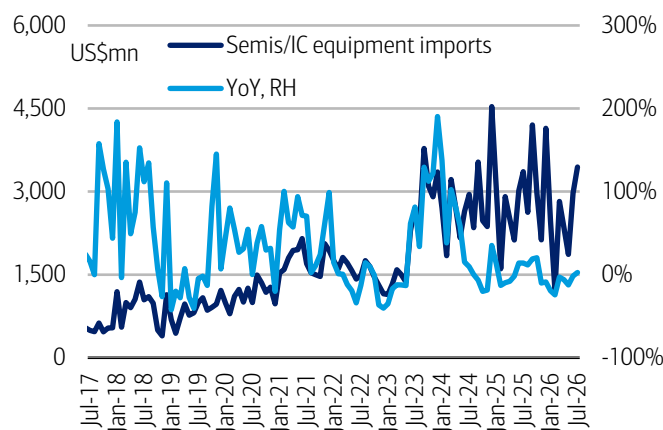
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Localization progress

China imported USD34bn/USD35bn worth semiconductor equipment (including both front-end wafer fabrication and back-end packaging/testing equipment) in 2024/25. This suggests the overall self-sufficiency ratio for semiconductor equipment in China was around 30% in the past 2 years. The ratio improved to nearly 40% in 1H26.

Exhibit 50: c.USD3bn worth semiconductor equipment imported to China per month since 2H23

China monthly semiconductor equipment import value

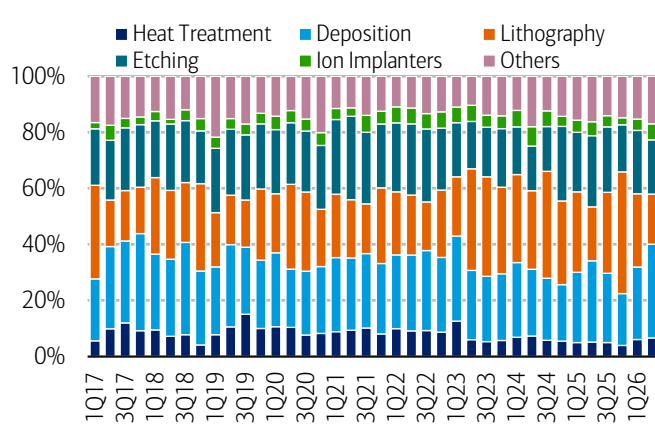


Source: China General Customs, BofA Global Research

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Exhibit 51: Lithography, deposition, and etching accounting for more than 70% of China's semiconductor equipment imports

China semiconductor equipment imports by type



Source: China General Customs, BofA Global Research

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Nevertheless, the localization progress differs significantly across equipment categories. In the wafer fabrication equipment (WFE) market, lithography, deposition, and etch equipment each account for more than 20% of total capital expenditure. For lithography equipment, media reports (*Financial Times*, 8 September 2026) indicate that several Chinese companies, including Shanghai Micro Electronics Equipment (SMEE), Yuliangsheng (a Huawei-backed equipment company), and Aishengna, are developing immersion DUV (deep ultraviolet) lithography systems capable of supporting 28nm and

more advanced process nodes. However, initial shipment volumes in 2026-27 are expected to remain very limited, with deliveries potentially reaching no more than a dozen tools in 2027. As a result, Chinese semiconductor manufacturers continue to rely heavily on overseas suppliers, particularly ASML, for lithography equipment.

In contrast, we estimate that the self-sufficiency rate for deposition and etch equipment has improved significantly, rising from c. 20% in 2023 to c. 40% in 2025, driven by technology advancements from domestic suppliers such as NAURA, AMEC, and Piotech. Local suppliers have also achieved substantial market penetration in the cleaning and CMP (chemical-mechanical planarization) segments, with domestic market share already exceeding 50%, led by companies such as ACM Research, NAURA, and Hwatsing Tech.

However, localization progress in metrology and inspection systems, ion implantation equipment, and photoresist track systems has generally lagged behind other equipment categories. In our view, domestic suppliers accounted for only low-teen percentage market share levels in these segments as of 2025.

Exhibit 52: Domestic suppliers secured notable market shares in etching, deposition, CMP, cleaning

China WFE localization progress by equipment type

Equipment	Local mkt share	Domestic players	Global leaders
Lithography	<5%	SMEI, Yuliangsheng	ASML, Nikon, Canon
Etching	near 40%	AMEC, NAURA, E-Town	Lam Research, TEL, AMAT
Deposition	>40%	NAURA, Piotech, Leadmicro, SiCarrier	AMAT, Lam Research, TEL
Ion implantation	10-15%	Wanye, CETC, NAURA	AMAT
PR development	>10%	Kingsemi	TEL
CMP	>50%	Hwatsing	AMAT, EBARA
Cleaning & PR stripping	>50%	NAURA, ACM Research, Kingsemi, E-Town	SCREEN, TEL, Lam Research
Metrology/inspection	>10%	Skyverse, Jingce, SiCarrier	KLA, AMAT

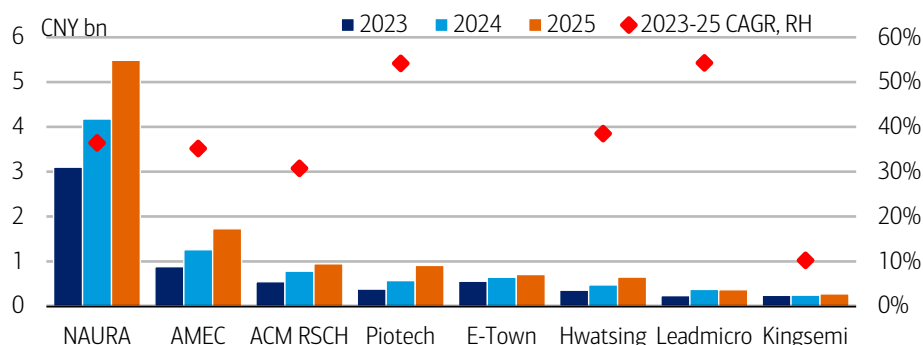
Source: BofA Global Research estimates

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Compared to the period prior to 2020 when China semiconductor equipment makers largely remained specialized in particular processes, the companies are increasingly pursuing a platformization strategy, expanding from single-process tools into multi-product portfolios that cover a broader range of wafer fabrication steps such as etch, deposition, cleaning, CMP, metrology, and ion implantation. The goal is to increase their addressable market, deepen relationships with key customers such as SMIC, CXMT, and YMTC, and leverage common technology platforms across multiple tool categories. While this strategy typically depresses profitability in the medium-/near-term due to heavy R&D spending, product development, and customer qualification costs, it can significantly enhance long-term profitability through higher revenue growth, stronger customer stickiness, better operating leverage, and recurring service income.

Exhibit 53: Most Chinese front-end equipment makers grew revenue 30-50% p.a. in 2023-25

China WFE suppliers' annual sales and growth

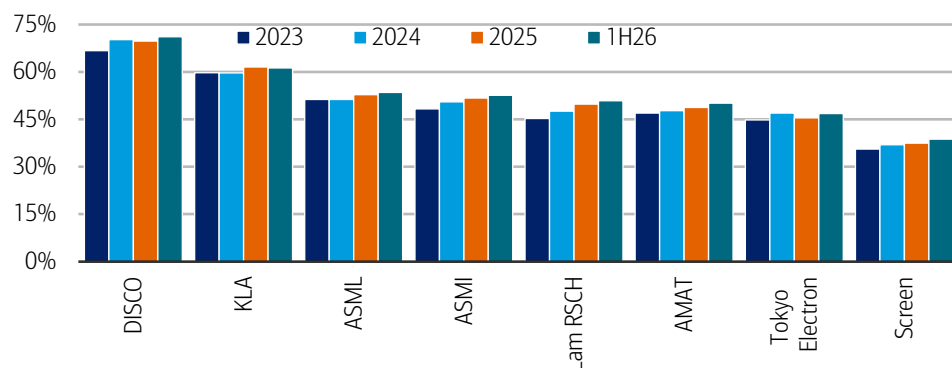


Source: Companies, BofA Global Research

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Exhibit 54: Global WFE suppliers witnessed gross margin improvement from 2023 to 1H26

Global WFE companies' gross margin (calendar year based)

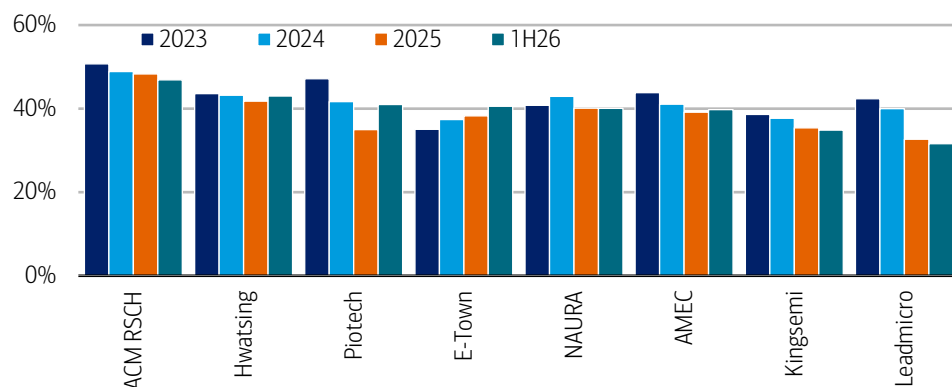


Source: Companies

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Exhibit 55: ...but Chinese companies' GPM mostly contracted due to competition, new product introduction cost, etc.

China WFE suppliers' gross margin

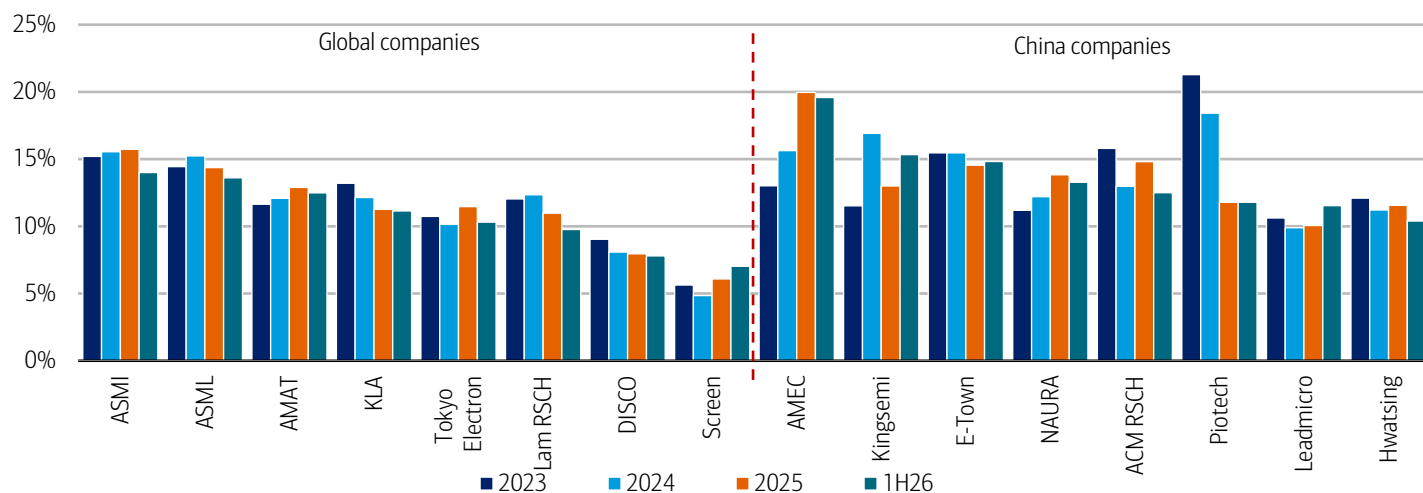


Source: Companies

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Exhibit 56: Global semiconductor equipment suppliers' R&D expenses account for low-teen percent of their total revenue on average; Chinese companies' ratios slightly higher due to their platformization and localization efforts

Global/China semis equipment companies' R&D expense of total sales



Source: Companies

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AI model

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In the past year, we observed 1) rapid evolvement of Made-in-China AI models with narrowed performance gap with the global peers; 2) significant user adoption of AI chatbot and more recently native AI agent products, and continuing AI features roll-out on top of existing platforms; 3) initial monetization of AI mainly via cloud/MaaS.

Looking ahead, our key views on China AI market/narrative development are:

AI segmentation: from frontier scale to smarter spend

A two-speed global AI market is emerging: US players are likely to lead in frontier capabilities, while Chinese models are gaining traction as cheap-and-capable performers. Despite compute constraint, Chinese players are optimizing for smarter spend—as pricing for frontier models increases, this cost-performance advantage becomes increasingly relevant globally, particularly for high-volume, non-critical use cases.

Hyper growth, neck-to-neck race

China's AI model landscape is vibrant and intensely competitive, with limited capability gaps across **incumbents (e.g. Alibaba, ByteDance)** and **independent AI labs (e.g. DeepSeek, Zhipu, MiniMax)**. In this environment, marginal model superiority is unlikely to drive durable profits—compute access, ecosystem integration, and distribution are the key moats, structurally favoring incumbents. With stronger balance sheets and integrated cloud stacks, incumbents can better tolerate low model layer profitability to drive compute cross-selling. Meanwhile, DeepSeek's improving capabilities and limited commercial focus are resetting industry benchmarks toward "high performance at low price," compressing profitability for standalone model providers.

Monetization: enterprise-led, cloud-centric

AI monetization remains enterprise-driven, with cloud and MaaS (Model-as-a-Service) as the clearest pathways. We expect China AI MaaS market to reach RMB676bn in 2030 (42% of cloud market), driving 38% CAGR of total cloud growth from 2026-30E. On compute build-out, based on media reports and company disclosure, **we examine the top 4 CAPEX spenders in China internet sector - collectively the combined CAPEX is likely to be up c.123% YoY in 2026E vs. 2025**, on top of c.46% YoY in 2025 vs. 2024, with ByteDance and Tencent contributing the most dollar amount growth.

Consumer AI: limited differentiation, still at infancy

In consumer AI, not all users monetize equally. Value is shifting toward high-intent "prosumers," while mass adoption monetization via subscription remains nascent. Ads/transaction-based models could offer longer-term revenue optionality. AI chatbot app represents a new entrance and interaction model for information discovery, hence has been a key battleground for incumbents. ByteDance currently leads on user scale (measured by app-only DAU), but **we note limited product differentiation and data-flywheel effect among China AI chatbots for now**, evident from daily user time spent showing no meaningful difference.

There are far **more questions than answers regarding the end-game of AI agents development in China** – would user behavior fixed to "one super agent" or "multiple agents"? Would smartphone operating systems launch their own AI agent which may disintermediate existing super apps? Would AI agent disrupt existing advertising monetization model? Given key internet content/service providers having strong incentive against disintermediation, our base case is that users will eventually interact with multiple agents, with segregate data domains.

Status quo of China AI model landscape

We summarize key developments of China AI model market in the past few months:

- Chinese AI models are improving rapidly on frontier-level capability, though gap remains in reasoning and agentic capability vs. global SOTA models.
- Chinese AI models are already dominating low-cost-high-performance market globally, with DeepSeek/Alibaba/Zhipu/Tencent as notable players in this field.
- Domestic leaders (incumbents & independent AI labs) are in a tight race of competition if measured by model intelligence level. Independent labs can stay (surprisingly) close to, or even lead, incumbents despite perceived compute disadvantages. This is likely due to AI-native focus, strong research talent, open-weight knowledge diffusion and efficiency-oriented architectures.

Exhibit 57: AI model landscape is intensely competitive in China

AI model details, benchmark scores, and pricing comparison table

	DeepSeek	Alibaba	Minimax	Zhipu AI	Tencent	Moonshot AI	ByteDance	Google	OpenAI	Anthropic
	DeepSeek		Minimax-					Gemini 3.8		Claude Fable
Flagship model	V4.1 Flash	Qwen-3.8 Max	M3.0	GLM-5.3	Hy4 preview	Kimi K3	Seed2.1 Pro	Flash	GPT-6 Astra	5.1
Country	China	China	China	China	China	China	China	US	US	US
Release date	9/10/2026	8/3/2026	6/1/2026	8/19/2026	8/28/2026	7/17/2026	6/23/2026	9/2/2026	9/3/2026	9/1/2026
Open/Proprietary	Open	Open weights	Open	Open	Open	Open	Proprietary	Proprietary	Proprietary	Proprietary
Model Architecture	MoE + hybrid	MoE + hybrid	MoE + hybrid	MoE + hybrid	MoE + hybrid	MoE + hybrid	MoE	Transformer	Transformer	Dense transformer
Context window	1mn	1mn	1mn	1mn	1mn	1mn	256k	1mn	1.05mn	1mn
Benchmark	DeepSeek V4.1 Flash (Max)	Qwen-3.8 Max	Minimax-M3.0	GLM-5.3 (max)	Hy4 preview	Kimi K3 (max)	Seed2.1 Pro	Gemini 3.8 Flash (high)	GPT-6 Astra (max)	Claude Fable 5.1 (Max)
AA Intelligence Index v4.3	40	40	30	45	NA	44	NA	41	53	53
AA Agentic Index	NA	50	31	53	NA	51	NA	41	52	58
AA Coding Agent Index	NA	43	NA	54	NA	52	NA	42	62	62
<i>Reasoning</i>										
GPQA Diamond	91%	93%	93%	92%	92%	94%	NA	95%	96%	94%
HLE	39%	43%	39%	42%	43%	47%	43%	48%	55%	59%
GDP.pdf	13%	20%	10%	11%	NA	22%	NA	21%	31%	26%
AA-LCR v1.1	84%	78%	83%	80%	NA	89%	NA	81%	81%	85%
GDPval-AA v2	50%	57%	40%	59%	NA	54%	NA	48%	54%	63%
CritPt	14%	20%	4%	19%	17%	23%	NA	18%	32%	30%
<i>Coding</i>										
SciCode	52%	53%	45%	56%	NA	59%	60%	57%	56%	56%
Terminal-Bench v2.1	91%	81%	65%	84%	85%	85%	71%	88%	88%	91%
<i>Knowledge</i>										
AA-Omniscience Accuracy	46%	32%	17%	34%	NA	48%	NA	55%	63%	67%
AA-Omniscience										
Hallucination Rate	96%	42%	18%	30%	NA	53%	NA	55%	51%	73%
<i>Pricing</i>										
Input (\$/1M)	0.2	2.0	0.3	1.4	0.9	3.0	0.9	0.8	10.0	10.0
Output (\$/1M)	0.6	6.0	1.2	4.4	2.6	15.0	4.4	3.8	50.0	50.0

Source: OpenRouter, Artificial Analysis, Vals AI, company websites, BofA Global Research (data as of September 11, 2026). *All prices are after discounts at off-peak hours and input price refers to input price (cache miss). Cell shading indicates relative performance across models for each benchmark. Darker green represents relatively stronger performance, while lighter green represents relatively weaker performance.

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Chinese AI models catch up quickly on intelligence, though gap remains vs. global SOTA models

On the Artificial Analysis Intelligence Index, which tracks progress toward AGI across mathematics, science, coding and reasoning, Zhipu's GLM-5.3 (max) ranked 7th globally with a score of 45, while Kimi K3 ranked 9th at 44. The leaderboard was led by Claude Fable 5.1 (max) and GPT-6 Astra (max) at 53, followed by Claude Opus 5 (max) at 51. Zhipu's performance was even stronger on agentic capabilities. On the Artificial Analysis

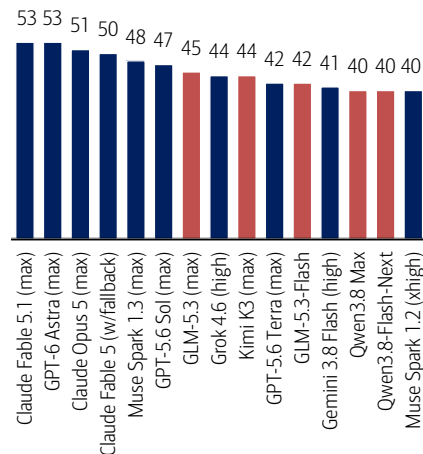


Agentic Index, which measures models' ability to complete multi-step agentic tasks, GLM-5.3 (max) ranked 4th globally with a score of 53, behind only Claude Fable 5.1 (max) (58), Claude Opus 5 (max) (56) and Muse Spark 1.3 (max) (56).

Chinese models also remained competitive on coding-agent benchmarks. On the Artificial Analysis Coding Agent Index, which evaluates end-to-end software engineering performance, GLM-5.3 (max) ranked 6th globally with a score of 54, while Kimi K3 ranked 7th at 52. The benchmark was led by Claude Fable 5.1 (max) and GPT-6 Astra (max) at 62.

Exhibit 58: Claude Fable 5.1 (max) was ranked highest in Intelligence index

Intelligence Index - Top 15 Models



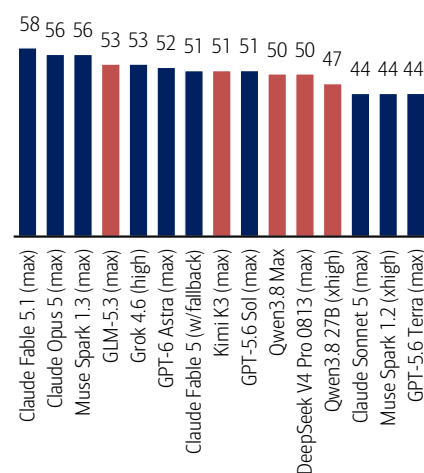
Source: Artificial Analysis as of Sep 11, 2026

*Ranking based on models selected by Artificial Analysis

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Exhibit 59: Claude Fable 5.1 (max) was ranked highest in Agentic index

Agentic Index - Top 15 Models



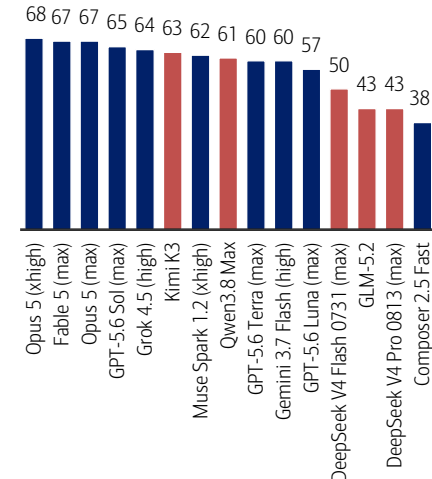
Source: Artificial Analysis and BenchLM as of Sep 11, 2026

*Ranking based on models selected by Artificial Analysis

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Exhibit 60: Claude Opus 5 (xhigh) was ranked highest in Coding Agent index

Coding Agent Index - Top 15 Models



Source: Artificial Analysis as of Sep 11, 2026

*Ranking based on models selected by Artificial Analysis

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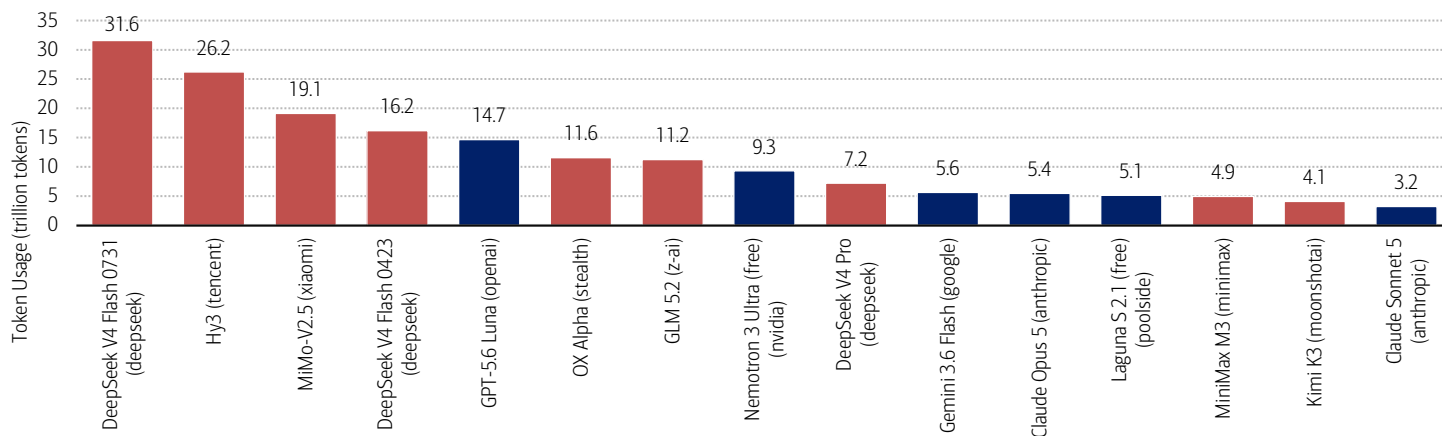
Chinese model already dominating low-cost-high-performance market globally

Token Usage - OpenRouter

Model usage trends from OpenRouter suggest strong adoption of lower-cost Chinese open models, with DeepSeek V4 Flash 0731 leading at 31.6tn tokens MTD, followed by Tencent's Hy3 at 26.2tn and Xiaomi's MiMo-V2.5 at 19.1tn, as of August 17.

Exhibit 61: DeepSeek V4 Flash and Tencent Hy3 lead in token usage

Top 15 models by token usage (August MTD, trillion token)

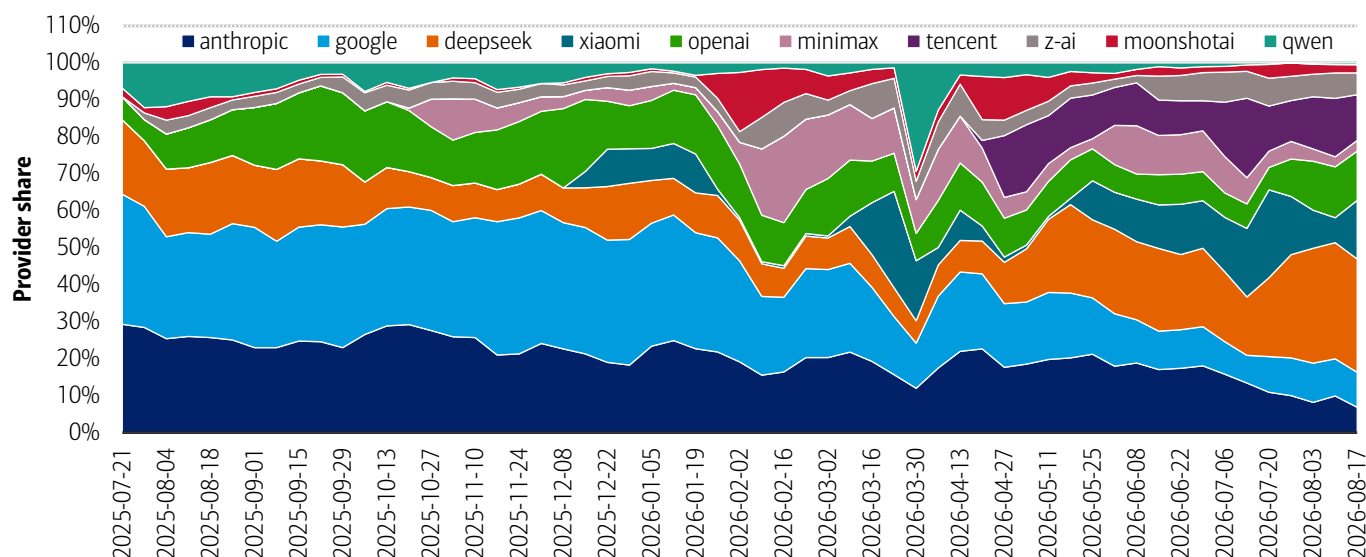


Source: Openrouter.ai data as of August 17

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Exhibit 62: DeepSeek led token usage share on OpenRouter, accounting for 22% weekly volume share in the week of August 17

Weekly token volume share by provider among OpenRouter's top 50 models



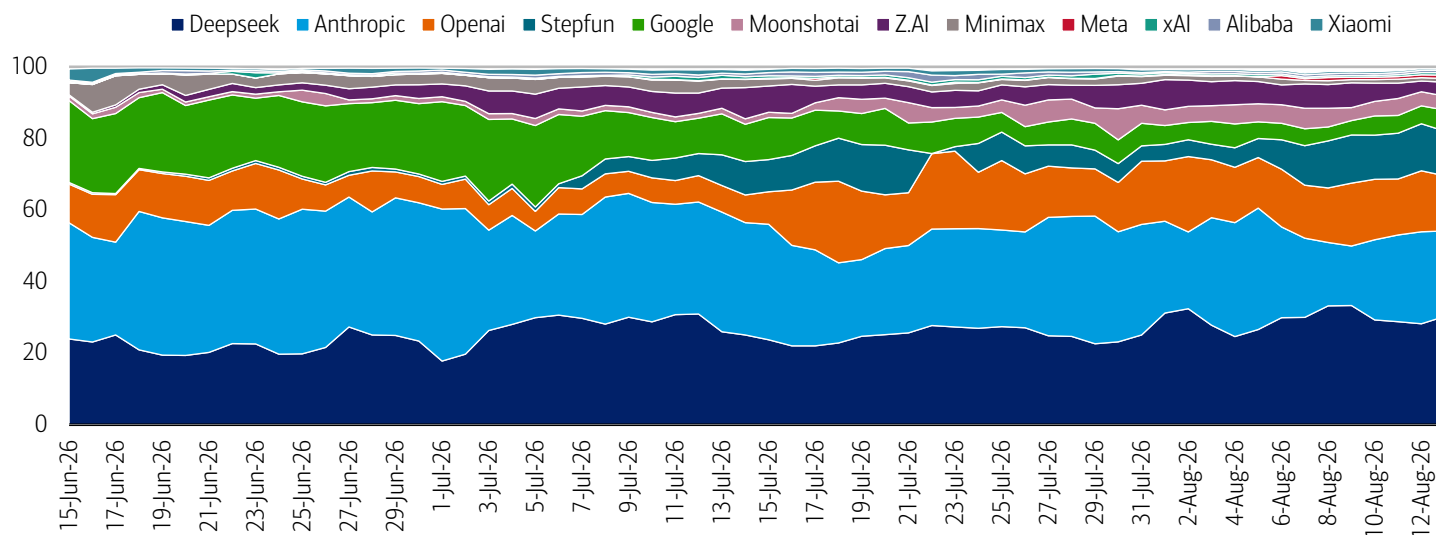
Source: Openrouter.ai data as of August 17. **Note:** OpenRouter is an AI model aggregation and routing platform that provides access to multiple LLMs through a single API. Its token-usage rankings reflect developer adoption, cost-performance preferences and commercial traction among workloads routed through OpenRouter, but do not represent total market usage, excluding direct API traffic, enterprise private deployments, self-hosted inference and hyperscaler cloud workloads.

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By providers, DeepSeek led the weekly token usage on OpenRouter at 20.3tn tokens, followed by Xiaomi's 10.5tn tokens, OpenAI's 8.8tn tokens, and Tencent's 8.2tn tokens in the week of August 17. Meanwhile, among the top 50 models, DeepSeek accounts for the largest token usage share at 22%, followed by Xiaomi at 11% and OpenAI's 9% in the week of August 17. Compared to the same week in July, the most notable share gains were DeepSeek (+4.7ppt) and OpenAI (+4.6ppt), while the largest declines were Xiaomi (down 7.8ppt), Anthropic (down 3.9ppt).

Exhibit 63: DeepSeek led AI token usage on Vercel, capturing nearly 25% share MTD in August (vs 30.1% in July)

Daily token volume share on Vercel AI platform (%)



Source: Vercel.com. **Note:** Vercel AI Gateway data tracks AI labs' share of routed token volume and spend, providing a proxy for developer adoption and commercial traction. Token share reflects usage volume, while spend share incorporates both usage and market-rate token pricing based on providers' published list prices. The data covers only workloads routed through Vercel AI Gateway and excludes direct API usage, enterprise private deployments and inference conducted through other cloud providers.

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Token volume share on Vercel AI platform

DeepSeek led the average MTD token usage share on the Vercel AI platform at 29.7% (vs 26.0% in July), followed by Anthropic's share at 24.7% (vs. 30.1% in July), OpenAI's 16.3% (vs. 12.3% in July) and Google 5.1% (vs. 11.5% in July). The most notable share gains were OpenAI (+4.1ppt) and DeepSeek (+3.7ppt), followed by StepFun (+3.3ppt), while the largest declines were Google (down 6.4ppt), Anthropic (down 5.4ppt), followed by MiniMax (down 1.3ppt).

US and China bring distinct AI model development strengths

Introducing our evaluation framework on AI model development

We believe the winning AI model businesses will be those that combine **training capability** — compute, talent and data — with **deployment advantage** — inference efficiency, use cases and customer access. As the industry shifts from a model-building race toward scalable and economically viable deployment, the key differentiator will be less about who can train the best model in isolation, and more about who can deploy AI at scale with clear monetization.

As shown below, we believe US and China bring distinct strengths to AI model development, each benefiting from different advantages across technology, infrastructure, talent, and applications.

Exhibit 64: US and China bring distinct AI model development strengths

Key factors, questions, and driving factors for AI model businesses

Factor	Key question	What drives winners	US advantage	China advantage	Current edge	Investment implication
Training compute	Can you build competitive foundation models?	GPU scale, training efficiency, model architecture	Greater access to frontier GPUs, hyperscale data centers, advanced networking, capital, and repeated large training runs	Stronger incentive to optimize around compute constraints using MoE, distillation, quantization, and efficient training architectures	US clearly ahead in compute; model-output gap much narrower	US has the higher scaling ceiling. China can remain competitive by generating more model capability per unit of compute, but frontier-chip restrictions remain the largest structural constraint.
Talent	Can you continuously improve and apply models?	Research capability + engineering execution	Concentration of elite frontier researchers, leading labs, global talent, and experience managing very large training clusters	Large STEM and engineering base, strong implementation speed, and close links among academia, technology companies, and industrial users	US marginally leads on frontier research; closer in engineering execution	US advantage is concentrated in a limited number of frontier labs. China can compete through broader engineering resources and faster diffusion of model improvements across companies.
Data	Do you have proprietary learning loops?	Unique, domain-specific data and feedback flywheel	Global, multilingual enterprise and professional data from cloud, SaaS, search, productivity, healthcare, and developer platforms	Large-scale consumer, payments, e-commerce, logistics, manufacturing, mobility, robotics, and other physical-world data	Split by domain	US has the stronger knowledge-work and global enterprise-data position. China has potentially stronger industrial and embodied-AI feedback loops, although regulation and platform silos can limit data sharing.
Inference compute	Can you deliver AI cheaply, at scale?	Low cost, low latency, high reliability	Superior GPUs, high-bandwidth memory, networking, cloud reliability, and mature software stacks	Potentially lower power, infrastructure, engineering, and operating costs; efficient MoE models and domestic accelerators for optimized workloads	US in performance; China potentially in cost	This is China's clearest potential advantage. China can achieve a lower cost per useful token if efficiency and utilization offset weaker chip performance and software.
Use case + client base	Can you monetize deployment?	Workflow depth, distribution, and paying customers	Higher enterprise willingness to pay and global distribution through cloud, SaaS, advertising, developer tools, and professional software	Large digitally integrated consumer base and strong demand from commerce, manufacturing, logistics, mobility, and public services	US in enterprise monetization; China in deployment volume	US companies should generate higher revenue per user and per token. China may generate larger usage volumes, but aggressive competition and lower pricing could constrain margins.
Deployment ecosystem	Can models be embedded into real products?	APIs, apps, agents, enterprise/consumer touchpoints	Global cloud platforms, APIs, operating systems, enterprise software, developer ecosystems, and international customer access	Dense integration with payments, super-apps, hardware supply chains, factories, vehicles, robots, and consumer electronics	US in software; China in consumer internet & physical-world AI	US likely leads in globally distributed software and enterprise agents. China could lead in embedding AI into hardware, factories, autonomous systems, and integrated consumer platforms.

Exhibit 64: US and China bring distinct AI model development strengths

Key factors, questions, and driving factors for AI model businesses

Factor	Key question	What drives winners	US advantage	China advantage	Current edge	Investment implication
Overall position	Can the full stack produce sustainable economic returns?	Combination of model quality, cost, distribution, utilization, and monetization	Controls more scarce upstream inputs, including frontier chips, capital, cloud infrastructure, elite labs, and global enterprise distribution	Strong cost optimization, rapid product iteration, large-scale deployment, manufacturing integration, and embodied-AI ecosystem	US still leads overall, but increasingly bifurcated	Likely outcome is US leadership in frontier training and premium software, alongside Chinese strength in efficient inference, open models, industrial AI, and embodied deployment.

Source: BofA Global Research

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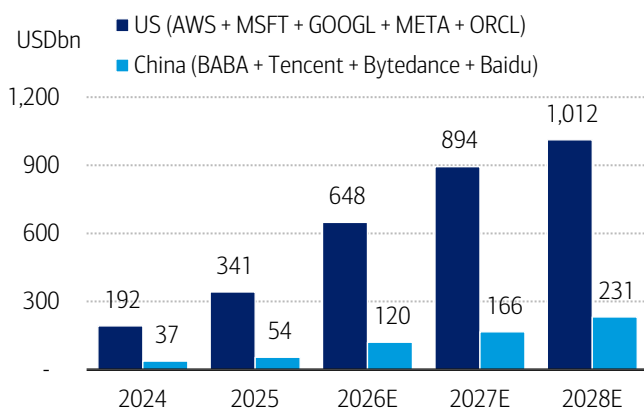
US leads on compute, capital and software ecosystem

China AI players face hardware access constraints and have lower capital intensity, while US hyperscalers and AI labs continue to invest aggressively in large-scale infrastructure. Given that model capability still scales meaningfully with compute, this divergence directly translates into likely a persistent performance gap.

We expect US frontier models to continue to lead globally on intelligence in the medium term, reflecting both higher cumulative training compute and faster iteration cycles. US frontier AI labs are leveraging their compute capacity advantage to maximize lead on model intelligence, which should support the token pricing premium (e.g. ARR) and profitability (e.g. inference margin) of US frontier models over Chinese models in the medium term.

Exhibit 65: We expect CAPEX gap between US hyperscalers and China AI players to widen

US vs. China top hyperscalers' CAPEX trend

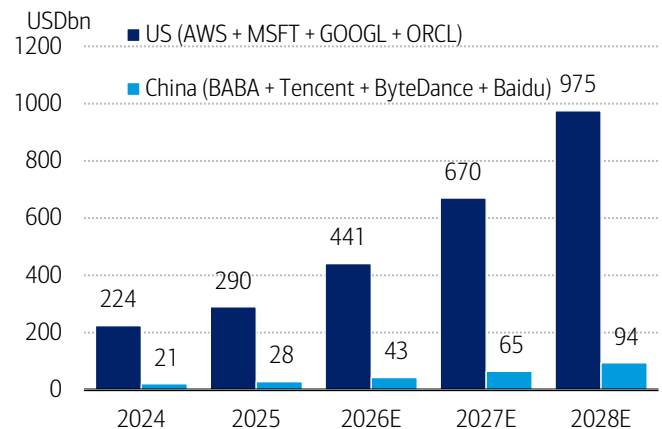


Source: Companies, BofA Global Research

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Exhibit 66: We expect US cloud monetization to outpace China

US vs. China top hyperscalers' cloud (IaaS/PaaS/MaaS) revenue trend



Source: Companies, BofA Global Research

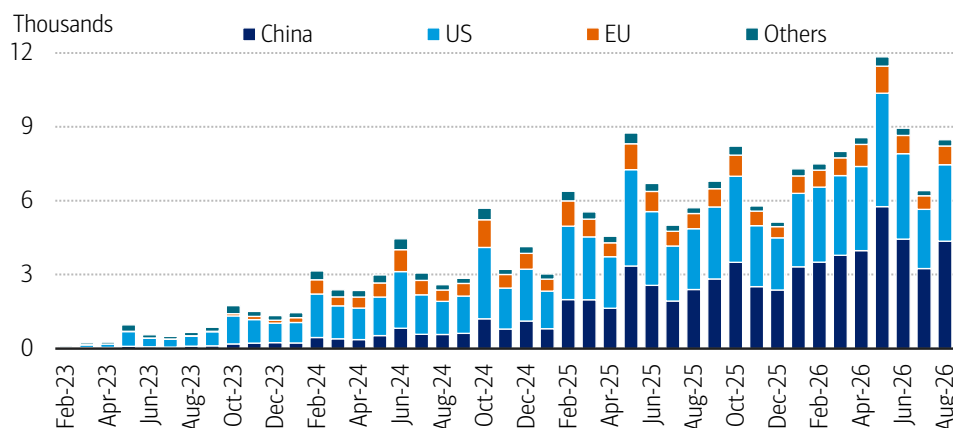
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China leads on infra, use cases and talents

On talent, China's strongest AI advantage may be talent scale and execution capability. While the US retains leadership in frontier research, China benefits from a deep engineering talent pool, strong STEM education, and an ability to rapidly commercialize and diffuse new technologies across companies and industries.

Exhibit 67: China has surpassed US in open-weight LLM-related paper publications since Nov 2025

Number of papers mentioning open-weight LLMs by model-provider region



Source: Interconnects Adoption Dashboard

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On use cases, China offers one of the world's richest environments for AI deployment, spanning e-commerce, payments, logistics, manufacturing, mobility, and public services. The breadth of real-world applications allows companies to iterate quickly, generate large volumes of usage, and build powerful feedback loops from actual customer behavior.

On infrastructure and ecosystem, although the US leads in frontier compute, China has advantages in deployment infrastructure, manufacturing, and hardware-software integration. Lower operating costs, dense supply chains, and leadership in industrial systems, robotics, and consumer hardware position China well to scale efficient AI inference and physical-world AI applications.

Exhibit 68: There have been many AI-supported use cases for China Internet companies

Key AI use cases for China Internet companies

Vertical	Key AI Use Cases	Leading Companies	Benefits / Impact
Advertising	Ad creation, targeting, recommendation, campaign optimization, performance analytics, AI agents, digital humans	Tencent, Kuaishou, Bilibili	Higher CTR, conversion rates, advertiser ROI, and advertising revenue growth
Gaming & Content	AI-powered NPCs, game balancing, content generation, creator tools, recommendation engines, video generation	Tencent, Kuaishou, Bilibili	Better user engagement, lower development costs, higher creator productivity, new monetization opportunities
E-commerce	Product search, store operations automation, shopping assistants, logistics optimization, cross-border commerce tools, AI livestreamers	Alibaba, JD.com, Kuaishou	Improved conversion rates, operational efficiency, fulfillment speed, and customer experience
Enterprise SaaS	Foundation models, AI copilots, agent frameworks, meeting transcription, content summarization, workflow automation	Alibaba (Qwen), Tencent (Hunyuan), Baidu (ERNIE)	Higher enterprise productivity, user retention, and AI-enabled software monetization
Vertical Services	Virtual consultations, medical AI assistants, property valuation, customer service, travel planning, route optimization, merchant tools	JD Health, Beike, Trip.com, Meituan	Better personalization, service quality, operational efficiency, and customer satisfaction

Source: Companies, BofA Global Research

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Smarter spend & AI segmentation: a cheap-and-capable market emerging

AI became popular fast – now it's becoming too expensive for some

As the highest value use case for AI thus far, AI coding tools are seeing rapid enterprise adoption, but early usage is exposing a key constraint: cost scalability. According to media reports (Verge/Fortune/The Information/Bloomberg) in May 2026, Microsoft has begun moving engineers away from direct Claude Code licenses toward GitHub Copilot CLI, only months after encouraging broad internal experimentation. Similarly, Uber reportedly exhausted its 2026 AI coding tools budget within four months, after actively promoting usage through internal adoption leaderboards. In June, Walmart seems to have imposed usage limits on its internal coding assistant Code Puppy after employees

flooded it with redundant requests. Amazon and Meta were also reported to roll back token-based adoption initiatives amid concerns about inflated token usage and rising AI cost governance.

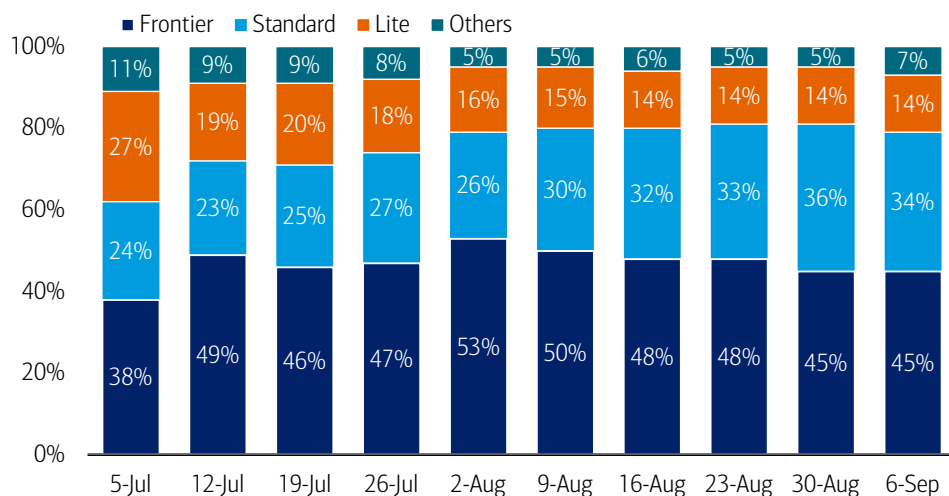
For investors, the takeaway is not that AI demand is weakening, but that monetization and ROI remain uneven. Heavy usage can drive meaningful productivity gains, yet compute and licensing costs may rise faster than expected, especially at enterprise scale. Nvidia's Bryan Catanzaro highlighted the challenge clearly, noting that for his team, compute costs exceed employee costs, in an interview in April 2026.

AI segmentation is clearly underway

Per RampAI who track US enterprise software/AI usage based on real data processed on its platform, the recent surge in enterprise AI usage has been driven primarily by low-cost, high-performance models such as GPT-5.6 Terra and Anthropic's Sonnet series, while adoption of frontier models has become more selective. Although leading-edge models like Opus, Fable, and Sol still account for a meaningful share of token consumption, many enterprises are increasingly enforcing company-wide defaults that steer users toward standard models. The rationale is simple: for most workflows, standard models deliver performance that is "good enough" at a materially lower cost, creating a natural bifurcation between premium models reserved for high-value use cases and cheaper models that handle the bulk of enterprise demand.

Exhibit 69: Token usage share of standard models improved since July

Share of token usage on Ramp AI by model tier



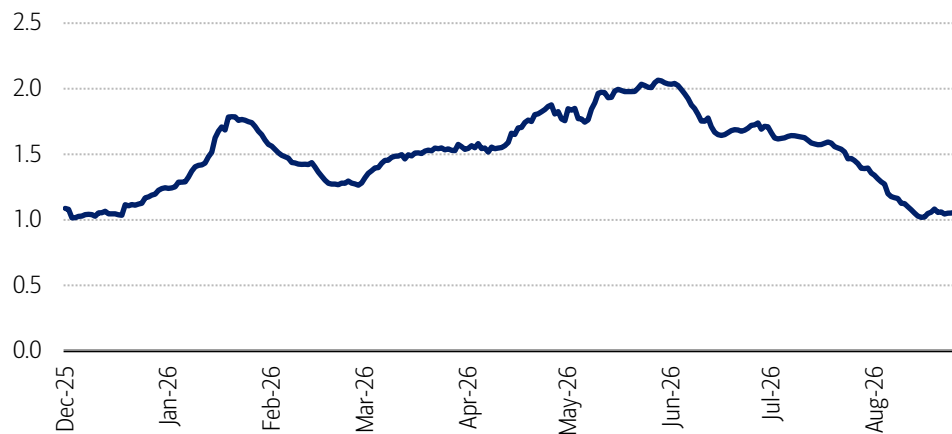
Source: Ramp AI Index; frontier includes expensive models (e.g. Opus and Sol); Standard includes mid-tier models (e.g. Terra and Sonnet)
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The early adoption data for Anthropic's Fable 5 reinforces this trend. Despite being widely regarded as the most capable model currently available, Fable 5 represented only 6% of Anthropic token volume and 11.4% of spending one month after launch, lagging even OpenAI's premium GPT-5.6 Sol in overall enterprise usage, according to RampAI. This suggests that enterprises are becoming increasingly price-sensitive and are placing practical limits on how much incremental performance is worth paying for.

In effect, Fable 5 may be revealing the current ceiling for enterprise AI willingness-to-pay: **beyond a certain point, model quality improvements alone are insufficient to drive broad adoption** unless they deliver clearly differentiated capabilities that competitors, including rapidly improving open-source models, cannot replicate. As a result, the AI market appears to be evolving toward a segmented structure, with premium frontier models serving specialized, high-value workloads while standard models capture the majority of volume.

Exhibit 70: LLM token expenditure index declined to c.USD1.07 in August

Silicon Data LLM Token Expenditure Index



Source: Bloomberg, Silicon Data LLM Token Expenditure Index, as of August 26, 2026.

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Silicon Data's LLM Token Expenditure Index provides a useful benchmark for tracking token costs. The index, created by Silicon Data—a data aggregator specializing in GPU and model usage—shows the average price paid per million LLM tokens, weighted by expenditure and independent of model type. This metric reflects the overall costliness of tokens.

Since June, the index has undergone noticeable correction. Silicon Data's commentary suggests this may signal a slowing migration toward premium closed models. In our view, this trend can be better explained by increasing cost sensitivity among end-users, leading to more balanced adoption across model tiers.

Chinese AI models likely dominate the “cheap and capable” segment globally

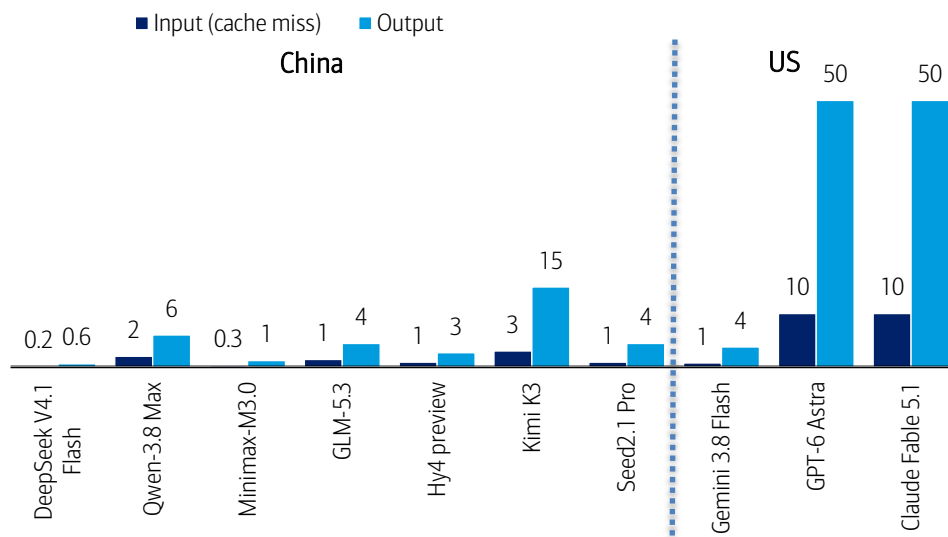
China's AI ecosystem is narrowing the capability gap with the US despite continued compute constraints. Recent model progress suggests Chinese developers are improving efficiency through architecture optimization, distillation, and lower-cost inference.

More importantly for investors, Chinese models are increasingly positioned at the “cheap and capable” frontier. We believe China is positioned to capture a substantial share of the global AI market within the “value-for-money” segment, particularly in cost-sensitive, non-mission-critical applications. Chinese AI models are witnessing international growth, both from established companies and independent laboratories, driven by competitive pricing and enhanced performance—especially in regions prioritizing affordability.

China AI models are priced at a significant discount to US models. GPT-6 Astra and Claude Fable 5.1 have the highest token prices among the models in our comparison, both at USD10/mn input tokens and USD50/mn output tokens. Within Chinese frontier models, Kimi K3 leads with input and output token prices of USD3/mn and USD15/mn, respectively, followed by Qwen-3.8 Max at USD2/mn and USD6/mn, significantly cheaper than US models. DeepSeek V4.1 Flash, released on Sep 10, is priced the cheapest at input and output token prices of USD0.2/mn and USD0.6/mn.

Exhibit 71: China frontier models have a relatively lower token pricing compared to US peers

Key models' input and output prices comparison (in USD/mn tokens)



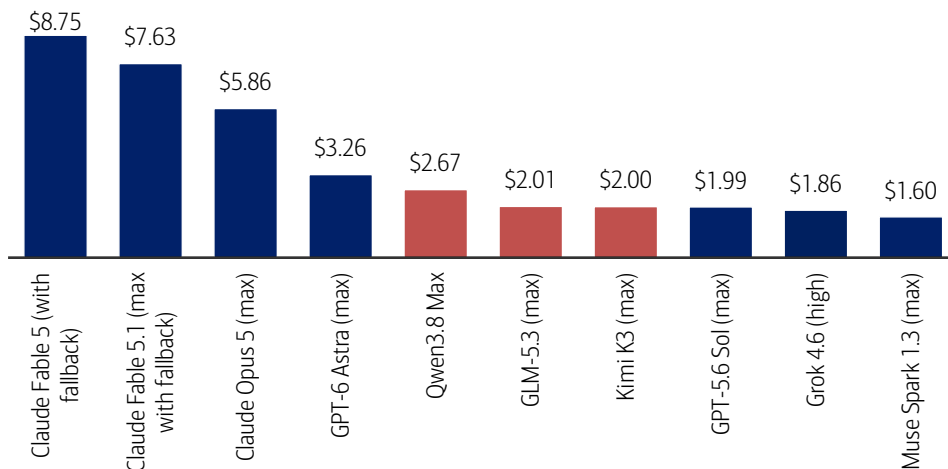
Source: DeepSeek, Alibaba Cloud, Minimax, Z.ai, Tencent, Kimi, Volcano Engine, OpenAI, Claude platform

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China AI models also have attractive cost-per-task vs. US models. According to Artificial Analysis as of Sep 11, Claude Fable 5 (w/fallback) has the highest cost per task among leading frontier AI models at USD8.75 per Intelligence Index task, followed by Claude Fable 5.1 (max w/fallback) at USD7.63 and Claude Opus 5 (max) at USD5.86. Qwen 3.8 Max is ranked 5th at USD2.67, GLM-5.3 Max is ranked 6th at USD2.01, and Kimi K3 is ranked 7th at USD2.00.

Exhibit 72: Claude Fable 5 (w/fallback) has the highest cost per task among leading AI models

Weighted average Dollar Cost per Intelligence Index task, by model



Source: Artificial Analysis as of Sep 11, 2026

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Artificial Analysis' Cost per Intelligence Index Task measures the effective cost of delivering a unit of AI capability, adjusting for model quality as captured by the Artificial Analysis Intelligence Index. The metric estimates the weighted average cost (USD) of completing a standardized AI task by incorporating all relevant inference costs, including input, cache, reasoning, and output tokens, and weighting each evaluation according to its contribution to the Intelligence Index. Lower values indicate greater cost efficiency, meaning a model can achieve a given level of intelligence at a lower inference cost.

Though at early stage, agentic adoption further drives China AI model usage in developed markets

Agent proliferation is driving a structural shift in AI demand toward high-frequency, multi-step workflows, where token consumption scales exponentially. In this environment, cost—not frontier capability—becomes the binding constraint, as many enterprise use cases prioritize throughput and efficiency over maximum performance.

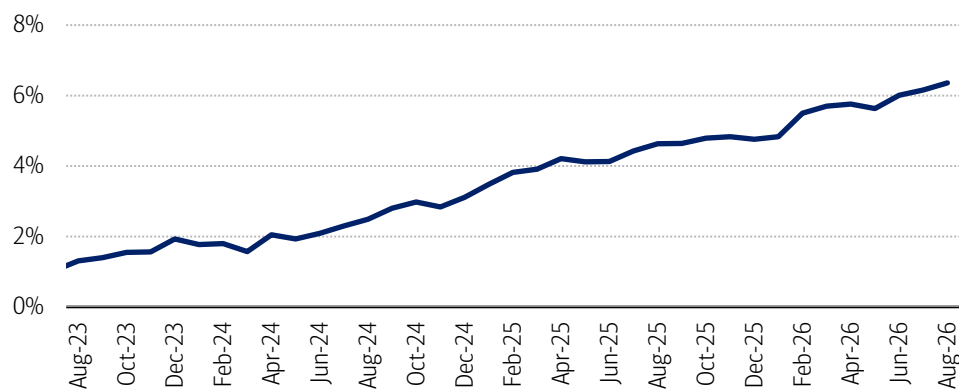
Chinese open-weight models are clear beneficiaries. With near-frontier performance at a fraction of the cost, they are increasingly deployed as the default compute layer for routine, large-scale tasks, particularly in coding and automation-heavy workflows. Critically, as noted by OpenRouter COO Chris Clark, this demand is “disproportionately heavy in agentic flows run by US firms,” underscoring that adoption is global and cost-driven rather than China-specific.

NBC News reported in late-2025 that more American AI startups are using open Chinese AI models, which increasingly compete with—and sometimes replace—costly U.S. systems. Interviews with industry insiders reveal that while U.S. models remain leaders in advanced capabilities, Chinese models have become cheaper, more customizable, and capable for many applications.

More recently in June 2026, according to SCMP, DeepSeek topped a major US business spending growth index, as companies opted for cost-effective alternatives to firms like OpenAI and Anthropic. RampAI, a New York-based corporate spending platform, reported that DeepSeek surpassed PheedLoop and Fireworks AI in first-time business purchases. US businesses made direct payments to DeepSeek, indicating data transfer through its China-hosted servers rather than internal hosting. This trend reflects a growing willingness among companies to send US data to Chinese providers for more affordable AI solutions.

Exhibit 73: 6.4% of AI spending businesses used model serving or inference platforms in August, up from 4.8% in Jan 2026

Share of businesses with any AI spend that used model serving and inference vendors



Source: Ramp AI Index, business spend data from Ramp

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Token export: more complicated than just low-power cost advantage

We believe China has visible advantages to becoming a competitive exporter of AI tokens, supported by lower data center construction costs, potentially cheaper accelerator procurement, greater access to low-cost power in selected regions and increasingly efficient domestic model architectures. However, these advantages should not be interpreted as evidence that token production in China is already structurally cheaper than in the US. The central issue is whether lower infrastructure costs can offset the weaker effective throughput of the domestic hardware and software stack.

Exhibit 74: China holds advantage on electricity price/IDC construction/Model architecture, but lags US on accelerator performance

Comparison of token cost production advantage/disadvantage between US and China

Factor	China vs US	Assessment	Token export implication
Electricity	Advantage	Lower-cost power is available in selected Chinese regions, but not uniformly nationwide	Supportive, but not the main cost driver
IDC construction	Advantage	Lower construction, equipment and labor costs	Reduces fixed infrastructure cost
Accelerator cost	Potential advantage	Domestic cards may be cheaper and offer better supply security	Can partly offset weaker performance
Accelerator performance	Disadvantage	US platforms retain an advantage in compute, memory, interconnect and software	More cards may be required for the same token output
Model architecture	Potential advantage	Chinese developers have emphasized efficient MoE models and hardware co-design	May compensate for weaker domestic silicon
Model size	Mixed	Lower active parameters reduce compute, while large total parameters raise deployment requirements	Net impact depends on actual throughput
Software ecosystem	Disadvantage	US inference frameworks and developer tools remain more mature	Limits conversion of theoretical performance into production throughput
Overseas delivery	Disadvantage	Latency, connectivity and data-residency requirements may restrict use cases	Favors batch and asynchronous workloads

Source: BofA Global Research

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Model architecture may offset part of the hardware disadvantage

Chinese model developers have pursued architectures designed to reduce the amount of computation required for each token. In principle, an MoE (Mixer of Expert) model can achieve high overall model capacity while activating only a subset of its parameters for each inference step. This creates an important potential advantage for China. A sufficiently efficient model can produce competitive tokens even when the underlying accelerator is less powerful.

However, MoE itself is not a uniquely Chinese advantage. American providers can also use MoE models on leading US accelerators. A fair comparison should therefore assess a Chinese MoE model on domestic accelerators against an American MoE model on American accelerators, rather than comparing Chinese MoE with an older US dense architecture.

GPU throughput is the decisive variable

The most important input in the analysis is effective production throughput. China can have lower electricity, facility and hardware costs but still produce more expensive tokens if each server generates substantially fewer commercially usable tokens.

Token export is likely to be workload-specific

China's token export proposition appears strongest in workloads where customers prioritize cost over immediate response time. These may include batch processing, coding workloads, document analysis, translation, data transformation and selected agentic applications.

The opportunity is less compelling for latency-sensitive consumer applications or regulated enterprise workloads that require local data residency. In these cases, network distance, compliance requirements and customer risk perception may outweigh a modest production cost advantage.

A hybrid delivery model may therefore be more commercially realistic. Core model inference could be concentrated in lower-cost Chinese facilities, while customer gateways, caching, safety systems and selected inference capacity are deployed closer to overseas users.

To conclude, China token export is not primarily a power-cost arbitrage. It is a model-hardware co-optimization thesis in which lower infrastructure costs and more efficient model architecture provide a buffer against weaker accelerator performance. The investment case turns on whether the domestic stack can narrow its effective throughput gap while maintaining comparable model quality and service standards.

AI monetization: enterprise-driven, cloud-centric, overseas ramping

How do Chinese internet/AI companies monetize AI?

We see three ways for China internet companies on monetizing AI, namely cloud (including both AI infrastructure and MaaS), AI applications and revenue/cost efficiency improvement. AI cloud remains the most immediate monetization opportunity within the sector.

AI cloud - AI infrastructure & MaaS

Key company: Alibaba, Bytedance, Tencent, Baidu, Zhipu, MiniMax

AI infrastructure—including GPU compute, training and inference systems, and AI Platform-as-a-Service—is currently the most rapidly expanding large-scale revenue segment, primarily due to generative AI adoption. MaaS is still at an early stage but structurally critical — it converts compute + models into API-based revenue. With sector token usage surging YTD 2026, this segment is seen as the fastest growing subsegment within AI cloud stack.

Overall we expect China AI infra cloud market to grow from RMB45bn in 2025 to RMB470bn in 2030E. We expect China AI MaaS market to grow from RMB21bn in 2025 to RMB676bn in 2030E. Within the MaaS market, we believe incumbents platforms (Alibaba, Bytedance, Tencent) to capture much of the incremental growth.

AI application

Key companies: Tencent, Bytedance, Alibaba, Kuaishou, MiniMax

Tencent, a leading Chinese internet company that leverages AI across its ecosystem in content, game, ad, cloud and social. Kuaishou, a leading content community and social platform in China, launched self-developed Kling, a top-tier AI video-generation large model, which has accumulated 45mn+ content creators and 20k+ business users.

Revenue/cost-efficiency gain for existing business model

For many Chinese internet companies, in addition to AI-native products which many take multiple years to generate material revenue, AI technology is already benefiting many existing business through better user engagement (more targeted content recommendation engine) and better advertising customer conversion & adoption (through higher click-through rate and easier advertising material generation). AI is also driving productivity gains within companies, such as shortening product development cycle with AI coding, that can support leaner headcount growth and streamline support functions, therefore driving long-term operating expense leverage.

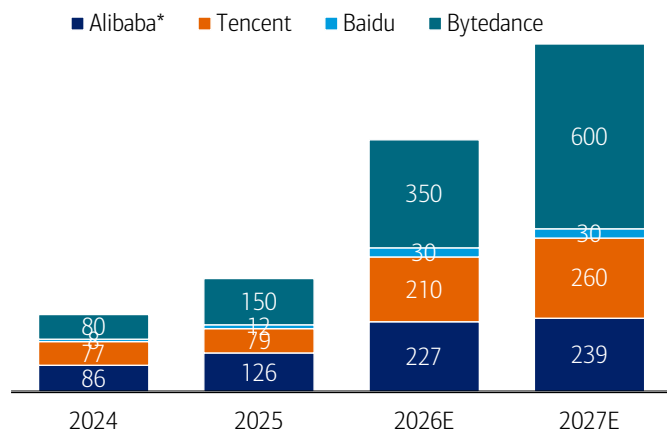
Incumbents: full-stack offering and strong compute on hand provides flexibility

Within China, the CAPEX gap among domestic peers is widening, as leading internet platforms ramp up investment in AI infrastructure since 2024 – top 4 China CAPEX spenders combined CAPEX came in at around RMB400bn in 2025, significantly higher than Zhipu & MiniMax combined. With stronger balance sheets and much bigger cash flow from existing businesses, incumbents are gaining relative compute advantage in

both model training and inference deployment, enabling them to scale AI services more effectively and explore new AI initiatives without worrying about near-term ROI. Comparing to independent AI labs, incumbents are likely to bear with lower model layer profitability given more cross-selling opportunities with existing cloud service.

Exhibit 75: Capex of major China internet companies is estimated to rise 123% YoY in 2026E

Annual CAPEX trend from 2024 to 2027E (RMBbn)



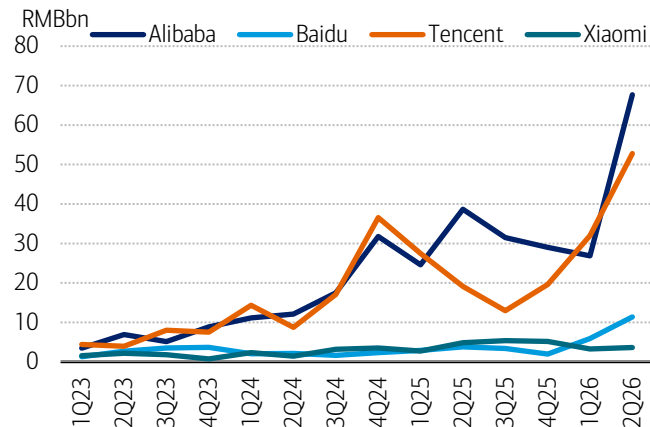
Source: company reports, SCMP, Reuters, Financial Times, BofA Global Research

*Alibaba 2024/2025E/2026E/2027E used FY2025/FY2026E/FY2027E/FY2028E

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Exhibit 76: China Internet companies have increased CAPEX significantly over the past two years

Quarterly CAPEX trend from 1Q23 to 1Q26 (RMBbn)



Source: company reports, BofA Global Research

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Alibaba: a clear example of ecosystem-level AI monetization

Full-stack capabilities broaden value capture, while Cloud leads near-term monetization and diversified revenue supports the long-term opportunity

Alibaba offers one of China's clearest examples of ecosystem-level AI monetization. Its competitive capabilities across chips, infrastructure, Cloud, models and applications create multiple avenues for value capture, with Cloud leading near-term monetization and a broader mix of hardware, model services, applications and overseas revenue supporting long-term opportunity. This full-stack position also gives Alibaba flexibility as AI economics and profit pools shift across the value chain. Combined with early investment and return discipline, we believe it strengthens Alibaba's ability to sustain investment through AI cycles and capture a durable share of long-term value creation.

Alibaba's full stack broadens ecosystem-level value capture

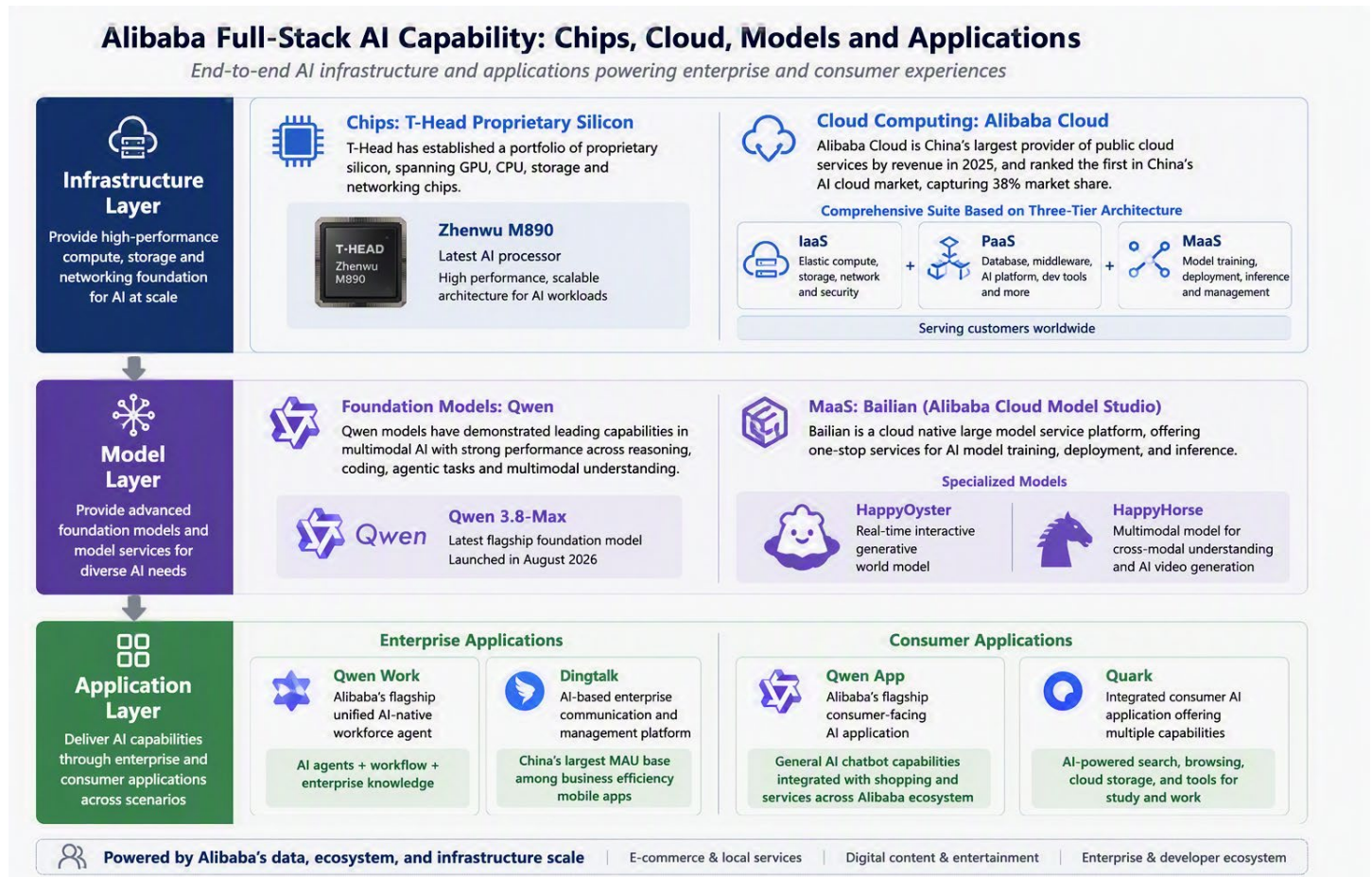
Alibaba's full-stack advantage reflects both breadth and depth. Competitive capabilities across the stack provide multiple routes to AI monetization, reduce reliance on the economics of any single model or product, support greater technological autonomy, and provide strategic flexibility as the industry evolves.

Multiple layers create multiple monetization pathways

Each layer of Alibaba's full stack offers a distinct route to value capture. Chips can support infrastructure supply and unit economics before direct sales become material; Cloud monetizes compute and managed infrastructure; Qwen drives adoption, Cloud workloads and applications, with potential licensing or revenue-sharing upside; and MaaS monetizes API access, customization, deployment and inference. Enterprise applications can monetize productivity and workflow integration, while consumer applications offer potential subscription, transaction, advertising and commerce revenue. Cloud is currently the largest and most visible revenue pool, but Alibaba has established monetization pathways across most of the AI value chain, creating a more economically complete offering than a standalone model business.

Exhibit 77: Leading capabilities across key layers broaden Alibaba's monetization opportunity and strategic flexibility

Alibaba's full-stack AI capabilities spanning chips, cloud, models and applications



Source: Company reports, BofA Global Research

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A measured path toward greater autonomy

Alibaba's proprietary chips, Cloud stack, models, applications, data assets and customer relationships provide a pathway toward greater technological autonomy. But, advanced-chip manufacturing and memory supply look constrained by external production capacity, meaning that localization is not yet complete. Hence, we see Alibaba as one of China's most advanced examples of progressive AI localization, rather than a fully self-sufficient stack. More immediately, this breadth allows Alibaba to control and monetize a major portion of the AI service chain while gradually reducing external dependencies.

Strategic flexibility as industry economics evolve

The final allocation of AI industry profits remains uncertain. Value could concentrate in scarce compute, managed infrastructure, model services, enterprise agents, consumer distribution or transaction-based applications. Alibaba's presence across these layers gives the company greater flexibility to respond as technology, pricing and customer behavior evolve. If value remains concentrated in infrastructure, Alibaba has Cloud scale and proprietary silicon. If value shifts toward model services, Alibaba has Qwen model family and specialized AI models. If enterprise or consumer interfaces become the primary value layer, Alibaba has direct access to business users, consumers, merchants and proprietary application scenarios.

AI monetization is increasingly visible, led by Alibaba Cloud

AI demand and initial monetization are increasingly visible in China, but revenue remains concentrated in infrastructure. Alibaba Cloud provides the clearest current evidence of value capture, while T-Head monetization is occurring primarily through Cloud services rather than direct external chip sales.

T-Head: monetizing through Cloud before external chip sales

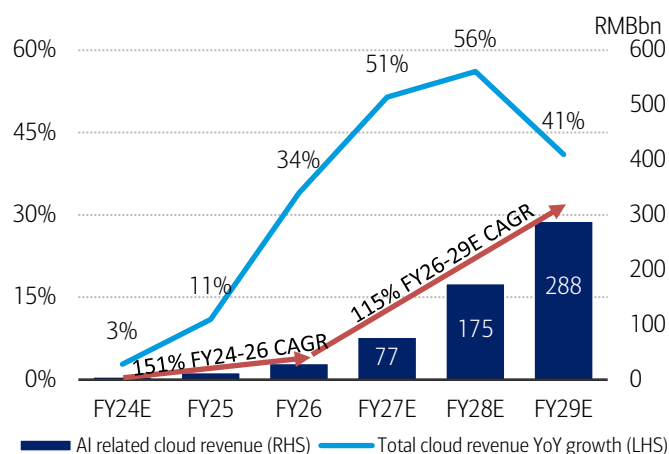
T-Head's direct external revenue remained immaterial as of June 2026 because limited supply was allocated primarily to Alibaba Cloud. Proprietary processors represented less than 10% of Alibaba's accelerator units, while more than 60% of T-Head-supported capacity served external Cloud customers across more than 20 industries and hundreds of customers. This means T-Head is already contributing economically through recurring Cloud services. Direct sales should emerge as domestic manufacturing capacity expands, but we remain cautious on the initial ramp because Alibaba Cloud is likely to retain priority access to scarce supply. Over time, increasing T-Head's share of accelerator units toward the 40-50% target could support capacity, unit economics and adjusted EBITA margins.

Alibaba Cloud: the clearest current AI monetization channel

Alibaba's recent Cloud performance provides the clearest evidence that AI demand is translating into revenue and profit. **Cloud revenue acceleration and margin expansion** in 1QFY27 were mostly volume-driven, supported by strong enterprise demand and constrained compute availability. Demand is broadening beyond internet companies and foundation-model developers into sectors including manufacturing, embodied AI, intelligent vehicles, quantitative investing, financial services and pharmaceutical research. **The economics of public Cloud are also attractive to customers in the current phase:** owned AI infrastructure is expensive and technically complex, while future utilization remains uncertain. Hence, renting capacity therefore gives enterprises and AI labs access to scarce compute without committing to a full proprietary build-out.

Exhibit 78: We expect accelerating AI-related cloud revenue to drive Alibaba cloud growth in FY27E-29E

Alibaba's AI-related cloud revenue and total cloud revenue YoY growth

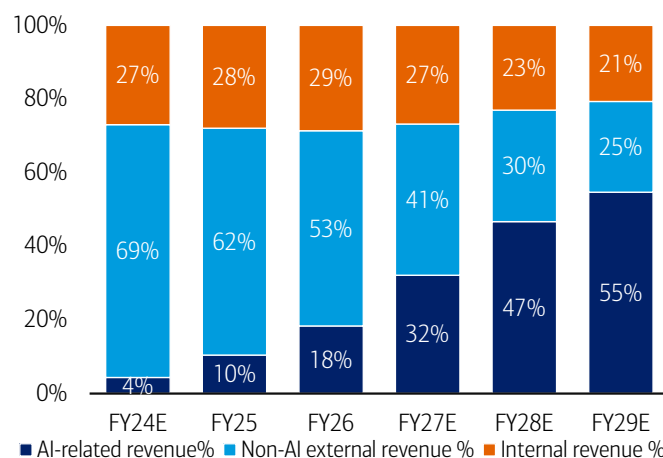


Source: Company reports, BofA Global Research

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Exhibit 79: We expect AI-related cloud revenue to account for 55% of Alibaba cloud revenue by FY29E

Alibaba cloud revenue breakdown by AI-related, non-AI external and internal



Source: Company reports, BofA Global Research

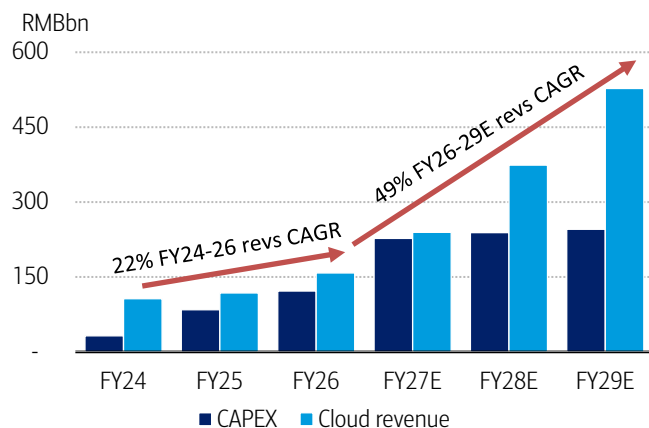
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Near-term demand is supported by aggressive model investment from major internet platforms and independent AI labs, together with early enterprise adoption of smaller, vertical-specific models. This creates a relatively favorable supply-demand backdrop for Alibaba Cloud. However, the quality of demand matters. Sustainable monetization requires workloads to move from experimentation into recurring production use, where

customers consume compute, storage, databases, networking, security and managed services over a longer period.

Exhibit 80: We expect Alibaba cloud revs to grow at 49% CAGR in FY26-29E, supported by CAPEX investment

Alibaba's CAPEX and cloud revenue trend



Source: Company reports, BofA Global Research

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Volume is driving the current phase, but monetization can deepen through pricing, utilization and workload mix. Management expects pricing to contribute more as annual contracts renew from 2QFY27. Inference is growing faster and increasing as a share of usage, while training demand remains meaningful as enterprises build and refine vertical models. Higher proprietary-chip penetration could provide another lever by expanding available capacity and potentially improving the cost profile of Alibaba Cloud. Direct chip sales would be a separate revenue stream and should not be conflated with recurring Cloud services.

What remains unproven

The debate has shifted to durability and returns. The key questions are whether workload growth can support Alibaba's investment through the utilization cycle, whether revenue and margins can continue to expand together, and whether China's cost-sensitive enterprises can sustain public AI Cloud demand without a large existing SaaS budget pool. Long-term monetization ultimately requires recurring production workloads and measurable productivity gains, rather than platform subsidies or continued AI-lab financing.

The Cloud return framework is defined; long-term monetization depends on revenue diversification

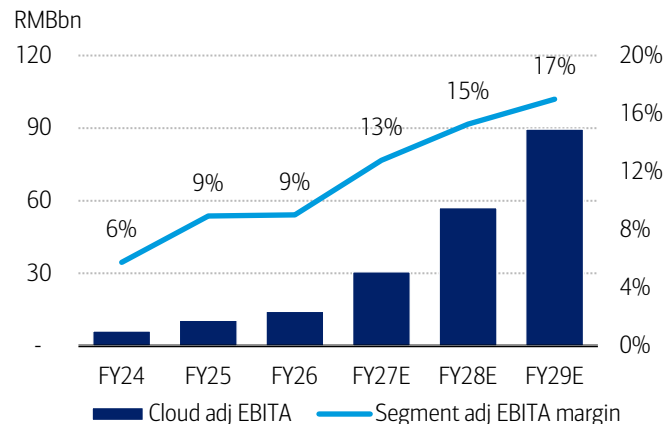
China's near-term AI infrastructure shortage supports continued Cloud investment, and management has established measurable return targets. Over the long term, however, Alibaba's external Cloud and AI ambitions require monetization to broaden beyond domestic compute capacity into chips, MaaS, models, applications and overseas markets.

Near-term Cloud returns: capacity investment is needed to capture a supply-constrained opportunity

Given the favorable near-term supply-demand backdrop discussed above, Alibaba Cloud remains capacity-constrained, making continued investment necessary to capture available demand. Management's targets of an approximately three-year cash-payback, 15-20% accounting ROIC after depreciation, and improving post-CAPEX group FCF, support our FY27-29E annual CAPEX forecasts of RMB227-246bn. The investment debate now centers on execution and capital discipline: whether Alibaba can deploy capacity on schedule, ramp utilization, improve pricing and margins, and adapt investment as market conditions evolve.

Exhibit 81: We expect Alibaba cloud margin to expand to 17% by FY29E

Alibaba cloud segment adj EBITA and margin trend



Source: Company reports, BofA Global Research

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Long-term monetization requires a broader AI revenue mix

Long-term infrastructure economics are less predictable than the near-term demand outlook. Supply, pricing, model efficiency and customer behavior can change quickly. Sustainable returns ultimately depend on AI creating measurable productivity and supporting a healthy application ecosystem. Domestic Cloud infrastructure can remain the largest revenue pool, but achieving Alibaba's FY31 ambition of USD100bn in external Cloud and AI revenue will likely require meaningful contributions from additional layers of the stack.

Exhibit 82: Alibaba's AI monetization pathways across the stack

Alibaba's AI revenue pools, current maturity, long-term drivers and key uncertainties

Revenue pool	Current stage	Long-term driver	Key uncertainty
Cloud infrastructure	Most developed	Volume, pricing and utilization	Duration of supply constraints and margin ceiling
T-Head chips	Early	Higher Cloud penetration and direct sales	Manufacturing capacity and effective throughput
Models and MaaS	Growing	API access, deployment, inference and revenue sharing	Open-model competition and pricing
Applications	Nascent	Productivity, workflows, transactions and advertising	Willingness to pay and product-market fit
Overseas Cloud and AI	Established, developing	Infrastructure, Chinese enterprise globalization and MaaS	Localization, regulation and execution

Source: Company reports, BofA Global Research

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• T-Head: from embedded Cloud economics to direct sales

Given limited domestic supply, Alibaba can currently capture more value from T-Head processors through recurring Cloud services than through direct hardware sales. Higher penetration, from below 10% of accelerator units toward management's 40% to 50% long-term target, could improve capacity availability and Cloud economics, while external sales provide long-term optionality. The key variables are manufacturing scale, effective throughput, software compatibility and utilization, as lower chip costs do not automatically result in lower inference costs.

• Models and MaaS: ecosystem reach, scalable revenue

Qwen's open-weight model limits standalone monetization because customers can self-host or access competing models through third-party platforms. However, its leading capabilities support developer adoption, enterprise experimentation and Alibaba Cloud workloads, while future licensing or revenue-sharing arrangements could provide incremental direct revenue.

MaaS provides a clearer and more scalable revenue model. Alibaba Cloud Model Studio monetizes API access, customization, deployment and inference across Qwen and third-party models, benefiting from both proprietary model improvements and the broader open-model ecosystem. MaaS ARR reached approximately USD2.4bn in August 2026, supported by growing enterprise API and inference demand.

Exhibit 83: Alibaba targets MaaS ARR of RMB30bn by March 2027

Alibaba's MaaS ARR milestones

Month	ARR in RMBbn	ARR in USDbn	Est. % of AI-related cloud
May-26	8	1.2	15%
Jun-26	10	1.5	
Aug-26	16	2.4	22%
By March-27	30	4.4	28%

Source: Company reports, BofA Global Research

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• Applications: adoption is developing ahead of a clear business model

Application monetization remains early, constrained by limited willingness to pay and intense competition. **Enterprise products** must demonstrate measurable productivity and workflow value to support subscriptions, agent services and integration revenue, while **consumer products** may need to move beyond subscriptions toward transactions, advertising, commerce enablement and service distribution. Qwen App's integration with Alibaba's commerce ecosystem provides a differentiated opportunity to convert AI usage into real-world services and transactions.



- **Global expansion broadens the AI revenue base**

Overseas expansion provides additional long-term upside. Alibaba has accelerated international AI infrastructure investment since FY26, with recent financing identifying overseas capacity as a key use of proceeds. Growth should be supported by local infrastructure serving international customers, Cloud services for Chinese enterprises expanding abroad, and cross-border MaaS and model API demand. Revenue-sharing arrangements with overseas Cloud providers could provide further upside for Qwen and other open-weight models. We expect overseas revenue, already above 20% of Alibaba Cloud revenue, to exceed 30% over the medium term.

Case study key takeaways

Alibaba's full-stack capabilities provide multiple routes to capture AI value, with Cloud driving near-term monetization and adjacent layers broadening the long-term opportunity. The investment case has shifted from demand validation to execution, particularly whether capacity investment can translate into sustained margins, cash returns and a more diversified AI revenue base.

Related reports:

[Alibaba Group: HK investor meeting takeaways: turning AI investment into cloud returns 07 September 2026](#)

[Alibaba Group: USD10bn Hong Kong share placement to fund AI spending 24 August 2026](#)

Independent AI labs: fast near-term growth, training first, eager for capital, looking overseas

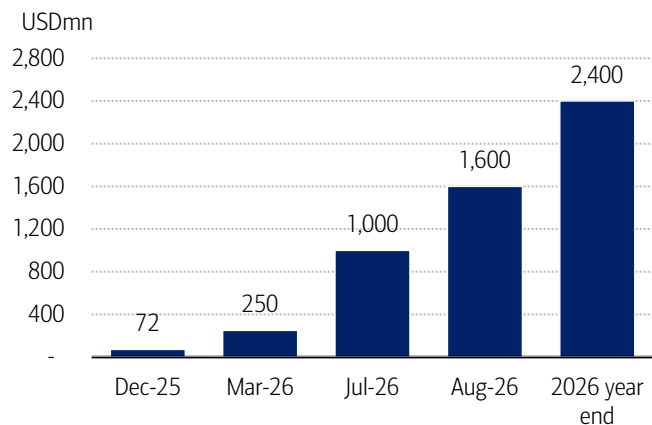
Despite likely faster near-term revenue growth vs. incumbents due to smaller base, we believe that Chinese independent AI labs face greater challenge on computing resources (more on inference), making them more inclined to seek outside funding to ensure long-term business viability. These independent AI labs are likely to focus on model training as a priority over inference needs when facing compute constraint, since staying competitive in the intelligence race is essential. They may depend more on internet/cloud platforms to distribute their models, which introduces the risk of being bypassed in the value chain.

Overseas markets will likely become a greater focus due to high enterprise and user willingness to pay, but on-going geopolitical headwinds may limit the adoption in countries such as US and EU. Zhipu management noted close to 30% of total cloud-based AI model revenue is generated from non-China market since early 2026, and we estimate that MiniMax has 30%+ overseas contribution for its API-based revenue in 1H26. Nevertheless, we expect major independent AI labs in China to heavily rely on China in terms of revenue generation over the next 3 years.

Separately, we believe DeepSeek is intensifying competition in the industry due to its strong AI research capabilities and lesser emphasis on commercial profit. DeepSeek is raising the standard for all Chinese AI models—those unable to keep pace with DeepSeek will find long-term success challenging.

Exhibit 84: We expect Zhipu to achieve ARR of USD2.4bn by end-2026

Zhipu Open platform API's ARR trend

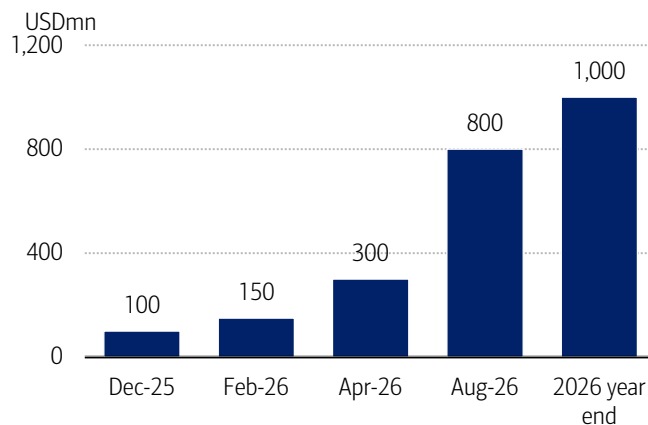


Source: Company report, BofA Global Research

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Exhibit 85: We expect MiniMax to achieve ARR of USD1bn by end-2026

MiniMax ARR trend



Source: Company report, BofA Global Research

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China cloud market forecasts

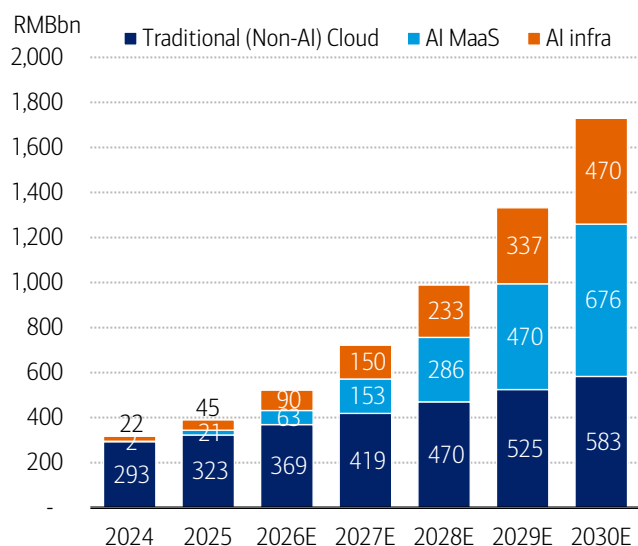
We summarize our key forecasts for China enterprise AI (B2B) monetization in this section. **Cloud, including AI MaaS/AI compute/non-AI cloud, remains the largest and most visible enterprise AI service revenue model** in China, with strong demand of compute power (for training and inference) and of AI model capability ("Model as a service").

We expect **major internet incumbents** (e.g. Alibaba, Tencent, Bytedance) with substantial cloud operations to gain more from the rise in AI-driven enterprise spending. Meanwhile, **independent AI labs** are likely to post faster growth between 2026E and 2030E due to their smaller revenue base and greater presence in international markets.

Overall, we expect China cloud market to grow from RMB389bn in 2025 to RMB1,729bn in 2030E, driven predominately by AI-related cloud business (MaaS and infra).

Exhibit 86: We expect China cloud market to reach RMB16tn by 2030E

China cloud market forecasts, 2024-2030E

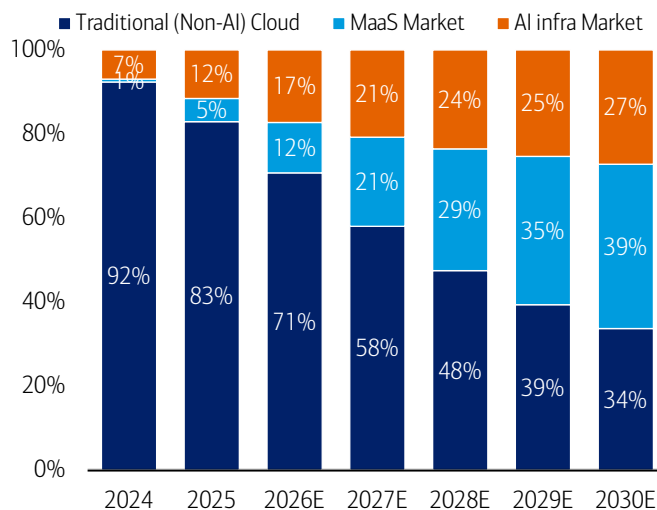


Source: company, IDC, Gartners, Omdia, BofA Global Research

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Exhibit 87: We expect AI-driven cloud (MaaS + infrastructure) to become the primary growth driver

China cloud market forecast breakdown by types, 2024-2030E



Source: Company, IDC, Gartners, Omdia, BofA Global Research

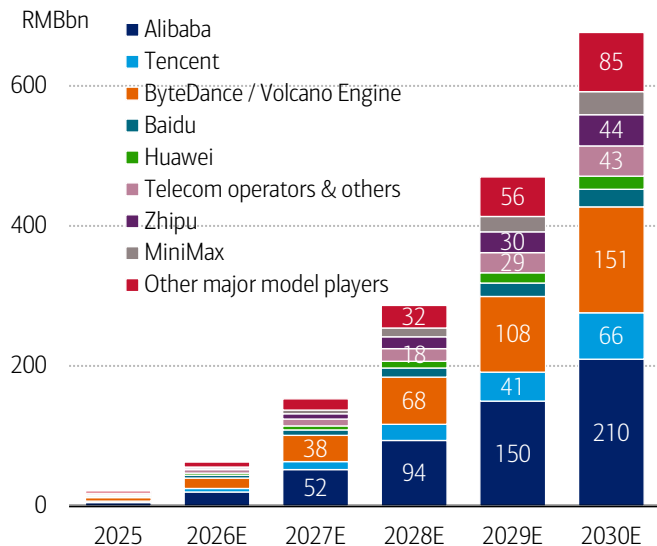
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AI MaaS (AI Model as a service)

Exhibit 88: We expect China AI MaaS market to reach RMB676bn in 2030E

China AI MaaS cloud market forecasts, 2025-2030E

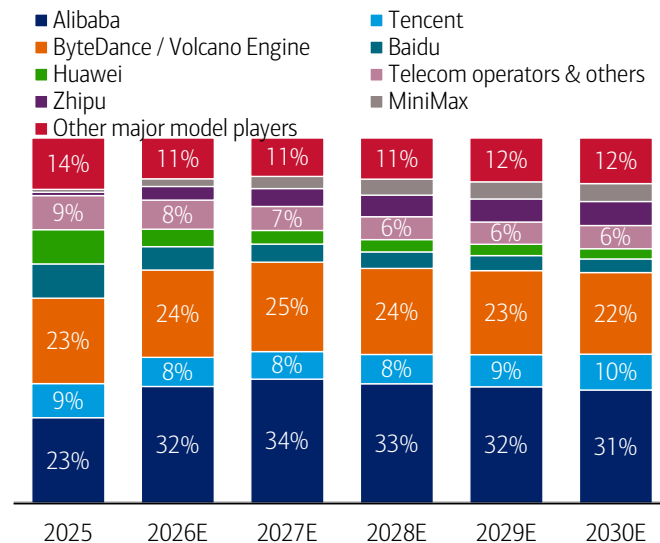


Source: Company, IDC, Gartner, Omdia, BofA Global Research

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Exhibit 89: We expect incumbents to capture much of incremental growth

China AI MaaS cloud market share forecasts by players



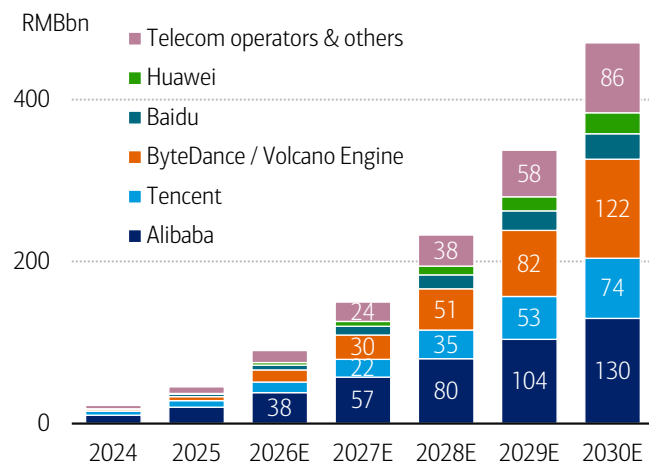
Source: Company, IDC, Gartner, Omdia, BofA Global Research

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AI infrastructure cloud

Exhibit 90: We expect China AI infra cloud market to reach RMB470bn by 2030E

China AI infra cloud market forecasts, 2024-2030E

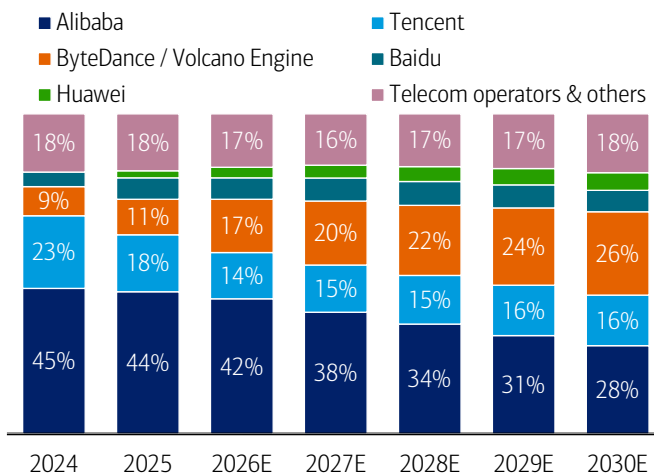


Source: Company, IDC, Gartner, Omdia, BofA Global Research

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Exhibit 91: We expect incumbents to capture much of incremental growth

China AI infra cloud market share forecasts by players



Source: Company, IDC, Gartner, Omdia, BofA Global Research

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Software

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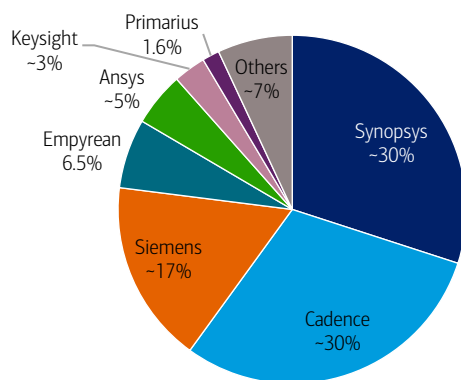
Import substitution - EDA software

Chinese EDA vendors have made significant progress but still lag global leaders in end-to-end flow integration, advanced-node production validation and ecosystem maturity. In terms of market share, Synopsys, Cadence and Siemens accounted for a combined share of nearly 80% of China's EDA market in 2025.

Overall, Chinese EDA vendors are progressing from point-tool substitution toward partial flow-level substitution. Domestic tools are already competitive, and in some cases, differentiated within selected verticals such as device modeling, circuit simulation, and display design. However, broad replacement of global platforms for advanced digital SoC design will require further improvements in tool completeness, PPA (performance, power, and area) / QoR (quality of results), large-scale design capacity, foundry and process-node certification, and accumulated customer validation.

Exhibit 92: Synopsys, Cadence and Siemens accounted for a combined share of nearly 80% of China's EDA market in 2025

China EDA market share (2025)



Source: Companies, ASKCI, BofA Global Research estimates

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Empyrean, the largest domestic vendor, has established relatively broad capabilities in analog/mixed-signal and custom IC design, flat-panel display design, foundry enablement and selected memory workflows. Empyrean's digital EDA product portfolio covers 80% of the major digital IC design tool categories. Specialist domestic vendors have also established competitive positions in selected niches. For example, Primarius has built competitive EDA products in device modeling and verification, as well as circuit simulation and verification. Its device-modeling platform has gained broad adoption among leading global foundries, such as TSMC and Samsung. Meanwhile, its circuit simulation and verification solutions have been adopted by leading semiconductor companies such as Samsung, SK Hynix and Micron.

Synopsys, Cadence and Siemens EDA remain materially ahead at the platform and ecosystem levels. They offer mature, tightly integrated workflows spanning architecture, synthesis, place-and-route, signoff, physical verification, manufacturing/test, advanced packaging and system-level analysis, supported by decades of silicon-proven deployment and validation across leading-edge process nodes. They also have stronger hardware-assisted verification, semiconductor IP and increasingly integrated AI-driven optimization capabilities.

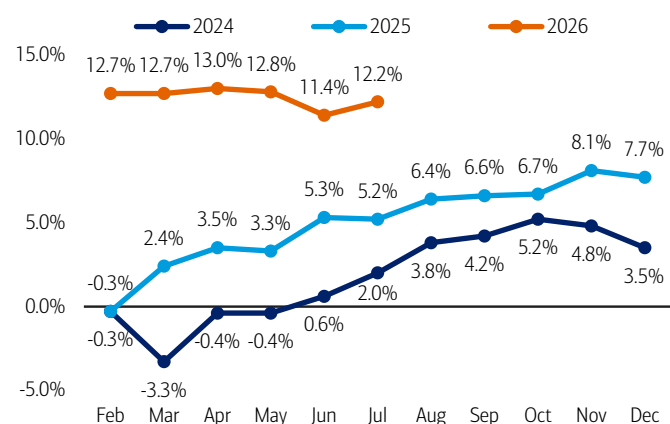
Opportunities for software overseas expansion

AI-driven globalization emerges as a new growth engine for Chinese software companies. Unlike the previous wave led by software outsourcing and internet services, the current cycle is increasingly centered on AI-powered applications and enterprise software. We see accelerating export growth in China software sector in 2026.

Companies such as Meitu and Kingsoft Office have expanded global presence through AI-enhanced content creation and tools, while enterprise software vendors such as Kingdee and Yonyou are gradually extending their ERP and enterprise management solutions into Southeast Asia, Latin America and the Middle East alongside Chinese multinationals. As overseas revenue contribution continues to rise, we believe international markets could evolve from a supplementary revenue source into a structural growth driver for leading Chinese software companies.

Exhibit 93: We see accelerated export growth for China software sector in 2026

China software sector export amount growth (Cumulative YTD)

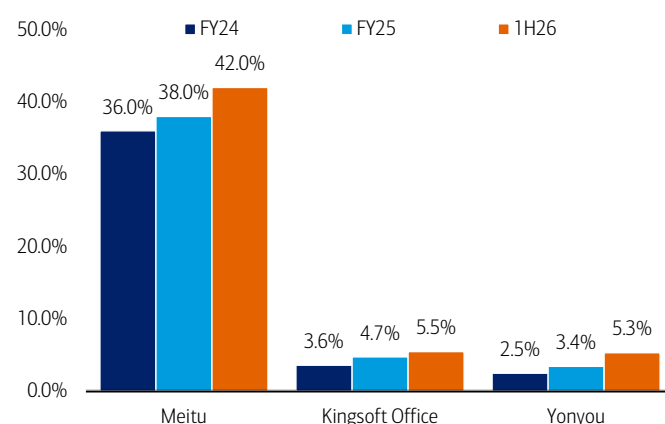


Source: MIIT, BofA Global Research

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Exhibit 94: We see rising overseas revenue contribution for Meitu, Kingsoft Office and Yonyou

Overseas revenue contribution for selected software companies



Source: Companies

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For consumer applications, Chinese software companies benefit from strong AI execution capabilities built in China's competitive digital ecosystem, enabling rapid iteration and commercialization. They also offer attractive cost-performance relative to overseas peers, supported by lower development costs and faster innovation cycles.

For enterprise software, Chinese vendors are increasingly embedding AI into ERP, office productivity, customer service and workflow automation solutions, while leveraging experience from China's large-scale digital transformation to offer cost-effective alternatives to global incumbents. China's position as the world's largest manufacturing hub has also enabled software companies to accumulate deep industry expertise across sectors such as electronics, automotive and industrial equipment, supporting AI applications in supply chain optimization, production planning, inventory management, quality control and industrial automation. We believe this manufacturing know-how positions Chinese enterprise software vendors well for overseas expansion, particularly in manufacturing-intensive economies across Southeast Asia, the Middle East and Latin America.

However, challenges remain substantial for expanding into developed markets, where Chinese software companies generally lag global incumbents like Microsoft, Salesforce, SAP, Oracle, and Adobe in brand recognition and ecosystem strength. Enterprise software vendors also face higher localization requirements than consumer applications, requiring local implementation, consulting, channel and compliance capabilities.

Data center

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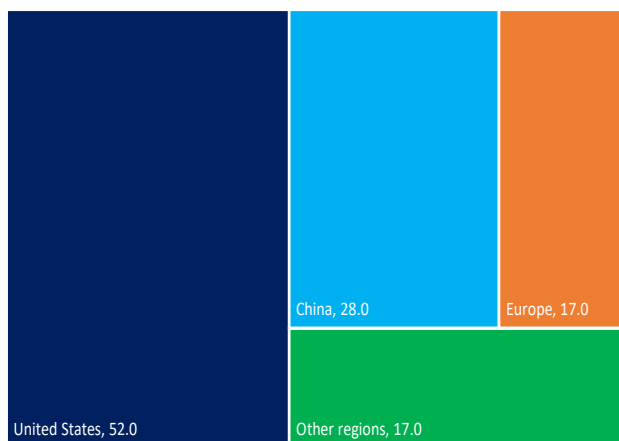
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Data center capacity comparison

According to IEA data, the US remains the dominant global data center market, with 52GW of installed capacity in 2025, accounting for 46% of worldwide capacity, compared with 28GW in China, comprising 25% of the global total. While China's installed capacity is nearly 54% of US levels, data centers account for only 2.3% of China's average power demand, versus 8.9% in the US. This suggests that the US has both a larger installed base and a significantly higher concentration of data center infrastructure within its electricity system, reflecting its leadership in hyperscale cloud and AI computing workloads.

Exhibit 95: US and China accounted for 46% and 25% of global installed data center capacity in 2025, respectively

Installed data center capacity by region (GW)

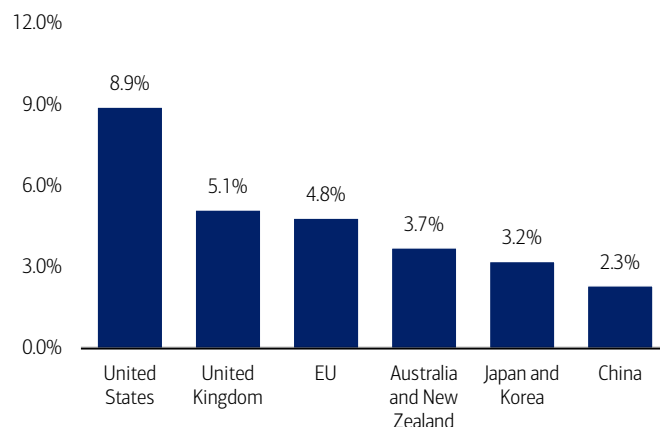


Source: IEA

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Exhibit 96: Data center installed capacity in the US and China was equivalent to c.8.9% and c.2.3% of average power demand in 2025

Data center installed capacity as share of regional average power demand (%)



Source: IEA; Note: Average power demand for a region refers to the average electricity demand in GW across each hour of the year

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Cost advantage in data center construction for China

China can build large-scale data centers at lower infrastructure cost than the US due to cheaper land, labor and power-related construction. While servers and GPUs account for the majority of AI data center CAPEX, supporting infrastructure remains substantially less expensive in China. We estimate that constructing a 1GW AI data center in China based on Huawei's CloudMatrix 384 (Ascend 910C) would require around USD30bn of CAPEX, versus roughly USD50bn for a comparable US facility based on Nvidia's GB200 NVL72 platform, implying a 45% per-GW CAPEX advantage. However, this cost advantage will be narrowed substantially when evaluated on a cost-per-unit-of-effective-compute basis as China's lack of access to advanced AI accelerators compared to overseas.

Land acquisition: China benefits from lower land costs and greater flexibility in deploying large-scale data centers in western provinces such as Inner Mongolia, Ningxia, Guizhou and Gansu, where industrial land is abundant. The "East Data, West Compute" initiative and local government support further reduce upfront development costs. In contrast, most US hyperscale projects are concentrated in regions such as Northern Virginia, Texas, Arizona and Ohio, where suitable sites with access to power become increasingly scarce.



Construction cost: Construction costs are lower in China due to cheaper labor, a competitive construction supply chain and extensive experience in large-scale industrial projects, resulting in lower costs for data center shells and supporting facilities.

Exhibit 97: Estimated CAPEX for a 1GW Huawei CloudMatrix 384 data center is ~45% lower than a comparable US Nvidia GB200 NVL72 facility

Construction cost comparison for 1GW data center in China and US

Construction cost comparison	China	US
	Huawei CloudMatrix 384 (Ascend 910C)	Nvidia GB200 NVL72
Computing system		
Operating power per system (kW)	559	130
Number of systems needed for 1GW data center	1,789	7,692
Price per system (USDmn)	8.0	3.1
Cost breakdown (USDbn)		
AI compute systems	14.3	23.8
Networking, storage & cluster infrastructure	7.7	12.8
Data center infrastructure (land, building, power & cooling)	7.3	15.7
Total CAPEX for 1GW AI data center	29.4	52.4
China vs. US	Around 45% lower	

Source: Financial Times, SemiAnalysis, Epoch AI, Orennia, BofA Global Research estimates

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Power infrastructure: China benefits from a well-established domestic supply chain for transformers, switchgear, cables, and other electrical equipment, helping lower procurement costs and shorten deployment lead times.

Compute efficiency partly offsets China's cost advantage

While China can often build the underlying data center shell, power systems, and cooling infrastructure at lower cost, compute performance per megawatt remains largely determined by the underlying hardware. When measured by effective AI compute capacity rather than physical power capacity, China's infrastructure cost advantage narrows materially, as US companies benefit from access to Nvidia's latest-generation GPU platforms (e.g., Blackwell-based platforms). Based on our assumptions regarding theoretical compute throughput and model FLOPs utilization (MFU), cost per effective EFLOPS is around 75-90% higher for a 1GW data center with Huawei CloudMatrix 384 in China than a 1GW data center with Nvidia GB200 VL72 in US.

Overall, while China maintains a meaningful advantage in infrastructure construction costs, overall economics increasingly depend on AI accelerator performance, networking efficiency and software ecosystems.

Exhibit 98: Cost per effective EFLOPS is around 75-90% higher for a 1GW data center with Huawei CloudMatrix 384 in China than a 1GW data center with Nvidia GB200 NVL72 in U.S., according to our estimate

Cost comparison per EFLOPS for 1GW data center in China and US

Computing efficiency comparison	China	US
	Huawei CloudMatrix 384 (Ascend 910C)	Nvidia GB200 NVL72
Computing power per system (PFLOPS, FP16)	307	180
Total computing power for 1GW AI data center (EFLOPS, FP16)	550	1,385
Cost per EFLOPS (US\$mn)	53.4	37.9
China vs. U.S.	Around 40% higher	
Model Flops Utilization (%) – for model training	30 - 40%	40 - 50%
Effective computing power for 1GW AI data center (EFLOPS, FP16)	165 - 220	554 - 692
Cost per effective EFLOPS (US\$mn)	134 - 178	76 - 95
China vs. US	Around 75-90% higher	

Source: Financial Times, SemiAnalysis, Epoch AI, Orennia, CoreWeave, BofA Global Research estimates

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Humanoid robot

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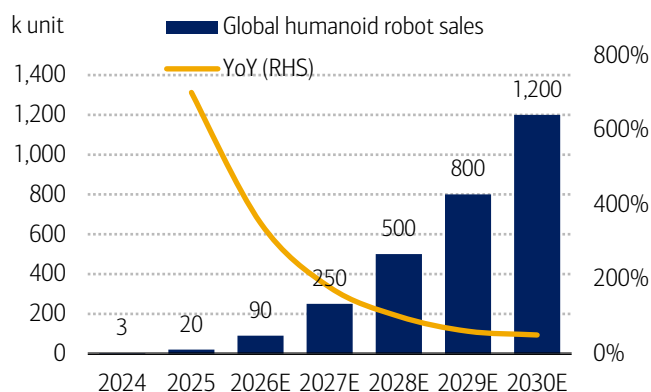
Robust shipment volume growth in 2026

We view the intelligence level of mainstream HR models currently remains at around L2 (i.e., basic multi-modal perception, execution, and reasoning capabilities), with locomotion (body movement) capability progressing faster than manipulation (hand operation). We expect HR to be gradually adopted in industrial & commercial scenarios in the coming 3-4 years, supported by better manipulation capabilities.

We forecast global humanoid robot shipment at 90k units in 2026, up from 20k in 2025. In 2030/35, we expect annual shipment volume of humanoid robots to reach 1.2mn/10.0mn units globally. This corresponds to 86% CAGR over 2025-35E. We are optimistic on long-term demand of humanoid robots, as massive adoption commences across household and service applications.

Exhibit 99: We expect 1.2mn annual shipment by 2030

BofA global humanoid robot shipment forecast

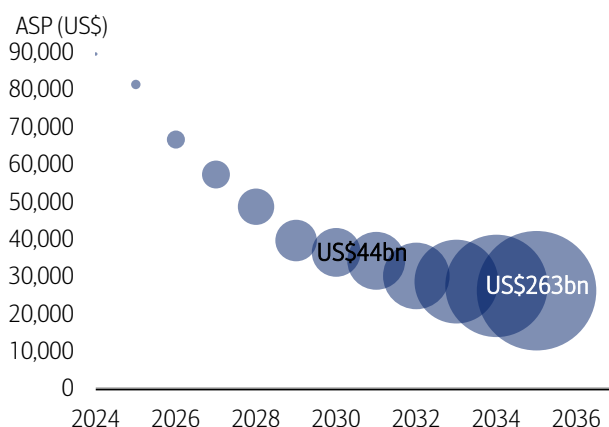


Source: BofA Global Research estimates

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Exhibit 100: Global market size to reach USD44bn/263bn by 2030/35

Global humanoid robot market size vs. ASP

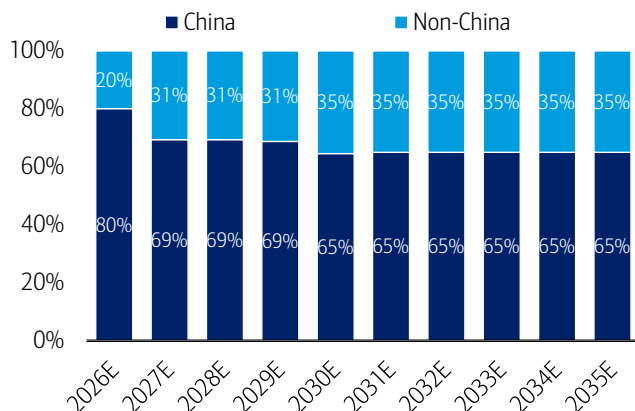


Source: BofA Global Research estimates; note: bubble size=market size

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Exhibit 101: We estimate China will represent 65% of global shipment in 2035E

Breakdown of global humanoid robot shipment volume

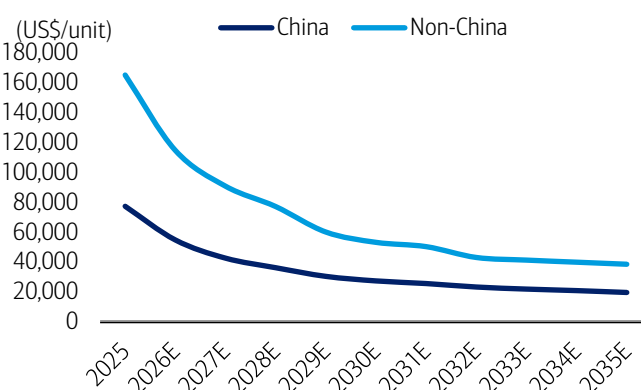


Source: BofA Global Research estimates

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Exhibit 102: We estimate ASP in China/non-China to decline to USD19.7k/USD38.6k in 2035E

ASP of humanoid robot in China & non-China markets



Source: BofA Global Research estimates

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We expect the global market size of humanoid robot to reach USD44bn/USD263bn by 2030/35, corresponding to a 66% CAGR over 2025-35E. This is based on our assumption of an ASP at USD37k/USD26k by 2030/35, vs. USD67k in 2026, supported by continuously falling BOM costs and manufacturing/development costs.

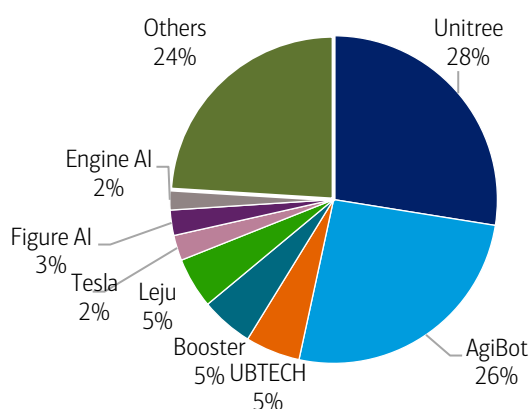
By area, we estimate China's humanoid robot market size to reach USD128bn by 2035E at a CAGR of 56% in 2025-35E, representing 49% of global market (by value). For non-China areas, we foresee a higher CAGR for market size (96% over 2025-35E) given faster growth in shipment volume (lower base).

Competition landscape: US and China take the lead

The US and China are leaders in developing humanoid-robot-related technologies. However, different companies in the US and China have different positioning along the value chain. US companies, such as **Tesla** and **Figure AI**, have advantages in software and algorithm resources, enabling them to develop general-purpose humanoid robots with internally developed embodied AI. On the other hand, Chinese companies, such as **Unitree** and **EngineAI**, focus more on motion control (small brain) and robot body. By leveraging the strong component supply chain in China, they can develop low-cost and high-mobility humanoid robots for secondary developers and system integrators.

Exhibit 103: Unitree recorded 28% global market share in 2025

Humanoid robot market share (by shipment volume) in 2025

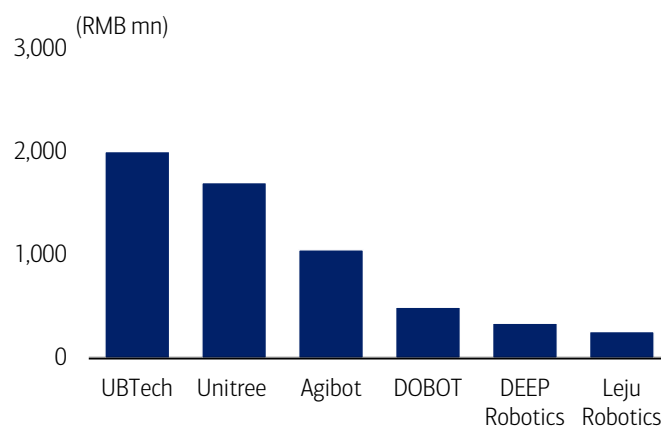


Source: BofA Global Research estimates, company reports

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Exhibit 104: UBTEch and Unitree have larger revenue size

Revenue for China's leading humanoid robot OEMs (2025)



Source: BofA Global Research estimates, company reports

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In recent years, some Chinese companies, such as **UBTech** and **Leju**, have also been seeking the integration of software (embodied AI algorithm) and hardware in specific applications, such as factories and warehouses. With data collected from real-world applications, companies can improve the capability of their embodied AI algorithms, further enabling wider adoption of their humanoid robots.

We see three types of companies in the field of humanoid robot, including (1) **robotics start-ups** such as Figure AI (US), Sanctuary AI (Canada), Agility (US), Unitree (China), UBTECH (China), Engine AI (China), Fourier Intelligence (China), etc.; (2) existing **auto OEMs and machinery companies**, including Tesla, XPeng, Xiaomi, BYD, Zoomlion, Estun, etc; and (3) **tech giants** (mainly providing embodied AI algorithms and training platforms for humanoid robots), such as NVIDIA, Meta, and ByteDance.

Exhibit 105: Leading names have accelerated product launches in 2025

Major humanoid robot models launched in recent years

Company	Country	Product name	Launching time	Height (m)	Weight (kg)	Degrees of freedom	Application	Selling price
AgiBot	China	A3	2026	1.73	55	>40	Industrial & commercial service	Not specified
XPeng	China	Iron	2025	1.78	70	82	Commercial service	Not specified
IX Technologies	Norway/US	NEO	2025	1.68	29.9	22 (hands)	Household assistance	USD20k
Unitree	China	H2	2025	1.80	70	31	Servicing, research and education	Not specified
Unitree	China	R1	2025	1.27	25	26	Service & industrial	USD5.6k
AgiBot	China	G2	2025	1.23-1.80	185	26	Manufacturing	Not specified
AgiBot	China	X2	2025	1.31	35	25	R&D, commercial service	Not specified
Figure AI	US	Figure 03	2025	1.73	61	Not specified	Household, mfg., commercial	Not specified
Fourier Intelligence	China	GR-3	2025	1.65	71	55	Interaction & companion	Not specified
Engine AI	China	SA02	2025	1.25	25	26+2 (fingers)	Companion	USD5.4k
UBTECH	China	Walker S2	2025	1.76	70	52	Manufacturing	Not specified
Sanctuary AI	Canada	Phoenix Gen 7	2024	1.70	70	20 (Hands)	Manufacturing	Not specified
Boston Dynamics	US	Atlas	2024	1.90	90	56	Industrial tasks	Not specified
Unitree	China	G1/G1 EDU	2024	1.32	35/35+	23/23-43	Research & education	USD16k (G1)
Tesla	US	Optimus Gen 2	2023	1.73	57	28 (body) + 11 per hand	Replace human in dangerous and repetitive tasks	USD20k-30k in the long run

Source: BofA Global Research estimates, companies

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Exhibit 106: US and China are home to the most-leading humanoid robot companies

US& China are market leaders



Source: BofA Global Research. HRIC: humanoid robot innovation center. *companies providing AI solutions for humanoid robots.

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From R&D to industrial & commercial applications

We estimate leading Chinese humanoid robot companies have received more than RMB2bn value of orders in 2025. Currently, the majority of orders come from R&D (tech giants, research institutions), data collection (local government-supported data collection centers), and manufacturing (mainly in the auto and industrial sectors).



Manufacturing applications are usually much more difficult than R&D and data collection, as they require a higher level of reliability, stability, and consistency, which translates into high requirements for the robustness and generalization capability in manipulation. Some humanoid robot companies such as Agility Robotics, UBTECH, and Leju have already deployed small batches of their humanoid robots in factories. According to UBTECH, the operating efficiency of its humanoid robot in handling parcels has reached 50% of that of a human worker, and the company targets to expand its application to sorting and inspection scenarios in 2026.

Exhibit 107: Major applications include R&D, data collection, and manufacturing

Summary of major orders of humanoid robot in China in 2025 and 2026 YTD

Date	Company	Order volume	Application	Customer
Mar-25	UBTECH	500 units	Guide service	Easyhome
Mar-25	UBTECH	20 units	Auto manufacturing	Dongfeng
Jul-25	UBTECH	RMB91mn	Auto manufacturing	Miyi (Shanghai)
Jul-25	AgiBot	RMB13mn	Data collection; performance	Zhuhai Embodied Intelligence Application Innovation Center
Jul-25	AgiBot	RMB78mn	Commercial service	China Mobile
Jul-25	Unitree	RMB46mn	Commercial service	China Mobile
Aug-25	AgiBot	Around 100 units	Manufacturing	Fulin Precision
Aug-25	Tiantai Robotics	10,000 units	Not specified	Shandong Future Robotics; Shandong Future Data; Gangzai Robotics
Sep-25	UBTECH	RMB30mn	Manufacturing	Miracle Automation
Sep-25	UBTECH	RMB250mn	Manufacturing	Leading domestic company
Sep-25	AstriBot	>1,000 units	Manufacturing; logistics	Shanghai Seer Intelligent
Sep-25	AI ² Robotics	RMB500mn	Manufacturing; logistics	Shenzhen Huizhi IoT
Oct-25	UBTECH	RMB126mn	Data collection	Guangxi Embodied Intelligence Data Collection and Testing Center
Oct-25	UBTECH	RMB32mn	Auto manufacturing; data collection	An automotive technology company
Oct-25	AgiBot	Nearly 1,000 units	Manufacturing	Longcheer
Oct-25	AgiBot	>RMB100mn	Manufacturing	Joyson Electronics
Nov-25	UBTECH	RMB159mn	Data collection	Zigong Humanoid Robot Data Collection Center
Nov-25	UBTECH	RMB264mn	Data collection; education	Fangchenggang Humanoid Robot Data Collection and Testing Center
Nov-25	UBTECH	RMB143mn	Data collection	Jiujiang Humanoid Robot Data Collection and Training Center
Dec-25	UBTECH	>RMB50mn	Integration & secondary development	Leading Chinese LLM company
Dec-25	UBTECH	RMB60mn	Data collection	Huizhou Huiyang Humanoid Robot Data Collection Center
Dec-25	UBTECH	RMB78mn	Data collection	Hohhot Embodied Intelligence Technology Center
Dec-25	EngineAI	3,000-4,000 units	Patrol & inspection	Not specified
Dec-25	Galbot	>1,000 units	Manufacturing	Zhejiang Baida Precision
Dec-25	Noetix	>1,000 units	Integration & secondary development	HCR
Jan-26	UBTECH	100 units	Manufacturing	Airbus
Feb-26	UBTECH	>RMB500mn	Auto manufacturing	BYD, Geely, Audi FAW, NIO etc.
Jun-26	UBTECH	>13,361 units	Emotional companionship	Commercial
Jun-26	Galbot	RMB236mn	Logistics, inspection & services	Hechuang Energy Management
Jun-26	Galbot	RMB39mn	Education	Taizhou Embodied Robotics Center (Phase II)

Source: Company report, BofA Global Research

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Wider application scenario for embodied AI

The recent WRC (World Robot Conference) has indicated improvement in monetization for humanoid robot. Compared with WRC 2025, where applications were primarily in R&D/industrial settings, WRC 2026 showcased a broader range of embodied AI use cases, extending into logistics, retail, services, and household scenarios. Among these, logistics emerged as one of the fastest-commercializing applications. Notably, X Square Robot demonstrated a parcel-sorting capacity of >1,800 parcels per hour using dual robotic arms, higher than the speed of human beings (1,200-1,400 parcels per hour). Many robotics companies (such as Galbot, Galaxea AI, Spirit AI, X Square Robot etc.) also demonstrated long-duration tasks in retail and household scenarios, such as folding clothes, preparing meals, and picking supermarket orders.

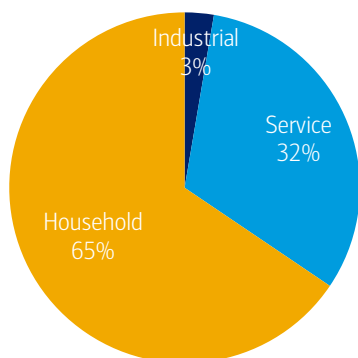
Total units in ownership to reach 3bn by 2060E

In the long term, assuming (1) humanoid robots can replace 20% of the workforce in the industrial sector (we believe the remaining industrial workforce could be replaced by industrial robots and mobile robots), and 50% of the workforce in the service sector; (2) a humanoid robot can replace 2.5/1.5 workers in the industrial/service industry; and (3) the penetration of humanoid robots can reach ~0.7 units per household in the steady

stage, we expect the total units in ownership (UIO) for humanoid robots to reach **3.0bn** globally. This corresponds to c.0.3 units per capita, slightly higher than that of passenger vehicles (0.2 units currently), but lower than that of smartphones (0.9 units currently). We expect service/household/industrial sector to account for 65%/32%/3% of the end-application of humanoid robot at the steady stage.

Exhibit 108: Consumer market to become the major end-market

Long-term application of humanoid robot breakdown

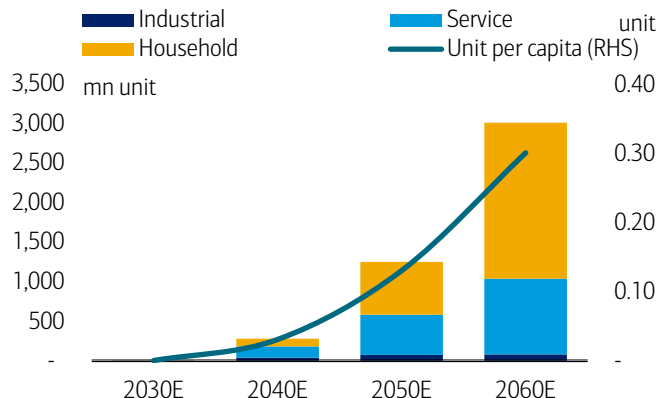


Source: BofA Global Research estimates

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Exhibit 109: We estimate total HR UIO to reach 3.0bn units by 2060

We estimate c. 0.3 units per capita by 2060



Source: BofA Global Research estimates

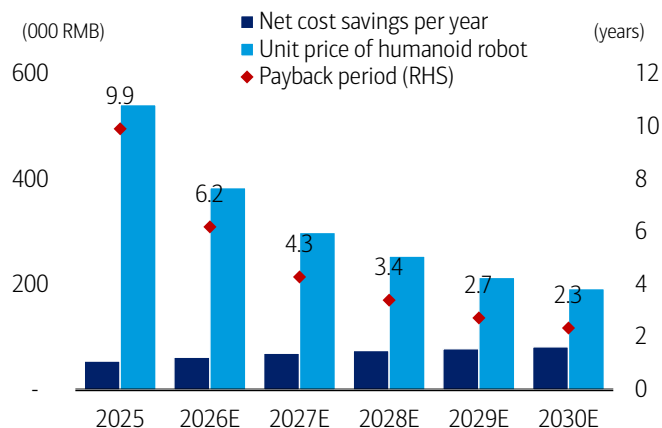
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Bottleneck #1 for monetization: cost & payback period

For 2026, we estimate the payback period for humanoid robot in China is around six years, due to the high initial purchasing cost (i.e., RMB384k/unit), lower efficiency level (i.e., c.45%) and lower labor cost. Looking ahead, we expect it to shorten to 2.3 years by 2030E, mainly driven by a decrease in ASP and an improvement in efficiency level. For developed markets, we estimate the payback period is lower given the high labor cost. We expect it to shorten to around 1 year by 2030E (vs. 3.2 years in 2026E).

Exhibit 110: Payback period in China to shorten to 2.3 years by 2030E

Payback period of humanoid robot in China

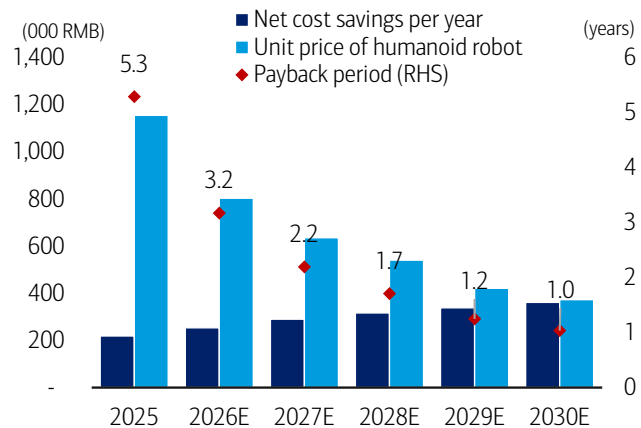


Source: BofA Global Research estimates

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Exhibit 111: Payback period in DM to shorten to 1 year by 2030E

Payback period of humanoid robot in developed markets (DM)



Source: BofA Global Research estimates

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Bottleneck #2 for monetization: real-world data crucial to model capability

Embodied AI algorithms remain the key bottleneck for humanoid robots over the next 3-5 years, as they lack robust generalization and adaptive capabilities. We believe the challenge of embodied AI algorithms mainly lies in (1) high-performance embodied AI algorithms, enabling robots to complete tasks in unseen situations and (2) availability of high-quality training data.

Exhibit 112: Major embodied AI models for humanoid robots

Key players include robot makers, tech giants, and embodied AI start-ups

Robot company		Tech giant		Embodied AI start-up/research institution	
Company	Embodied AI model	Company	Embodied AI model	Company	Embodied AI model
Tesla	Optimus AI stack (developed based on FSD & Grok)	Google DeepMind	RT-2, Gemini Robotics, Gemini Robotics-ER	Physical Intelligence	n0.7
Unitree	UnifoLM (UnifoLM-WMA-0, UnifoLM-VLA-0)	Nvidia	GR00T N1, DreamZero	Stanford	OpenVLA
AgiBot	GO-1	Alibaba	Qwen3.5	Skild AI	Skild Brain
Figure AI	Helix 02	ByteDance	GR-3	Galaxea	G0 Plus
1X Technologies	1X World Model			Spirit AI	Spirit-v1.5
UBTECH	Thinker			X Square	Wall-oss-v0.1
Galbot	GroceryVLA, NavFoM, GraspVLA etc.			GigaAI	GigaBrain-0.1
Robot Era	ERA-42			Dexmal	DM0
LimX	LimX VGM			Generalist AI	GEN-0

Source: BofA Global Research, company data

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Based on our takeaways from WRC, World action models (WAM) are increasingly adopted by embodied AI companies, which enables higher generalization capability and performs well in long-period tasks. Leading robotics companies are seeking to develop a unified base model that can control different body forms (wheeled, bipedal etc.). On the data front, the importance of high-quality real-world data has become a consensus among embodied AI developers. Egocentric data is increasingly used in pre-training, thanks to its cheaper cost and larger scale, while high-quality real-world robot data (teleoperation/failure-recovery trajectory) are used in post-training.

Exhibit 113: Galaxea AI's object sorting and delivery solution can achieve >95% success rate in retail scenario

Galaxea AI's demonstration in retail scenario

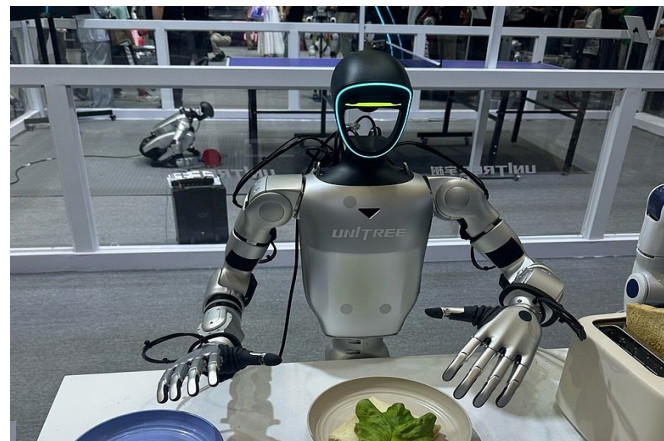


Source: BofA Global Research

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Exhibit 114: Unitree demonstrates object sorting in household scenario

Unitree's demonstration in household scenario



Source: BofA Global Research

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Currently, data availability is the key bottleneck for embodied AI developers. We estimate a general-purpose embodied AI model with high generalization capability would require 10mn-100mn hours of real-world data. Leading robotics companies now target to collect >1mn hours of real-world data by end-2026. Third-party data service providers (e.g. Lightwheel) also emerge to provide Egocentric/UMI data and simulation platform for model training.

BOM cost trend: to reach USD16k in China in 2030E

The structure of a typical humanoid robot can be divided into three major layers: AI system, motion control system, and robot body.

- **AI system** is the “**brain**” of the humanoid robot, mainly comprising AI chip and embodied AI algorithm. It oversees high-level information processing and decision making (including task decomposition, understanding surrounding environment and model inference, etc.), as well as interaction with humans.
- **Motion control system** is the “**little brain**” of the humanoid robot, mainly including controller and motion control algorithm. It is mainly in charge of motion coordination, body balancing, and route navigation.
- **Body** contains major hardware used for environmental data collection and execution of movement, including the vision system, sensory system, actuators, dexterous hand, energy system and structure materials, etc.

Notably, an increasing number of humanoid robots are adopting a unified embodied AI model to control the locomotion and manipulation of the robot (i.e., loco-manipulation), which blurs the boundary between the “brain” and “small-brain”.

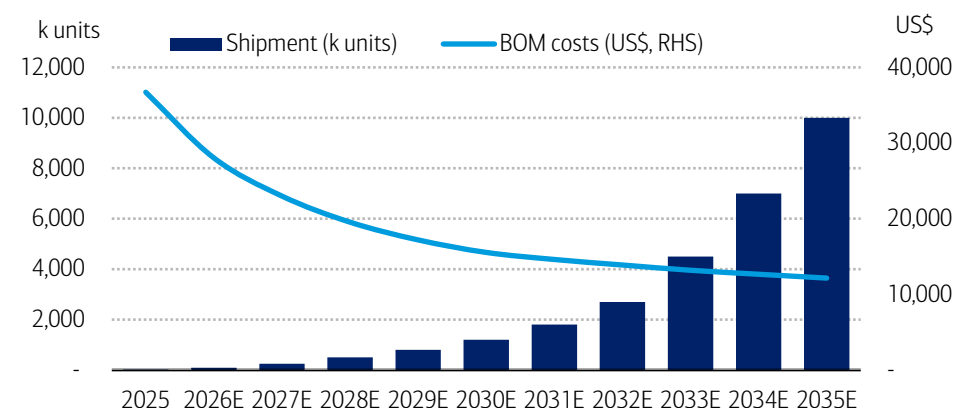
45% BOM cost reduction potential in 2026-30E

Cost reduction at humanoid robots has been faster than our original expectation in 2025, thanks to the rapid cost reduction of key components such as planetary roller screw and coreless motor upon mass production and localization. Based on our estimate, the total bill of materials (BOM) cost (or hardware cost) of a typical humanoid robot would be **USD28k** per unit by end-2026 (-24% YoY), if most of the components are made in China. This is based on the assumptions that: (1) a humanoid robot uses 16 rotary actuators and 14 linear actuators; (2) its rotary actuator uses harmonic reducers; (3) its linear actuator is based on planetary roller screw; (4) its dexterous hand has 12 DoFs, supported by direct-drive actuators; (5) its vision system includes one depth camera and one LiDAR; and (6) the humanoid robot mainly uses Chinese-made components, if possible, in order to save costs.

Looking ahead, we expect the BOM costs of a humanoid robot to fall below **USD16k/13k** per unit by 2030/35, thanks to economies of scale and improvements to component design. This corresponds to a 45% decline in the BOM cost over 2026-30, equivalent to a ~14% CAGR of decline. The CAGR of humanoid robot BOM cost decline during 2026-35 will be 9%, according to our estimate.

Exhibit 115: We expect humanoid robot's BOM cost to fall below USD16k by 2030, when annual shipment reaches 1.2mn

Humanoid robot shipment vs. BOM cost (in China)



Source: BofA Global Research estimates

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Rotary actuator/dexterous hand to have the highest content value by 2030

We believe **rotary actuators** will account for the largest portion of the BOM cost by 2030, with a 25% share of the total BOM cost, followed by dexterous hand (19%) and linear actuator (18%). By component, **processor chip** would have the highest content value per robot by 2030, with a 13% share in the total BOM cost, followed by coreless motors (9%), reducers (8%), encoders (8%), and planetary roller screw (8%). That said, for robots that adopt all-rotary actuator design (such as UBTECH's Waker S2 and Unitree G1), reducer will be the component with the highest content value in a humanoid robot. Among all components, we expect transmission components, including **reducer**, **planetary roller screw**, and **gear**, to see the greatest cost reduction in 2026-30 (c.50% cost decline), efficiency gains, and scaling effects.

Among major components of a humanoid robot, we believe **reducer**, **planetary roller screw**, and **sensor** have the highest entry barriers, apart from AI chips, given their high precision requirement in design and manufacturing, and difficulties in procuring manufacturing equipment.

Exhibit 116: We expect BOM cost to decline 45% by 2030 (the cost assumes > 80% key components are made in China)

Detailed BOM cost breakdown of a typical humanoid robot (updated in April 2026)

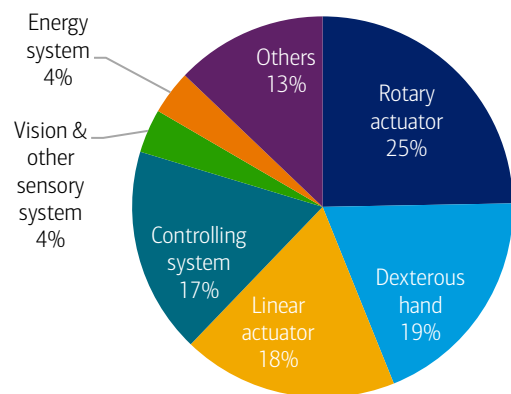
Category	Item	Unit cost (USD, 2025)	Unit cost (USD, 2026)	Unit cost (USD, 2030)	Number per robot	Total cost (USD, 2025)	Total cost (USD, 2026)	Total cost (USD, 2030)	Entry barrier
Rotary actuator	Frameless torque motor	79	64	36	16	1,257	1,029	571	Medium
	Reducer	171	157	79	16	2,743	2,514	1,257	High
	Servo drive	50	46	29	16	800	731	457	Medium
	Encoder	43	39	21	32	1,371	1,234	686	Medium
	Bearing	19	16	11	16	297	251	183	Relatively high
	1D torque sensor	114	100	43	16	1,829	1,600	686	Medium
	Subtotal					8,297	7,360	3,840	
Linear actuator	Frameless torque motor	79	64	36	14	1,100	900	500	Medium
	Planetary roller screw	571	214	86	14	8,000	3,000	1,200	High
	Servo drive	50	46	29	14	700	640	400	Medium
	Bearing	16	14	10	14	220	200	140	Relatively high
	Encoder	43	39	21	14	600	540	300	Medium
	Force sensor	57	50	21	14	800	700	300	Medium
	Subtotal					11,420	5,980	2,840	
Dexterous hand	Coreless motor	357	257	114	24	4,286	3,086	1,371	Relatively high
	Gear	29	25	11	24	686	600	257	Relatively high
	Mini screw	43	37	21	24	1,029	891	514	High
	Encoder	16	14	10	24	377	343	240	Medium
	Tactile sensor	103	97	50	12	1,234	1,166	600	High
	Subtotal					7,611	6,086	2,983	
Vision & other sensory system	Depth camera	514	500	357	1	514	500	357	High
	LiDAR	857	471	143	1	857	471	143	High
	IMU	143	121	79	1	143	121	79	Relatively high
	Subtotal					1,514	1,093	579	
Controlling system	Communication module	1,029	929	720	1	1,029	929	720	Medium
	Processor chip	2,999	2,857	2,000	1	2,999	2,857	2,000	High
	Subtotal					4,028	3,786	2,720	
Energy system	Battery	329	378	246	1	329	378	246	Medium
	Heat management	514	457	334	1	514	471	334	Medium
	Subtotal					843	849	581	
Structural material & others						3,000	2,880	2,000	
Total						36,713	28,034	15,542	

Source: BofA Global Research estimate; BOM cost does not include labor/overhead cost and software/algorithm/AI development cost

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Exhibit 117: Rotary actuators accounted for 25% of BOM cost

BOM cost breakdown (by module/system, 2030)

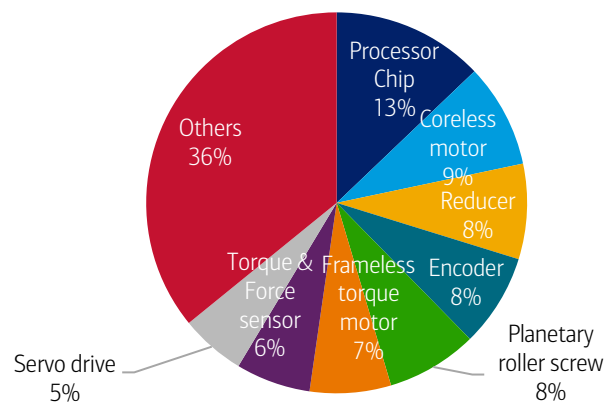


Source: BofA Global Research estimates

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Exhibit 118: Processor chip has the highest content value

BOM cost breakdown (by component, 2030)

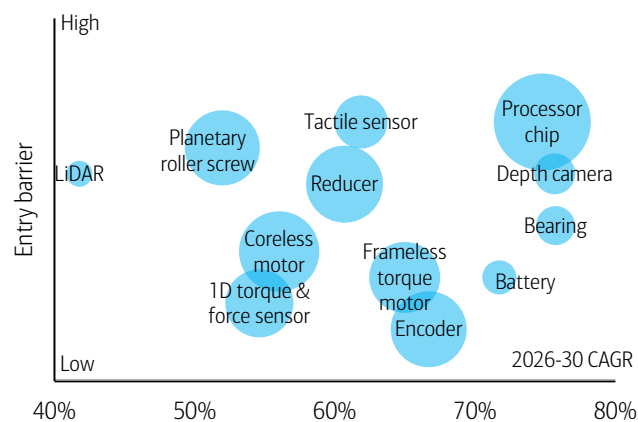


Source: BofA Global Research estimates

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Exhibit 119: Processor chip to witness the largest market size by 2030

Market size & growth comparison (by component)

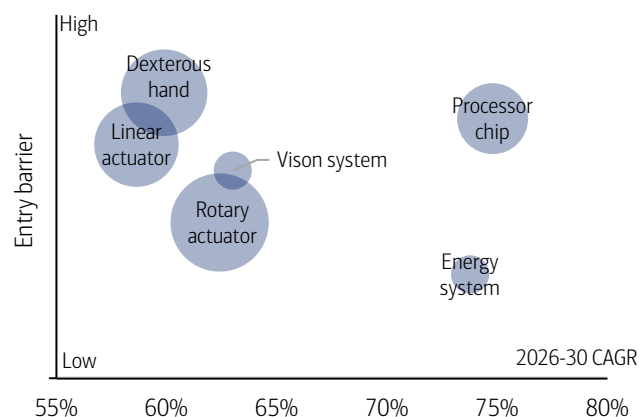


Source: BofA Global Research estimates. Bubble size represents 2030 market size (USDmn).

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Exhibit 120: Dexterous hand has the highest entry barrier

Market size & growth comparison (by subsystem)



Source: BofA Global Research estimates. Bubble size represents 2030 market size (USDmn).

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Glossary

ADTV: Average Daily Trading Value
 AGI: Artificial General Intelligence
 API: Application Programming Interface
 ARPU: Average Revenue Per User
 ARR: Annual Recurring Revenue
 ASP: Average Selling Price
 B2B: Business-to-Business
 BF16: Brain Floating Point 16-bit
 BOM: Bill of Materials
 CAGR: Compound Annual Growth Rate
 CAPEX: Capital Expenditure
 CCL: Copper Clad Laminate
 CLI: Command-Line Interface
 CMP: Chemical Mechanical polishing
 CPO: Co-Packaged Optics
 CPU: Central Processing Unit
 CSP: Cloud Service Provider
 CUDA: Nvidia's Compute Unified Device Architecture system
 DAU: Daily Active Users
 DDR: Double Data Rate
 DRAM: Dynamic Random-Access Memory
 DUV: Deep Ultraviolet
 EBITA: Earnings Before Interest, Taxes and Amortization
 EDA: Electronic Design Automation
 EFLOPS: Exa Floating-Point Operations per Second
 ERP: Enterprise Resource Planning
 FCF: Free Cash Flow
 FP16: 16-bit Floating Point
 FP32: 32-bit Floating Point
 FP4: 4-bit Floating Point
 FP64: 64-bit Floating Point
 FP8: 8-bit Floating Point
 FSD: Full Self-Driving
 GPU: Graphics Processing Unit
 HBM: High-Bandwidth Memory
 HPC: High-Performance Computing
 IC: Integrated Circuit
 IDC: Internet Data Center
 IMU: Inertial Measurement Unit
 INT16: 16-bit Integer
 INT8: 8-bit Integer
 IP: Intellectual Property
 JV: Joint Venture
 LLM: Large Language Model
 LPDDR: Low-Power Double Data Rate
 MFU: Model FLOPs Utilization
 MLC: Multi-Level Cell

MXFP4: Microscaling 4-bit Floating Point
NAND: NOT-AND Flash Memory
NOR: NOT-OR Flash Memory
OCS: Optical Circuit Switch
OSAT: Outsourced Semiconductor Assembly and Test
PCB: Printed Circuit Board
PFLOPS: Peta Floating-Point Operations per Second
POPS: Peta Operations per Second
PPA: Power, Performance and Area
QLC: Quad-Level Cell
SOTA: State of the Art
SRAM: Static Random-Access Memory
TDP: Thermal Design Power
TF32: TensorFloat-32
TFLOPS: Tera Floating-Point Operations per Second
TLC: Triple Layer Cell
TOPS: Tera Operations per Second
VLA: Vision-Language-Action
WFE: Wafer Fabrication Equipment

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Investment rating	Total return expectation (within 12-month period of date of initial rating)	Ratings dispersion guidelines for coverage cluster ^{R1}
Buy	≥ 10%	≤ 70%
Neutral	≥ 0%	≤ 30%
Underperform	N/A	≥ 20%

^{R1} Ratings dispersions may vary from time to time where BofA Global Research believes it better reflects the investment prospects of stocks in a Coverage Cluster.

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